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Part I

Quantum Symmetries and Spacetime Groups

Introduction

1 Symmetry in Quantum Physics

Symmetry transformations are operations carried out on the devices which prepare the states and the apparatuses which measure the observables of a quantum system having the property of not changing the results of experiments. For instance, space translations are symmetry transformations since the uniform translation of all preparations devices and measuring apparatuses does not change the results of experiments due to the homogeneity of space. Likewise, space rotations are symmetry transformations since the uniform rotation of all preparations devices and measuring apparatuses does not change the results of experiments due to the isotropy of space. In relativity, Lorentz boosts are symmetry transformations since as the result of experiments are the same in a given inertial frame and any inertial frame moving at uniform speed with respect to the first because the isotropy of spacetime.

In general, symmetry transformations gather together in families with special properties. Let G be one such family. Carrying out successively two symmetry transformations g, g' of G yields a third symmetry transformation $g' \cdot g$ of G called the **product** of g, g'. The effect of a symmetry transformation g of G can be undone by another symmetry transformation g^{-1} of G called the **inverse** of g. There exists a distinguished symmetry transformation e of G that has no effect called **neutral**. The multiplication and inversion operations and neutral element of G have the following properties:

- **Associativity**: for any three symmetry transformations g, g', g'' of G we have
$$(g'' \cdot g') \cdot g = g'' \cdot (g' \cdot g). \tag{1.0.1}$$
- **Neutrality**: there exists a symmetry transformation e of G such that for any symmetry transformation g of G we have
$$e \cdot g = g \cdot e = g. \tag{1.0.2}$$
- **Invertibility**: for any symmetry transformation g of G there exists a symmetry transformation g^{-1} of G such that
$$g^{-1} \cdot g = g \cdot g^{-1} = e. \tag{1.0.3}$$

These relations characterize G as an algebraic structure called a **group**. Since G is constituted by symmetry transformations, G is called more specifically a **symmetry group**. A group G is called **finite** if it is constituted by a finite number of elements, else it is called **infinite**. A group G is called **Abelian** if multiplication turns out to be commutative, that is

$$g \cdot g' = g' \cdot g, \quad \forall g, g' \in G. \tag{1.0.4}$$

Else, it is called non Abelian. The groups encountered in quantum physics are bot finite and infinite and Abelian and non Abelian.

TODO: Complete the example

Example (Space Translation Symmetry). [...]

TODO: Complete the example

Example (Space Rotation Symmetry). [...]

1.1 | Symmetry Groups and Operator Realization

In quantum theory, states and observables of a system are represented respectively by normalized vectors defined up to a multiplicative phase in a Hilbert space $\mathcal{H}$ and selfadjoint operators in the operator algebra $L(\mathcal{H})$. The invariance of experimental results under the action of a symmetry group reduces to the invariance of expectation values: if Ψ is a state, O is an observable and g is a symmetry transformation of G, ${}^g\Psi$ is the transformed state of Ψ and ${}^gO$ is the transformed observable of O, one must have

$$\langle {}^g\Psi|^g\, O\, |{}^g\Psi\rangle = \langle \Psi|\, O\, |\Psi\rangle\,. \tag{1.1.1}$$

Let us now analyze the implications of the invariance of expectation values under the action of a symmetry group G with a simple example.

Example (“yes–no” observables). As is well–known, there is a class of elementary observables, the “yes–no” observables, defined by taking the value 1, 0 according to whether the system is in a certain state Φ or not. The associated operator is given by

$$O_\Phi \Psi = \Phi \,\langle \Phi|\Psi\rangle\,.$$

It is natural to assume that under the symmetry transformation g of G the observable O_Φ transforms accordingly with the state Φ as follows:

$${}^gO_\Phi = O_{{}^g\Phi}.$$

The expectation value of O_Φ in the state Ψ' is given by the transition probability from Ψ' to Φ:

$$\langle \Psi'|\, O_\Phi\, |\Psi'\rangle = |\,\langle \Phi|\Psi'\rangle\,|^2,$$

such that the **invariance of expectation values** under the action of the symmetry group G implies that

$$|\,\langle {}^g\Phi|{}^g\Psi'\rangle\,|^2 = |\,\langle \Phi|\Psi'\rangle\,|^2.$$

This means that the symmetry transformation g of G must be implemented by a unitary or antiunitary operator U_g on the Hilbert space $\mathcal{H}$ of the system such that

$$|{}^g\Psi\rangle = U_g\,|\Psi\rangle\,, \quad {}^gO = U_g O U_g^{-1}.$$

In this example we have highlighted the importance of the **Wigner’s theorem**:

Theorem 1.1 (Wigner’s theorem)**.** *Let $\mathcal{H}$ be a Hilbert space and let G be a symmetry group of a quantum system represented on $\mathcal{H}$. Then, for each symmetry transformation g of G, there exists a unitary or antiunitary operator U_g on $\mathcal{H}$ such that*

$$\begin{cases} {}^g\Psi = U(g)\Psi, \\ {}^gO = U_g O U_g^\dagger, \end{cases} \tag{1.1.2}$$

since U_g is unitary or antiunitary, $U_g^{-1} = U_g^\dagger$, and so we can write the transformation of observables as $O \mapsto U_g O U_g^\dagger$. This implies that the expectation value of any observable O in any state Ψ is invariant under the action of the symmetry group G.

This is a fundamental result in quantum theory since it allows us to represent symmetry transformations as unitary or antiunitary operators on the Hilbert space of the system.

Recall that a unitary operator U on a Hilbert space $\mathcal{H}$ is a linear operator that preserves the Hilbert inner product,

$$U(\Phi\, a,\, \Phi'\, a') = U(\Phi)a,\, U(\Phi')a'$$
$$\langle U\Phi|U\Phi'\rangle = \langle\Phi|\Phi'\rangle\,,$$

while an antiunitary operator A on a Hilbert space $\mathcal{H}$ is an antilinear operator that preserves the Hilbert inner product,

$$A(\Phi\, a,\, \Phi'\, a') = A(\Phi)a^*,\, A(\Phi')a'^*$$
$$\langle A\Phi|A\Phi'\rangle = \langle\Phi|\Phi'\rangle^*\,.$$

in both cases the adjoint operator $U^\dagger$ (or $A^\dagger$) of a unitary (or antiunitary) operator U (or A) are equal to the inverse operator U^{-1} (or A^{-1}):

$$\begin{cases}\langle\Psi'|U\Psi\rangle = \left\langle U^\dagger\Psi'\middle|\Psi\right\rangle, \\ \langle\Psi'|A\Psi\rangle = (\left\langle A^\dagger\Psi'\middle|\Psi\right\rangle)^*,\end{cases} \qquad \Longrightarrow \qquad \begin{cases}U^{-1} = U^\dagger, \\ A^{-1} = A^\dagger,\end{cases}$$

respectively for the linear and antilinear case. In what follows, we leave aside the antiunitary case, which occurs only for symmetry transformations involving time reversal, and concentrate on the unitary one. Letting the symmetry transformation g vary in the group G, we obtain a family of unitary operators $U(g)$ contained in the operator algebra $L(\mathcal{H})$.

Let us come back to relation $(1.1.2)_1$. We recall again that state vectors are normalized vectors defined up to a multiplicative phase factor. The content of $(1.1.2)_1$ is that phase choice can be adjusted so that for any symmetry transformation g the state correspondence $\Psi \mapsto^g \Psi$ is codified by a unitary operator $U(g)$. This however does not completely fix $U(g)$: we are still free to replace $U(g)$ by

$$\hat{U}(g) = \alpha(g)U(g), \tag{1.1.3}$$

where $\alpha(g)$ is a complex number of unit modulus, which is called a **phase factor**. This involves a phase redefinition

$${}^g\Psi \to \alpha(g)\,{}^g\Psi,$$

in $(1.1.2)_1$, which does not change the physical content of the transformation since states are defined up to a multiplicative phase factor. By a suitable choice of the phase factors $\alpha(g)$ many relations simplyfy.

It is natural and possible to require

$${}^e\Psi = \Psi, \quad \forall\Psi \in \mathcal{H},$$

so that the neutral element e of the symmetry group G is represented by the identity operator

$$U(e) = \mathbb{I},$$

on the Hilbert space $\mathcal{H}$ of the system. With this choice, we can now analyze the implications of the group properties of G on the family of unitary operators $U(g)$. Considering two symmetry transformations g, g' of G, and a physical state Ψ, we have ${}^{g'}({}^g\Psi)$ and ${}^{g'\cdot g}\Psi$ defined only up to a conventional choice of a multiplicative **phase factor** as seen above, one cannot expect that a relation of the form ${}^{g'}({}^g\Psi) ={}^{g'\cdot g}\Psi$ to hold strictly in general, but only up to some phase factor

$$\gamma(g, g', \Psi) \in \mathbb{C}, \quad |\gamma(g, g', \Psi)| = 1,$$

depending on the symmetry transformations g, g' and the state Ψ. Thus we can write

$$^{g'}({}^{g}\Psi) = {}^{g'\cdot g}\,\Psi\,\gamma(g, g', \Psi).$$

Let us then express an important result concerning the phase factor $\gamma(g, g', \Psi)$ appearing in the above relation.

Proposition 1.2. *The phase factor $\gamma(g, g', \Psi)$ can be chosen to be independent of the state Ψ, thus obtaining a relation of the form*

$$U(g \cdot g') = \gamma(g, g')U(g)U(g'), \tag{1.1.4}$$

which is called a ***projective representation*** *of the symmetry group G. If the phase factor $\gamma(g, g')$ can be chosen to be 1 for all g, g' in G, then we have a* ***unitary representation*** *of the symmetry group G.*

Proof. Proof... □ **TODO:** Complete the proof if needed

The phase factors $\gamma(g, g')$ appearing in the projective representation of the symmetry group G are not arbitrary, but must satisfy some consistency conditions, for example the *2-cocyclicity*:

Proposition 1.3. *Given three symmetry transformations g, g', g'' of G, the following relation must hold:*

$$\gamma(g, g' \cdot g'')\gamma(g', g'') = \gamma(g \cdot g', g'')\gamma(g, g'), \tag{1.1.5}$$

where functions $\gamma : G \times G \to U(1)$ satisfying the 2-cocyclicity condition are called ***2-cocycles*** *of G.*

The 2-cocycle condition is a consequence of the associativity property of the group multiplication of G.

Proof. Proof... □ **TODO:** Complete the proof if needed

Proposition 1.4. *The G 2-cocycle $\gamma(g, g')$ enjoys a few special properties, which albeit technical are often useful at various points:*

$$\begin{aligned} \gamma(e, g) = \gamma(g, e) = 1, \\ \gamma(g, g^{-1}) = \gamma(g^{-1}, g), \end{aligned} \tag{1.1.6}$$

for any symmetry transformation g of G.

The first property is a consequence of the neutrality property of the group multiplication of G, while the second one is a consequence of the invertibility property of the group multiplication of G.

Proof. Proof... □ **TODO:** Complete the proof if needed

Relation (1.1.5) entails also that, for any symmetry transformation g, the operators $U(g^{-1})$ and $U(g)^{-1}$ differ by a g-dependent phase factor. Specifically, the relation between them takes the form

$$U(g^{-1}) = \varpi(g)U(g)^{-1}, \quad \varpi(g) = \gamma(g, g^{-1})^{-1}.$$

Proof. Proof... □ **TODO:** Complete the proof if needed

In general it is not possible to absorb the phase $\gamma(g', g)$ by redefining the operators $U(g)$ into the operators $\hat{U}(g)$ given in (1.1.3) with $\alpha(g)$ a suitable phase function. By doing so $\gamma(g', g)$ gets replaced by

$$\hat{\gamma}(g, g') = \gamma(g, g')\alpha(g \cdot g')\alpha(g)^{-1}\alpha(g')^{-1}, \tag{1.1.7}$$

and in general this cannot be made equal to 1 by a judicious choice of $\alpha(g)$. A phase function $\gamma(g', g)$ of the form

$$\gamma(g, g') = \alpha(g \cdot g')\alpha(g)^{-1}\alpha(g')^{-1}, \tag{1.1.8}$$

for some function $\alpha : G \to U(1)$ is called a **2-coboundary** of G.

Proposition 1.5. *A G 2-coboundary is a G 2-cocycle, that is, any function $\gamma(g, g')$ of the form* (1.1.8) *satisfies the 2-cocyclicity condition* (1.1.5).

TODO: Complete the proof if needed

Proof. Proof... □

The G 2–cocycle $\hat{\gamma}(g', g)$ in (1.1.7) can be rendered equal to 1 precisely when the G 2–cocycle $\gamma(g', g)$ is a G 2–coboundary of the form (1.1.8). This may or may not happen. The study of these matters is the object of a branch of algebra known as group cohomology. Some special results will be obtained below. At any rate, if $\hat{\gamma}(g', g)$ can be made equal to 1, the representation $\hat{U}(g)$ becomes genuinely unitary, so that

$$\hat{U}(g \cdot g') = \hat{U}(g)\hat{U}(g'),$$

for all symmetry transformations g, g' of G. Then the unitary family $U(g)$ forms a genuine **unitary representation** of the symmetry group G.

1.2 | Dynamical Symmetry Groups

Suppose that for a symmetry group G, the following is observed: by acting with any symmetry transformations of G on the devices that prepare the states of the system but not the apparatus that measures energy, the results of energy measurements do not change. The symmetry is then called **dynamical**, because *energy is the infinitesimal generator of time shifts*.

We note that also by acting with any symmetry transformations of G on both the devices preparing the states of the system and the apparatus measuring energy, the results of energy measurements do not change. Therefore, when G is a dynamical symmetry group, then the system's Hamiltonian H satisfies for any symmetry element g of G the relation

$$^{g}H = H. \tag{1.2.1}$$

If G acts on th Hilbert space through a unitary representation $U(g)$, then the invariance of the Hamiltonian under the action of G can be written as

$$U(g)HU(g)^{\dagger} = H, \quad \forall g \in G, \tag{1.2.2}$$

by virtue of Wigner's theorem $(1.1.2)_2$. This means that the Hamiltonian H commutes with all the unitary operators $U(g)$ representing the symmetry transformations of G:

$$[H, U(g)] = 0, \quad \forall g \in G.$$

This is a non trivial property severely constraining the form of H and thus providing useful structural information about the form of the interactions the system is involved in. Let us illustrate this point with a few examples taken from quantum theory.

Example (The Hydrogen Atom). The hydrogen atom is a system constituted by a proton and an electron interacting through the Coulomb potential. The Hamiltonian of the hydrogen atom is given by

$$H = \frac{\mathbf{p}^2}{2m}V(\mathbf{q}).$$

Direct observation shows that all proper rotations of the devices preparing the states of the atom leave the results of all energy measurements unchanged. Therefore, the proper rotation group SO(3) is dynamical. The atom has spherical symmetry, so from (1.2.2) we have that the Hamiltonian ha to remain unchanged under the action of any proper rotation $\mathbf{R}$ of SO(3):

$$U(\mathbf{R})HU(\mathbf{R})^{\dagger} = H, \quad \forall \mathbf{R} \in \mathrm{SO}(3).$$

It is sensible to assum that $^{\mathbf{R}}\mathbf{q} = \mathbf{R}^{-1}\mathbf{q}$ and $^{\mathbf{R}}\mathbf{p} = \mathbf{R}^{-1}\mathbf{p}$ under the action of a proper rotation. Then, we can write

$$U(\mathbf{R})\mathbf{q}U(\mathbf{R})^{\dagger} = \mathbf{R}^{-1}\mathbf{q}, \quad U(\mathbf{R})\mathbf{p}U(\mathbf{R})^{\dagger} = \mathbf{R}^{-1}\mathbf{p},$$

which means that the position and momentum operators transform as vectors under the action of SO(3).[1]

Using this last relation along with the invariance of the Hamiltonian under the action of SO(3) implies that

$$\frac{(\mathbf{R}^{-1}\mathbf{p})^2}{2m} + V(\mathbf{R}^{-1}\mathbf{q}) = \frac{\mathbf{p}^2}{2m} + V(\mathbf{q}), \quad \forall \mathbf{R} \in \mathrm{SO}(3).$$

[1] This is a consequence of the fact that they transform according to the fundamental representation of SO(3): $\mathbf{q} \mapsto \mathbf{R}^{-1}\mathbf{q}$, $\mathbf{p} \mapsto \mathbf{R}^{-1}\mathbf{p}$.

In particular, knowing that the rotations preserve the length of vectors, we have $(\mathbf{R}^{-1}\mathbf{p})^2 = \mathbf{p}^2$, so that the invariance of the Hamiltonian under the action of SO(3) reduces to

$$V(\mathbf{R}^{-1}\mathbf{q}) = V(\mathbf{q}) \qquad \Longrightarrow \qquad V(\mathbf{q}) = v(|\mathbf{q}|),$$

so that the potential V must be a function of the length of the position vector $v(|\mathbf{q}|)$ only. This is a non trivial constraint on the form of the potential V and thus on the form of the interactions the system is involved in; it is not sufficient to yield the Coulombic expression of the potential, but it is a necessary condition for it. The spherical symmetry of the hydrogen atom is a consequence of the dynamical symmetry of SO(3).

TODO: Complete the example

Example (A System of Identical Particles).

TODO: Complete the example

Example (Conduction Electrons in a Crystal Lattice).

1.3 | Energy Eigenvalues Classification

The energy spectrum of a quantum system typically contains many energy values. The problem then arises of devising a classification method for the energy eigenvalues. There does not exist any a priori ideal classification scheme. However, symmetry provides a natural one. In this section, we show that when the system is characterized by a dynamical symmetry group G, the energy eigenvalues can be efficiently classified based on the unitary representations of G according to which the energy eigenstates transform under the action of G.

We assume that the system's Hamiltonian H has a **completely discrete spectrum** and that each energy eigenvalue has **finite degeneracy**. Let E be an energy eigenvalue of H and let $\mathcal{H}_E$ be the corresponding eigenspace, which is d_E-dimensional given a degeneracy d_E; this subspace is spanned by all vectors Ψ which respect the eigenvalue equation

$$H\Psi = E\Psi.$$

Let now Ψ be a vector of $\mathcal{H}_E$, then for any symmetry transformation g of G we have

$$HU(g)\Psi = U(g)H\Psi = EU(g)\Psi,$$

as a direct consequence of (1.2.2). This means that $U(g)\Psi$ is also an eigenvector of H with the same eigenvalue E. Thus, the eigenspace $\mathcal{H}_E$ is invariant under the action of G. The unitary operators $U(g)$ representing the symmetry transformations of G on the Hilbert space $\mathcal{H}$ of the system restrict to unitary operators on the eigenspace $\mathcal{H}_E$, thus providing a unitary representation of G on $\mathcal{H}_E$. The energy eigenvalue E can be classified according to the unitary representation of G according to which the energy eigenstates in $\mathcal{H}_E$ transform under the action of G.

1.3.1 | Unitary Representations on the Eigenspaces

If we now consider $\{\Phi_i\}$ an ON basis for the subspace $\mathcal{H}_E$, then these vectors respect the eigenvalue equation and the orthonormality condition

$$\begin{cases} H\Phi_i = E\Phi_i, \\ \langle \Phi_i | \Phi_j \rangle = \delta_{ij}. \end{cases}$$

Since the unitary representation of G on $\mathcal{H}_E$ is given by the restriction of the unitary representation of G on $\mathcal{H}$, we have that for any symmetry transformation g of G and any vector Φ_i of the ON basis $\{\Phi_i\}$ of $\mathcal{H}_E$, the transformed vector $U(g)\Phi_i$ is a linear combination of the vectors of the ON basis $\{\Phi_i\}$:

$$U(g)\Phi_i = \sum_{j=1}^{d_E} \Phi_j D_{ji}(g), \tag{1.3.1}$$

where $D_{ji}(g)$ are the complex, g-dependent matrix elements of the unitary representation of G on $\mathcal{H}_E$. We aim to prove that the family of matrices $D(g)$ with elements $D_{ji}(g)$ forms a unitary representation of the symmetry group G on the eigenspace $\mathcal{H}_E$.

Proposition 1.6. *For each element g of the symmetry group G, there exists a $d_E \times d_E$ complex matrix $D(g) = (D_{ji}(g))$ which is unitary:*

$$\begin{gathered} D(g \cdot g') = D(g)D(g'), \\ D(g)^\dagger D(g) = D(g)D(g)^\dagger = \mathbb{I}, \end{gathered} \tag{1.3.2}$$

for all g, g' in G. The family of unitary matrices $D(g)$ forms a unitary representation of the symmetry group G on the eigenspace $\mathcal{H}_E$ corresponding to the energy eigenvalue E.

Proof. As $U(g)$ is a unitary representation of G on the Hilbert space $\mathcal{H}$ of the system, we know that for any g, g' in G we have

$$U(g \cdot g')\Phi_i = U(g)U(g')\Phi_i.$$

Since the vectors Φ_i form a basis for the eigenspace $\mathcal{H}_E$, the above equation implies that the matrix elements of $U(g)$ in this basis satisfy the same multiplication rule as the matrices $D(g)$, from (1.3.1):

$$U(g \cdot g')\Phi_i = \sum_{j=1}^{d_E} \Phi_j D_{ji}(g \cdot g'),$$

and

$$\begin{aligned} U(g)U(g')\Phi_i &= U(g)\sum_j \Phi_j D_{ji}(g') = \sum_j U(g)\Phi_j D_{ji}(g') \\ &= \sum_j \left(\sum_k \Phi_k D_{kj}(g) \right) D_{ji}(g') = \sum_k \Phi_k \left(\sum_j D_{kj}(g) D_{ji}(g') \right). \end{aligned}$$

Now if we put together the above two equations, we can prove the first relation in (1.3.2):

$$D_{ji}(g \cdot g') = \sum_k D_{kj}(g) D_{ji}(g'),$$

which is the multiplication rule for the matrices $D(g)$.

The second relation in (1.3.2) is a consequence of (1.3.1) and the orthonormality of the basis $\{\Phi_i\}$:

$$\begin{aligned} \delta_{ij} &= \langle \Phi_i | \Phi_j \rangle = \langle U(g)\Phi_i | U(g)\Phi_j \rangle \\ &= \left\langle \sum_k \Phi_k D_{ki}(g) \middle| \sum_l \Phi_l D_{lj}(g) \right\rangle = \sum_{k,l} D_{ki}(g)^* \langle \Phi_k | \Phi_l \rangle D_{lj}(g) \\ &= \sum_k D_{ik}(g)^\dagger D_{kj}(g) = (D(g)^\dagger D(g))_{ij}, \end{aligned}$$

where in the last steps we have transposed the indices and used the definition of the adjoint matrix $D(g)^\dagger$. This proves that $D(g)^\dagger D(g) = \mathbb{I}$. □

Having proven (1.3.2), we know $g \mapsto D(g)$ to be a unitary representation of the symmetry group G on the eigenspace $\mathcal{H}_E$ corresponding to the energy eigenvalue E, and from (1.3.1) we know the basis vectors Φ_i to transform under the action of G through $D(g)$.

The energy eigenvalue E can thus be classified according to the unitary representation $D(g)$, where the energy eigenstates in $\mathcal{H}_E$ transform under the action of G. The problem is that there are infinitely many orthonormal bases of the energy eigenspace $\mathcal{H}_E$ of E. The representation $D(g)$ depends on the chosen basis $\{\Phi_i\}$. So $D(g)$ by itself cannot be used to distinguish E from other energy eigenvalues. Let us analyze this point in greater detail.

Proposition 1.7. *Let $\{{\Phi^1}_i\}$ and $\{{\Phi^2}_i\}$ be two ON bases for the eigenspace $\mathcal{H}_E$ corresponding to the energy eigenvalue E. Then, let the corresponding unitary representations be $D^1(g)$ and $D^2(g)$, defined by (1.3.1) with respect to the bases $\{{\Phi^1}_i\}$ and $\{{\Phi^2}_i\}$, respectively.*

There exist a $d_E \times d_E$ unitary matrix A such that its elements A_{ij} relate the two bases $\{\Phi^1{}_i\}$ and $\{\Phi^2{}_i\}$ as

$$\Phi^2{}_i = \sum_{j=1}^{d_E} \Phi^1{}_j A_{ji}, \tag{1.3.3}$$

and the two unitary representations $D^1(g)$ and $D^2(g)$ are related by the similarity transformation

$$D^2(g) = A^{-1} D^1(g) A, \tag{1.3.4}$$

for all g in G. Thus, the two unitary representations $D^1(g)$ and $D^2(g)$ are equivalent representations of the symmetry group G.

Proof. Using (1.3.1), we can develop (1.3.3) as

$$\begin{aligned} U(g)\Phi^2{}_i &= U(g) \sum_{j=1}^{d_E} \Phi^1{}_j A_{ji} = \sum_{j=1}^{d_E} U(g)\Phi^1{}_j A_{ji} = \sum_{j=1}^{d_E} \left(\sum_{k=1}^{d_E} \Phi^1{}_k D^1_{kj}(g) \right) A_{ji} \\ &= \sum_{k=1}^{d_E} \Phi^1{}_k \left(\sum_{j=1}^{d_E} D^1_{kj}(g) A_{ji} \right) = \sum_{k=1}^{d_E} \Phi^1{}_k (D^1(g)A)_{ki}, \end{aligned}$$

on one hand, and

$$\begin{aligned} U(g)\Phi^2{}_i &= \sum_{k=1}^{d_E} \Phi^2{}_k D^2_{ki}(g) = \sum_{k=1}^{d_E} \left(\sum_{j=1}^{d_E} \Phi^1{}_j A_{jk} \right) D^2_{ki}(g) \\ &= \sum_{j=1}^{d_E} \Phi^1{}_j \left(\sum_{k=1}^{d_E} A_{jk} D^2_{ki}(g) \right) = \sum_{j=1}^{d_E} \Phi^1{}_j (AD^2(g))_{ji}. \end{aligned}$$

Putting together the above two equations, we have

$$(D^1(g)A)_{ki} = (AD^2(g))_{ki}, \quad \text{i.e., } D^1(g)A = AD^2(g),$$

which shows that the two unitary representations $D^1(g)$ and $D^2(g)$ are related by the similarity transformation $D^2(g) = A^{-1}D^1(g)A$. This proves that the two unitary representations $D^1(g)$ and $D^2(g)$ are equivalent representations of the symmetry group G. □

Thus two unitary representations $D^1(g)$ and $D^2(g)$ of the symmetry group G of the same dimension d_E, related as in (1.3.4), for some unitary matrix A, are said to be **unitarily equivalent**. Thus the unitary representation $D(g)$ in (1.3.1) is independent of the choice of the ON basis $\{\Phi_i\}$ for the eigenspace $\mathcal{H}_E$ up to unitary equivalence.

We can in this way characterize the energy eigenvalue E by a *unitary equivalence class of unitary representations*[2] of G and use this class to group E with other eigenvalues sharing the very same class. In this way, **a symmetry based classification of the energy spectrum is achieved**. Since however there are infinitely many representations of any given group, the classification scheme obtained may not be so useful. Fortunately, generic representations can be decomposed as combinations of a smaller set of more basic representations, which can then be used to refine the scheme.

[2] It is a class of equivalence of unitary representations, where the representations are unitarily equivalent.

1.3.2 | Reducibility and Irreducibility

Suppose that we are able to chose the orthonormal basis $\{\Phi_i\}$ of the energy eigenspace $\mathcal{H}_E$ to be of indexed $\Phi_{\alpha r}$ by two indices α, r in such a way that (1.3.1) takes the form

$$U(g)\Phi_{\alpha r} = \sum_s \Phi_{\alpha s} D_{\alpha s r}(g),$$

for every element g of G, for certain g-dependent coefficients $D_{\alpha s r}(g)$. Therefore, the vectors $\Phi_{\alpha r}$ for fixed α mix among themselves under the action of G. This means that, for fixed g, the matrix $D(g)$ has the block diagonal form

$$D(g) = \begin{pmatrix} D_1(g) & 0 & \cdots & 0 \\ 0 & D_2(g) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & D_a(g) \end{pmatrix},$$

and let us assume that the blocks $D_\alpha(g)$ are of dimension d_α, which is the smallest non trivial dimension for each block (i.e. the blocks cannot be further decomposed into diagonal blocks by a furhter change of basis $\Phi_{\alpha r}$). Then, the unitary representation $g \mapsto D(g)$ of G on $\mathcal{H}_E$ is said to be **reducible**, and its blocks $D_\alpha(g)$ are said to define an **irreducible** representations $g \mapsto D_\alpha(g)$ of G (*irreducible components*).

The energy eigenvalue E can then be classified through the irreducible representations $D_\alpha(g)$ of G. The former way of proceeding provides a classification scheme of the energy spectrum relying on a more restricted set of elements and so potentially more useful than that furnished by the latter.

We can resume the results obtained in this section as follows:

- Each energy eigenvalue E is characterized by a set of irreducible unitary representations D_α of G, unique up to unitary equivalence, describing how the eigenvectors belonging to E transform under the action of G.

- The energy eigenvalues can be grouped in subsets of eigenvalues sharing the same representation content D_α providing a classification scheme of the energy spectrum.

Let us illustrate the above results with a few examples.

Example (The One-Dimensional Harmonic Oscillator). The parity transformation P is space reflection about the origin. Observation show that the parity group $C_2 = \{\mathbb{I}, P\}$ is a dynamical symmetry group of the one–dimensional harmonic oscillator.

As well known, the energy eigenvalues of the one–dimensional harmonic oscillator are given by

$$H\Phi_n = E_n\Phi_n. \quad E_n = \hbar\omega(n + 1/2),$$

with $n = 0, 1, 2, \ldots$, and are non degenerate. The energy eigenstates of the one–dimensional harmonic oscillator transform under the action of the parity transformation P as

$$U(P)\Phi_n = \Phi_n(-1)^n.$$

Thus the parity group C_2 has two irreducible unitary representations: the trivial representation $D_+(P) = 1$ and the sign representation $D_-(P) = -1$. The energy eigenvalues of the one–dimensional harmonic oscillator are classified according to these two irreducible unitary representations of C_2: the energy eigenvalues with even n transform according to the trivial representation D_+, while those with odd n transform according to the sign representation D_-. Thus we classify them as even or odd energy eigenvalues according to the parity of n.

Example (The Hydrogen Atom). It can be shown that the proper rotation group SO(3) is a dynamical symmetry group of the hydrogen atom. The energy eigenvalues of the hydrogen atom are known to be given by

$$H\Phi_{nlm} = E_n\Phi_{nlm}, \quad E_n = -\frac{mc^2\alpha^2}{2n^2}.$$

The energy eigenvalues of the hydrogen atom are degenerate, with degeneracy $d_{E_n} = n^2$:

1. $n = 0, 1, 2, \dots,$
2. $l = 0, 1, \dots, n-1,$
3. $m = -l, -l+1, \dots, l-1, l,$

and the eigenstates Φ_{nlm} are also eigenstates of the total angular momentum operator L^2 and of the component along the quantization axis of the angular momentum operator L_z:

$$\begin{cases} L^2\Phi_{nlm} = \hbar^2 l(l+1)\Phi_{nlm}, \\ L_0\Phi_{nlm} = \hbar m\Phi_{nlm}. \end{cases}$$

Furthermore, the other spherical coordinates of the angular momentum $L_\pm$ act on the eigenstates Φ_{nlm} as

$$L_\pm\Phi_{nlm} = \hbar\sqrt{l(l+1) - m(m\pm 1)}\,\Phi_{nl(m\pm 1)},$$

thus acting as ladder operators on the index m.

Since $\mathbf{L}$ is the **infinitesimal generator** of the proper rotation group SO(3) (which will be discussed in the next section 2), the energy eigenstates Φ_{nlm} transform under the action of SO(3) according to the irreducible unitary representation $D_l(\mathbf{R})$ of SO(3), of dimension $d_l = 2l+1$:

$$U(\mathbf{R})\Phi_{nlm} = \sum_{m'=-l}^{l} \Phi_{nlm'} D_{lm'm}(\mathbf{R}),$$

so that $D_l(\mathbf{R})$ is $(2l+1)\times(2l+1)$ dimensional, and the mappings $\mathbf{R} \mapsto D_l(\mathbf{R})$ can be shown to be unitary irreducible representations of SO(3). Thus for each energy eigenvalue E_n of the hydrogen atom, the energy eigenstates Φ_{nlm} transform under the action of SO(3) according to the n irreducible $(2l+1)$-dimensional unitary representation $D_{l=0,1,\dots,n-1}$ of SO(3). In this way, the symmetry based classification of the energy spectrum is achieved for the hydrogen atom.

Example (The Sulfur Trioxide Molecule SO_3). The molecule of sulfur trioxide SO_3 has the shape of a planar equilateral triangle, with the sulfur atom at the center and the three oxygen atoms at the vertices. The symmetry group of this molecule is the dihedral group D_3, which is a subgroup of SO(3).

TODO: put image of the molecule

A planar equilateral triangle has the following symmetries:

- Three rotations of 120° about the axis perpendicular to the plane of the triangle, which we call C_3 rotations:

$$0^\circ : \begin{pmatrix} 123 \\ \downarrow \\ 123 \end{pmatrix}, \quad 120^\circ : \begin{pmatrix} 123 \\ \downarrow \\ 231 \end{pmatrix}, \quad 240^\circ : \begin{pmatrix} 123 \\ \downarrow \\ 312 \end{pmatrix}.$$

- Three reflections about the axes of symmetry of the triangle, which we call σ_v reflections:

$$\sigma_v^1 : \begin{pmatrix} 123 \\ \downarrow \\ 132 \end{pmatrix}, \quad \sigma_v^2 : \begin{pmatrix} 123 \\ \downarrow \\ 321 \end{pmatrix}, \quad \sigma_v^3 : \begin{pmatrix} 123 \\ \downarrow \\ 213 \end{pmatrix}.$$

Putting together the above symmetries, we find a total of six symmetries, which form the dihedral group D_3, which coincides with the symmetric group S_3 of permutations of three objects. The energy eigenvalues of the sulfur trioxide molecule SO_3 can be classified according to the irreducible unitary representations of D_3.

This group has three *unitary equivalence classes of irreducible unitary representations*: the trivial representation D_+, the sign representation D_- and a two-dimensional representation D_2. They are of dimensions $d_+ = 1$, $d_- = 1$ and $d_2 = 2$, respectively, and we can deduce their dimensionality through the following theorem:

Theorem 1.8. *The sum of the squares of the dimensions d_α of the irreducible unitary representations D_α of a finite non-abelian group G is equal to the order $|G|$ (the number of elements) of the group:*

$$\sum_\alpha d_\alpha^2 = |G|.$$

In our case the only way to satisfy the above relation is to have $d_+^2 + d_-^2 + d_2^2 = 1^2 + 1^2 + 2^2 = 6$, which is the order of D_3. The energy eigenvalues of the sulfur trioxide molecule SO_3 can be classified according to these three irreducible unitary representations of D_3.

2 Lie Groups, Algebras and Representations

A Lie group G is an infinite group that can be parametrized in a neighborhood of the neutral element by continuous parameters θ^a, $a = 1, \dots, d$, organized as a vector $\underline{\theta}$, in one–to–one fashion:

$$\begin{cases} \underline{\theta} \in D \subseteq \mathbb{R}^d, \\ g \,:\, D \mapsto G, \\ \underline{\theta} \mapsto g(\underline{\theta}), \end{cases}$$

where D is an open neighborhood of the origin in $\mathbb{R}^d$ ($\underline{0} \in D$ and $g(\underline{0}) \in G$). The number d of parameters is called the dimension of G, seen as a differential map. The group operations of multiplication and inversion are smooth functions of the parameters.

If the parametrization is smooth, as we assume, G can be thought geometrically as a group manifold. Below, we treat G as a matrix group for definiteness.

We assume that 0 belongs to the parameter space and that

$$g(\underline{0}) = 1,$$

which can be read as a normalizing condition. By virtue of the parametrization, for any pair of parameter vectors $\underline{\theta}$, $\underline{\theta}'$, there exists a third parameter vector $\underline{\theta}''$ such that

$$g(\underline{\theta}')g(\underline{\theta}) = g(\underline{\theta}'').$$

As the parametrization is one–to–one,[1] $\underline{\theta}''$ is also uniquely determined by $\underline{\theta}$, $\underline{\theta}'$. So, it can be expressed as a function of them

$$\underline{\theta}'' = \underline{m}(\underline{\theta}', \underline{\theta}),$$

which is a function of the **group multiplication**. The function $\underline{m}$ is smooth as the group multiplication is smooth. The inverse of $g(\underline{\theta})$ is given by

$$g(\underline{\theta})^{-1} = g(\underline{\theta}'),$$

for some $\underline{\theta}' = \underline{j}(\underline{\theta})$, which is unique since the parameterization is one-to-one. The function $\underline{j}$, implementing the group inversion, is smooth as the group inversion is smooth.

We have thus found two smooth functions $\underline{m}$, $\underline{j}$ that encode the group multiplication and inversion in terms of the parameters $\underline{\theta}$:

$$\begin{cases} \underline{m} : D \times D \to D, \\ \underline{j} : D \to D, \end{cases} \quad \text{such that} \quad \begin{cases} g(\underline{\theta}')g(\underline{\theta}) = g(\underline{m}(\underline{\theta}', \underline{\theta})), \\ g(\underline{j}(\underline{\theta})) = g(\underline{\theta})^{-1}. \end{cases} \tag{2.0.1}$$

[1] This means that each element of the group corresponds to a unique set of parameters: $g(\underline{\theta}) = g(\underline{\theta}') \implies \underline{\theta} = \underline{\theta}'$.

The group properties of associativity, invertibility and neutrality, which are satisfied by the group multiplication and inversion, translate into properties of the functions $\underline{m}$, $\underline{j}$ as follows:

Proposition 2.1. *The following relations must hold:*

$$\begin{aligned} m(m(\underline{\theta},\underline{\theta}'),\underline{\theta}'') &= m(\underline{\theta},m(\underline{\theta}',\underline{\theta}'')), \\ m(\underline{\theta},j(\underline{\theta})) &= m(j(\underline{\theta}),\underline{\theta}) = \underline{0}, \\ m(\underline{\theta},\underline{0}) &= m(\underline{0},\underline{\theta}) = \underline{\theta}. \end{aligned} \tag{2.0.2}$$

Given any choice of three parameter vectors $\underline{\theta}$, $\underline{\theta}'$, $\underline{\theta}''$. These identities constrain the form of the group multiplication function m and the inversion function j to an important extent.

Proof. The first identity is a consequence of the associativity of the group multiplication: using the definition of the multiplication function m along with the associativity of the group multiplication, we find

$$\begin{aligned} g(m(m(\underline{\theta},\underline{\theta}'),\underline{\theta}'')) &= g(m(\underline{\theta},\underline{\theta}'))g(\underline{\theta}'') = g(\underline{\theta})g(\underline{\theta}')g(\underline{\theta}'') \\ &= g(\underline{\theta})g(m(\underline{\theta}',\underline{\theta}'')) = g(m(\underline{\theta},m(\underline{\theta}',\underline{\theta}''))), \end{aligned}$$

which implies the first identity in (2.0.2)$_1$ by the one-to-one nature of the parametrization (otherwise we would have $g(\underline{\theta}) = g(\underline{\theta}')$ for some $\underline{\theta} \neq \underline{\theta}'$). The second identity is a consequence of the invertibility of the group multiplication: using the definition of the inversion function j along with the invertibility of the group multiplication, we find

$$\begin{aligned} g(m(\underline{\theta},j(\underline{\theta}))) &= g(\underline{\theta})g(j(\underline{\theta})) = g(\underline{\theta})g(\underline{\theta})^{-1} = 1 = g(\underline{0}), \\ g(m(j(\underline{\theta}),\underline{\theta})) &= g(j(\underline{\theta}))g(\underline{\theta}) = g(\underline{\theta})^{-1}g(\underline{\theta}) = 1 = g(\underline{0}), \end{aligned}$$

proving both sides of the second identity in (2.0.2)$_2$. The third identity is a consequence of the neutrality of the group multiplication: using the definition of the multiplication function m along with the neutrality of the group multiplication, we find

$$g(m(\underline{0},\underline{\theta})) = g(\underline{\theta}), \quad g(m(\underline{\theta},\underline{0})) = g(\underline{\theta}),$$

proving both sides of the third identity in (2.0.2)$_3$. ☐

2.0.1 | Local Structure of the Group

The idea now is to understand the structure of G by looking at the behavior of $g(\underline{\theta})$ in a neighborhood of the neutral element. This is done by Taylor expanding in power series around $\underline{0}$ as:

$$g(\underline{\theta}) = g(\underline{0}) + \left.\frac{\partial g}{\partial\theta^a}\right|_{\underline{0}}\theta^a + \frac{1}{2}\left.\frac{\partial^2 g}{\partial\theta^a\partial\theta^b}\right|_{\underline{0}}\theta^a\theta^b + \mathrm{O}(\theta^3).$$

Now computing individual terms of the expansion, we have

$$g(\underline{0}) = 1, \quad t_a = \left.\frac{\partial g}{\partial\theta^a}\right|_{\underline{0}}, \quad t_{ab} = \left.\frac{\partial^2 g}{\partial\theta^a\partial\theta^b}\right|_{\underline{0}} = t_{ba},$$

so that

$$g(\underline{\theta}) = 1 + t_a\theta^a + \frac{1}{2}t_{ab}\theta^a\theta^b + \mathrm{O}(\theta^3). \tag{2.0.3}$$

Let us highlight the fact that the coefficients t_a, t_{ab} are matrices, as $g(\underline{\theta})$ is a matrix group. t_{ab} in particular is symmetric in its indices a, b since the order of differentiation does not matter.

We can now think of G as an **hypersurface in the space of matrices**, containing the identity matrix. In the neighborhood of the identity, there is D, subset of our hypersurface, which is diffeomorphic to an open neighborhood of the origin in $\mathbb{R}^d$ (the domain of our parametrization).

If we vary $g(0,\ldots,0,\theta^a,0,\ldots,0)$, we get a curve in D passing through the identity (when $\theta^a = 0$). The tangent vector to this curve at the identity is given by $t_a = \frac{\partial g}{\partial \theta^a}\big|_{\underline{0}}$. Varying all parameters θ^a we get d curves in D passing through the identity and their tangent vectors at the identity are given by t_a, $a = 1,\ldots,d$. These tangent vectors are **linearly independent** since the parametrization is one–to–one, thus they span a vector space of dimension d.

The tangent space to G at the identity is more than a vector space of dimension d (spanned by the matrices t_a, $a = 1,\ldots,d$, costituing a basis). We denote this tangent space by $\mathfrak{g}$ and call it the **Lie algebra** of G. It is a vector space of dimension d and it is closed under the commutator of matrices: the Lie algebra $\mathfrak{g}$ encodes all the information about the local structure of G around the identity element, but this will be clarified in a moment.

We could now think to Taylor expand the group multiplication functions m, j around $\underline{0}$ and try to find relations among the coefficients of the expansion which encode the group structure. Starting from the multiplication function, we find that the coefficients of the expansion of m must satisfy

$$m^a(\underline{\theta},\underline{\theta}') = \theta^a + \theta'^a + m^a{}_{bc}\theta^b\theta'^c + \mathrm{O}(\underline{\theta}^3), \tag{2.0.4}$$

while for the inversion property we find

$$j^a(\underline{\theta}) = -\theta^a + j^a{}_{bc}\theta^b\theta^c + \mathrm{O}(\underline{\theta}^3), \tag{2.0.5}$$

where $m^a{}_{bc}$, $j^a{}_{bc}$ are the coefficients of the expansions. These coefficients are not independent, as it can be proven that they satisfy:

$$j^a{}_{bc} = m^a{}_{bc} + m^a{}_{cb}.$$

If we now recall the Taylor expansion of $g(\underline{\theta})$ around $\underline{0}$

$$g(\underline{0}) = 1 + t_a\theta^a + \frac{1}{2}t_{ab}\theta^a\theta^b + \mathrm{O}(\theta^3),$$

we can note how the coefficients t_a define another member of the Lie algebra $\mathfrak{g}$: the Lie algebra $\mathfrak{g}$ is the tangent space to G at the identity, so it is spanned by the matrices t_a, $a = 1,\ldots,d$, which we have said to be closed under the commutator of matrices. Let us now define the commutator of two basis elements t_a, t_b of $\mathfrak{g}$ as follows:

Definition 2.1 (Lie Bracket)**.** The **commutator**, or **Lie bracket**, of two basis elements t_a, t_b of $\mathfrak{g}$ is defined as follows:

$$[t_a, t_b] = t_a t_b - t_b t_a = f_{ab}{}^c t_c \in \mathfrak{g},$$

where $f_{ab}{}^c$ are the **structure constants** of the Lie algebra $\mathfrak{g}$, encoding the properties of the Lie algebra $\mathfrak{g}$ and thus of the local structure of the Lie group G around the identity element.

Having defined the Lie bracket of two basis elements t_a, t_b of $\mathfrak{g}$, we can now prove the following proposition:

Proposition 2.2. *The structure constants are antisymmetric in their lower indices a, b since the commutator is antisymmetric. These structure constants are related to the coefficients $m^a{}_{bc}$ of the expansion of the group multiplication function by*

$$f^a{}_{bc} = m^a{}_{bc} - m^a{}_{cb}.$$

If we change parametrization, the structure constants change accordingly, but the Lie algebra $\mathfrak{g}$ remains the same.

Proof. Taking again the definition of the multiplication function

$$g(\underline{\theta}')g(\underline{\theta}) = g(m(\underline{\theta}', \underline{\theta})),$$

we can Taylor expand both sides around $\underline{0}$ and find the following relation between the coefficients of the expansion of g and the structure constants:

$$\begin{aligned}&\left[1 + \theta^a t_a + \frac{1}{2}\theta^a\theta^b t_{ab} + \mathrm{O}(\theta^3)\right]\left[1 + \theta'^a t_a + \frac{1}{2}\theta'^a\theta'^b t_{ab} + \mathrm{O}(\theta'^3)\right] \\ &= 1 + \left(\theta^a + \theta'^a + m^a{}_{bc}\theta^b\theta'^c\right) t_a + \frac{1}{2}\left(\theta^a + \theta'^a\right)\left(\theta^b + \theta'^b\right) t_{ab} + \mathrm{O}(\theta^2, \theta') + \mathrm{O}(\theta, \theta'^2).\end{aligned}$$

Relating the coefficients of the expansion of both sides in $\theta^a\theta'^b$, we find the following relations (renaming summation indices):

$$t_a t_b = t_{ab} + m^c{}_{ab} t_c,$$
$$t_b t_a = t_{ba} + m^c{}_{ba} t_c,$$

but since $t_{ab} = t_{ba}$, if we subtract the second from the first, we find

$$[t_a, t_b] = (m^c{}_{ab} - m^c{}_{ba})t_c = f_{ab}{}^c t_c,$$

so that the structure constants $f_{ab}{}^c$ of the Lie algebra $\mathfrak{g}$ are related to the coefficients $m^c{}_{ab}$ of the expansion of the group multiplication function m by

$$f^c{}_{ab} = m^c{}_{ab} - m^c{}_{ba}.$$

□

Thus to sum up:

- The Lie algebra $\mathfrak{g}$ of a Lie group G is the tangent space to G at the identity element, spanned by the matrices t_a, $a = 1, \dots, d$.
- The basis elements t_a of the Lie algebra $\mathfrak{g}$ satisfy the commutation relations

  $$[t_a, t_b] = f_{ab}{}^c t_c,$$

  where $f_{ab}{}^c$ are the structure constants of the Lie algebra $\mathfrak{g}$.
- The structure constants $f_{ab}{}^c$ of the Lie algebra $\mathfrak{g}$ are related to the coefficients $m^c{}_{ab}$ of the expansion of the group multiplication function m by

  $$f^c{}_{ab} = m^c{}_{ab} - m^c{}_{ba}.$$

The Lie algebra, the tangent space to the Lie group at the identity, is not only a vector space, but also an algebra, since it is closed under the commutator of matrices. It encodes all the information about the local structure of the Lie group around the identity element. In particular, the structure constants $f_{ab}{}^c$ determine the commutation relations of the basis elements t_a of the Lie algebra, which in turn determine the local structure of the Lie group G around the identity element.

The commutator of the basis elements t_a of the Lie algebra $\mathfrak{g}$ has some properties, which are the **Jacobi identity** and the **antisymmetry** of the commutator:

$$\begin{cases} [t_a,\, t_b] + [t_b,\, t_a] = 0, \\ [t_a,\, [t_b,\, t_c]] + [t_b,\, [t_c,\, t_a]] + [t_c,\, [t_a,\, t_b]] = 0. \end{cases}$$

These two properties can be expressed in terms of the structure constants ${f_{ab}}^c$ as follows:

$$\begin{cases} {f_{ab}}^c + {f_{ba}}^c = 0, \\ {f_{ad}}^e {f_{bc}}^d + {f_{bd}}^e {f_{ca}}^d + {f_{cd}}^e {f_{ab}}^d = 0, \end{cases}$$

which are the **antisymmetry** of the structure constants in their lower indices and the **Jacobi identity** for the structure constants, respectively. This means that not every set of numbers ${f_{ab}}^c$ can be the structure constants of a Lie algebra, but only those that satisfy the antisymmetry and Jacobi identity properties.

2.0.2 | Exponential Parametrization of Lie Groups

Given the parametrization of a Lie group G in a neighborhood of the identity element

$$g \,:\, D \mapsto G, \quad \underline{\theta} \in D \mapsto g(\underline{\theta}) \in G,$$

is it possible to find a global parametrization of G in terms of the elements of its Lie algebra $\mathfrak{g}$? The answer is yes, and the way to do it is through the *exponential map*.

If we consider another parametrization of G in a neighborhood of the identity element defined by

$$\tilde{g}(\underline{\theta}) = \lim_{N\to\infty} g\left(\tfrac{\underline{\theta}}{N}\right)^N,$$

which if the group is suitably closed under the group multiplication, is well defined and provides a global parametrization of G in terms of the elements of its Lie algebra $\mathfrak{g}$:

$$\tilde{g}(\underline{\theta}) = e^{\theta^a t_a}.$$

This new parametrization is called the **exponential parametrization** of G and the map $\tilde{g} : \mathbb{R}^d \to G$ is called the **exponential map**.

Proof. Let us show that the limit defining $\tilde{g}(\underline{\theta})$ exists and is equal to $e^{\theta^a t_a}$. Starting from the definition of $\tilde{g}(\underline{\theta})$, we can Taylor expand $g\left(\underline{\theta}/N\right)$ around $\underline{0}$ as follows:

$$g\left(\frac{\underline{\theta}}{N}\right)^N = \left[1 + \frac{t_a\theta^a}{N} + \frac{1}{2}\left(\frac{t_a\theta^a}{N}\right)^2 + \mathrm{O}\left(\frac{1}{N^3}\right)\right]^N \sim \left(1 + \frac{t_a\theta^a}{N}\right)^N \xrightarrow{N\to\infty} e^{t_a\theta^a},$$

which is the exponential of a matrix, defined by Euler's formula as the power series expansion

$$e^x = \lim_{n\to\infty} (1 + \frac{x}{n})^n = \sum_{k=0}^{\infty} \frac{x^k}{k!}.$$

$\square$

Thus, the exponential map provides a global parametrization of the Lie group G in terms of the elements of its Lie algebra $\mathfrak{g}$ as follows:

$$\tilde{g}(\underline{\theta}) = e^{t_a\theta^a} = 1 + \tilde{t}_a\theta^a + \frac{1}{2}\theta^a\theta^b\tilde{t}_{ab} + \mathrm{O}(\theta^3),$$

which is the same expansion did in (2.0.3). But when comparing the coefficients to the ones of the expansion of $e^{i\theta^a t_a}$, we get the following relations:

$$\tilde{t}_a = t_a, \quad \tilde{t}_{ab} = \tfrac{1}{2} t_a t_b + t_b t_a.$$

The same goes for the he group multiplication function $\underline{m}$ and the inversion function $\underline{j}$. We can define new functions $\underline{\tilde{m}}$, $\underline{\tilde{j}}$ as

$$\tilde{g}(\underline{\tilde{m}}(\underline{\theta}, \underline{\theta}')) = \tilde{g}(\underline{\theta}')\tilde{g}(\underline{\theta}),$$
$$\tilde{g}(\underline{\tilde{j}}(\underline{\theta})) = \tilde{g}(\underline{\theta})^{-1},$$

which can be thought as the group multiplication and inversion functions in the exponential parametrization. If we now Taylor expand these new functions around $\underline{0}$ we find that for the inversion function it is as the previous expansion (2.0.5), but all the orders higher than the first vanish:

$$\tilde{j}(\underline{\theta}) = -\theta^a, \quad \tilde{j}^a_{\ bc} = 0$$
$$e^{\underline{\theta}^a t_a} = \tilde{g}(\tilde{j}(\underline{\theta})) = \tilde{g}(\underline{\theta})^{-1} = e^{-t_a \theta^a}.$$

For the multiplication function we have instead the **Campbell-Baker-Hausdorff function** so that the expansion of the multiplication function in the exponential parametrization is given by

$$\tilde{m}^a(\underline{\theta}, \underline{\theta}') = \theta^a + \theta'^a + \tilde{m}^a_{\ bc}\theta^b\theta'^c + \mathrm{O}(\underline{\theta}^3),$$

which is the same of (2.0.4) but with different coefficients $\tilde{m}^a_{\ bc}$ of the expansion, which are related to the structure constants $f^a_{\ bc}$ of the Lie algebra $\mathfrak{g}$ by

$$\tilde{m}^a_{\ bc} = \tfrac{1}{2} f^a_{\ bc}.$$

These new structure constants are not different from the original ones, since the exponential map is a reparametrization of the Lie group G in terms of the elements of its Lie algebra $\mathfrak{g}$: $\tilde{t}_a = t_a$ implies that

$$f^c_{\ ab} t_c = [t_a,\, t_b] = \left[\tilde{t}_a,\, \tilde{t}_b\right] = \tilde{f}^c_{\ ab}\tilde{t}_c = \tilde{f}^c_{\ ab} t_c,$$

thus the structure constants of the Lie algebra $\mathfrak{g}$ are the same in both parametrizations, as expected since the Lie algebra is an intrinsic property of the Lie group G and does not depend on the choice of parametrization:

$$f^c_{\ ab} = \tilde{f}^c_{\ ab}.$$

Thus we have

$$\tilde{m}^a_{\ bc} = \frac{1}{2}\tilde{f}^a_{\ bc} = \frac{1}{2} f^a_{\ bc},$$

and if we expand the CBH function in power series we find[2]

$$\tilde{m}^a(\theta, \theta') = \theta^a + \theta'^a + \frac{1}{2} f^a_{\ bc}\theta^b\theta'^c + \mathrm{O}(\theta^3, \theta'^3),$$

which is like a polynomial expansion of the group multiplication function m in terms of the structure constants $f^a_{\ bc}$ of the Lie algebra $\mathfrak{g}$. The exponential map thus provides a global parametrization of the Lie group G in terms of the elements of its Lie algebra $\mathfrak{g}$ and allows us to express the group multiplication function m in terms of the structure constants $f^a_{\ bc}$ of the Lie algebra $\mathfrak{g}$.

[2]There exists a closed form expression for the CBH function, but it is not important for our purposes: remind only that exact complicated expressions for computing all the teerms exist, but we are only interested in the first few terms of the expansion.

Example (Phase Group U(1)). Let us define the group of phase transformations U(1) as the group of complex numbers of unit modulus under multiplication:

$$\mathrm{U}(1) = \{z \in \mathbb{C} : |z| = 1\} = \{e^{i\theta} : \theta \in \mathbb{R}\}.$$

It is an abelian Lie group of dimension 1 which represents the Gauge group of QED. The elements of U(1) can be represented by an angle θ with an exponential parametrization as follows:

$$\tilde{g}(\theta) = e^{i\theta^a t_a} = e^{i\theta},$$

since the Lie algebra of U(1) is one-dimensional and spanned by the identity matrix $t = 1$. The structure constants of the Lie algebra of U(1) are zero since the commutator of any two elements of the Lie algebra is zero:

$$[t, t] = 0 \implies f_{ab}{}^c = 0.$$

Example (Special Unitary Group SU(2)). The special unitary group SU(2) is the group of 2×2 unitary matrices with determinant 1:

$$\mathrm{SU}(2) = \{U \in \mathbb{C}^{2\times 2} : U^\dagger U = \mathbb{I}_2, \det(U) = 1\}.$$

It is a non-abelian Lie group of dimension 3: $\underline{\theta} \in \mathbb{R}^3$. The elements of SU(2) can be parametrized using the Pauli matrices σ_i as follows:

$$\tilde{g}(\theta) = e^{i\theta^i \frac{\tau(\theta)}{2}},$$

where

$$\tau(\theta) = \begin{pmatrix} \theta^3 & \theta^1 - i\theta^2 \\ \theta^1 + i\theta^2 & -\theta^3 \end{pmatrix} = \theta^1 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \theta^2 \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + \theta^3 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \theta^i \sigma_i,$$

Thus $t_a = \frac{1}{2}\tau_a$ and the structure constants of the Lie algebra $\mathfrak{su}(2)$ are given by the Levi-Civita symbol ϵ_{ijk}:

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k \implies f_{ij}{}^k = 2\epsilon_{ijk}.$$

Example (Special Orthogonal Group SO(3)). The special orthogonal group SO(3) is the group of 3×3 real orthogonal matrices with determinant 1:

$$\mathrm{SO}(3) = \{R \in \mathbb{R}^{3\times 3} : R^T R = \mathbb{I}_3, \det(R) = 1\}.$$

It is a non-abelian Lie group of dimension 3. The elements of SO(3) can be parametrized using the generators of rotations J_i as follows:

$$\tilde{g}(\theta) = e^{i\theta^i J_i},$$

where the generators J_i satisfy the commutation relations

$$[J_i, J_j] = i\epsilon_{ijk} J_k.$$

[-....-] to be finished

2.1 | Unitary Representations of Continuous Symmetries

If the symmetry group G is a Lie group, then the quantum unitary operators representing the group action on states and observables can be expressed as functions of the parameter vector $\underline{\theta}$ of G:

[...]

$$U(m(\theta,\theta')) = U(g(m(\theta,\theta'))) = U(g(\theta)g(\theta')) = \gamma(g(\theta),g(\theta'))U(g(\theta))U(g(\theta')),$$

where $\gamma(g(\theta),g(\theta'))$ is a phase factor which can be written as $\gamma(g(\theta),g(\theta')) = e^{i\omega(g(\theta),g(\theta'))}$ for some real function ω. Thus

$$U(m(\theta,\theta')) = e^{i\omega(g(\theta),g(\theta'))}U(g(\theta))U(g(\theta')).$$

Example (EXCERCISE).

??? photo

When you have a Lie Group and a projected unitary representation in the Hilbert space of states, you can always get to this relation with the phase factor γ by redefining the unitary operators $U(g)$ with a suitable phase factor, all given to the properties of the Lie group G and the properties of the projective representation U.

Since the Lie groups act homogeneously, we can always translates what happens at the identity element to any other element of the group. Thus, by looking at the properties of the group around the identity element, we can understand the properties of the whole group. By Taylor expanding $U(\underline{\theta})$ near the neutral element $\underline{0}$ of the group, we get

$$U(\underline{\theta}) = 1 - i\theta^a Q_a + \frac{1}{2}\theta^a\theta^b Q_{ab} + \mathrm{O}(\theta^3), \quad Q_{ab} = Q_{ba},$$

where Q_a are the generators of the representation U of the Lie group G. The generators Q_a are essentially the tangent vectors to the representation U of the Lie group G at the identity element

$$Q_a = i\frac{\partial U(\underline{\theta})}{\partial\theta^a}\Big|_{\underline{\theta}=\underline{0}}.$$

We are now going to prove that the generators Q_a of the representation U of the Lie group G are self-adjoint operators:

$$Q_a^\dagger = Q_a.$$

We can start from the adjoint

$$\begin{aligned}
Q_a^\dagger &= \left(i\frac{\partial U(\underline{\theta})}{\partial\theta^a}\Big|_{\underline{\theta}=\underline{0}}\right)^\dagger \\
&= -i\frac{\partial U^\dagger(\underline{\theta})}{\partial\theta^a}\Big|_{\underline{\theta}=\underline{0}} \\
&= -i\frac{\partial U^{-1}(\underline{\theta})}{\partial\theta^a}\Big|_{\underline{\theta}=\underline{0}} \\
&= +iU(\underline{\theta})^{-1}\frac{\partial U(\underline{\theta})}{\partial\theta^a}U(\underline{\theta})^{-1}\Big|_{\underline{\theta}=\underline{0}} = Q_a,
\end{aligned}$$

where we have used the unitarity of U and the fact that $U(\underline{0}) = 1$. [comments]

We have another property to prove: we have to normalize the phase function ω

$$\omega(\underline{\theta}, \underline{0}) = \omega(\underline{0}, \underline{\theta}) = 0,$$

so that we remember that

$$\omega(\underline{\theta}, \underline{\theta}') = \theta^a \theta'^b \omega_{ab} + \mathrm{O}(\theta^3),$$

so that the phase factor γ is normalized to 1 at the identity element of the group, since

$$e^{i\omega(\underline{0},\underline{\theta})} = \gamma(g(\underline{0}), g(\underline{\theta}')) = 1 = e^{i\omega(\underline{\theta},\underline{0})} = \gamma(g(\underline{\theta}), g(\underline{0})) = 1.$$

???

[...]

$$[Q_a, Q_b]$$

... as we did for $g(\theta)g(\theta')$ we can multiply the taylor expansion of $U(\theta)$ and $U(\theta')$ and compare the result with the taylor expansion of $U(m(\theta, \theta'))$ to get

$$U(\underline{\theta})U(\boldsymbol{\theta'}) = e^{-i\omega(\underline{\theta},\boldsymbol{\theta'})} U(m(\underline{\theta}, \boldsymbol{\theta'})).$$

If we insert the taylor exapnsions of

- $U(\underline{\theta}) = 1 - i\theta^a Q_a + \frac{1}{2}\theta^a \theta^b Q_{ab} + \mathrm{O}(\theta^3)$
- $\omega(\underline{\theta}, \boldsymbol{\theta'}) = \theta^a \theta'^b \omega_{ab} + \mathrm{O}(\theta^3)$
- $m(\underline{\theta}, \boldsymbol{\theta'}) = \underline{\theta} + \boldsymbol{\theta'} + \frac{1}{2} f^c_{ab} \theta^a \theta'^b + \mathrm{O}(\theta^3)$

we get a mess, but if we compute everything and compare the coefficients of the terms of order $\theta^a \theta'^b$ we get

$$Q_a Q_b = i m^c_{ab} Q_c + i\omega_{ab}\mathbb{I} - Q_{ab},$$

and since we wanted to compute the commutator $[Q_a, Q_b]$ we can compute

$$Q_b Q_a = i m^c_{ba} Q_c + i\omega_{ba}\mathbb{I} - Q_{ba},$$

so that subtracting the two we get

$$[Q_a, Q_b] = i(m^c_{ab} - m^c_{ba})Q_c + i(\omega_{ab} - \omega_{ba})\mathbb{I} = i f^c_{ab} Q_c + i\kappa_{ab}\mathbb{I},$$

where we have recalled the structure constants of the Lie algebra $\mathfrak{g}$ of the Lie group G and we have defined $\kappa_{ab} = \omega_{ab} - \omega_{ba}$. This last coefficients κ_{ab} are called the **central charges** of the representation U of the Lie group G. A central algebra is an algebra where the central charges are zero, so that the commutator of the generators Q_a of the representation U of the Lie group G is just

$$[Q_a, Q_b] = i f^c_{ab} Q_c.$$

The properties of the commutators of the generators Q_a of the representation U can be expressed as

- $[Q_a, Q_b] = i f^c_{ab} Q_c + i\kappa_{ab}\mathbb{I}$
- $\kappa_{ab} = \omega_{ab} - \omega_{ba}$

- $[Q_a, Q_b] + [Q_b, Q_a] = 0 \implies \kappa_{ab} + \kappa_{ba} = 0$
- $[Q_a, [Q_b, Q_c]] + [Q_b, [Q_c, Q_a]] + [Q_c, [Q_a, Q_b]] = 0$ (Jacobi identity)

The structure constants respects those identity in the following form

- $f^a_{bc} + f^a_{cb} = 0$
- $f^d_{ab} f^e_{dc} + f^d_{bc} f^e_{da} + f^d_{ca} f^e_{db} = 0$ (Jacobi identity)

and along with the central charges they respect the following identity

$$\kappa_{ad} f^d_{bc} + \kappa_{bd} f^d_{ca} + \kappa_{cd} f^d_{ab} = 0,$$

which is a consequence of the Jacobi identity for the generators Q_a of the representation U of the Lie group G. A set of coefficients f^c_{ab} and κ_{ab} which respect the above identities is called a **Lie algebra $\mathfrak{g}$ 2-cocycle**. This naming is consistent since the central charges κ_{ab} are related to the phase factor γ of the projective representation U of the Lie group G by the relation $\kappa_{ab} = \omega_{ab} - \omega_{ba}$.

...

$$U(g, g') = \gamma(g, g') U(g) U(g'),$$
$$U(g) \to \hat{U}(g) = \alpha(g) U(g),$$
$$\gamma(g, g') = \alpha(g, g') \alpha(g)^{-1} \alpha(g')^{-1},$$

so that we can redefine the unitary operators $U(g)$ of the projective representation U of the Lie group G by a suitable phase factor $\alpha(g)$ to get a new projective representation $\hat{U}$ of the Lie group G with a new phase factor $\hat{\gamma}$ which is related to the old one by the relation

$$\hat{\gamma}(g, g') = \alpha(g, g') \alpha(g)^{-1} \alpha(g')^{-1} \gamma(g, g'),$$

so that if we can find a suitable phase factor $\alpha(g)$ such that $\hat{\gamma}(g, g') = 1$ for all $g, g' \in G$, then we can redefine the unitary operators $U(g)$ of the projective representation U of the Lie group G by a suitable phase factor $\alpha(g)$ to get a new projective representation $\hat{U}$ of the Lie group G with a trivial phase factor $\hat{\gamma}$, so that the new projective representation $\hat{U}$ of the Lie group G is actually a linear representation of the Lie group G. However, in general, it is not always possible to find such a phase factor $\alpha(g)$, so that there are some projective representations of the Lie group G which cannot be redefined to get a linear representation of the Lie group G.

???

$$Q_a \to \hat{Q}_a = Q_a + k_a \mathbb{I},$$

where k_a are the central shifts of the generators Q_a of the representation U of the Lie group G. These shifts are related to the central charges κ_{ab} by the relation

$$\kappa_{ab} = f^c_{ab} k_c,$$

which defines a **2-coboundary** Lie algebra. A 2-coboundary Lie algebra is special case of a 2-cocycle Lie algebra, where the central charges κ_{ab} can be expressed as a linear combination of the structure constants f^c_{ab} with coefficients given by the central shifts k_a. This is **Lie Algebra Cohomografy**

2.1.1 | Exponential Parametrization

Starting from the exponential parametrization of the Lie group G

$$\tilde{g}(\underline{\theta}) = \lim_{n\to\infty} g(\frac{\theta}{n})^n = e^{\theta^a T_a},$$

while for the inverse and the multiplication we have

$$\begin{cases} \tilde{j}(\underline{\theta}) = -\theta^a, \\ \tilde{m}(\underline{\theta}, \underline{\theta}') = \underline{\theta} + \underline{\theta}' + \frac{1}{2} f^c_{ab}\theta^a\theta'^b + \mathrm{O}(\theta^3), \end{cases}$$

where the last function is the Baker-Campbell-Hausdorff formula (BCH) for the multiplication of the Lie group G in the exponential parametrization.

We can now use this exponential parametrization of the Lie group G to get an exponential parametrization of the projective representation U of the Lie group G:

$$\tilde{U}(g(\underline{\theta})) = U(\tilde{g}(\underline{\theta})) = U(e^{\theta^a T_a}) = 1 - i\theta^a Q_a + \frac{1}{2}\theta^a\theta^b Q_{ab} + \mathrm{O}(\theta^3).$$

For the phase function ω we have

$$\tilde{\omega}(\underline{\theta}, \underline{\theta}') = \tilde{\omega}_{ab}\theta^a\theta'^b + \mathrm{O}(\theta^3).$$

At the linear level there are no differences between g and $\tilde{g}$, they only manifest from the second order in the parameters θ^a. Thus we can derive

$$Q_a = \tilde{Q}_a,$$

and for the phase function ω we have

$$\tilde{Q}_{ab} = \frac{1}{2}((\omega_{ab} + \omega_{ba})\mathbb{I} - Q_a Q_b - Q_b Q_a).$$

Now we are interested in knowing if the following is true:

$$\tilde{U}(\theta) = \exp(i\theta^a Q_a),$$

which is not the case: this is a projected representation, so this will be proven to be true only up to a phase factor. We can start from the exponential parametrization of the projective representation U of the Lie group G

$$\tilde{U}(\underline{\theta}) = U(\tilde{g}(\underline{\theta})) = \lim_{n\to\infty} U(g(\frac{\theta}{n}))^n = \lim_{n\to\infty} U((\frac{1}{n^2})^n) \neq \lim_{n\to\infty} U((\frac{1}{n^2}))^n,$$

where we highlighted the difference between this projected and a true representation. Computing further we get

$$\tilde{U}(\underline{\theta}) = \exp\left(i\sum_k^n \omega(\frac{\theta}{n}, \frac{k\theta}{n}) \right) U(\frac{\theta}{n})^n,$$

where the last term, as we take the limit $n \to \infty$, becomes

$$\lim_{n\to\infty} U(\frac{\theta}{n})^n = (1 - i\frac{\theta^a}{n}Q_a)^n = e^{-i\theta^a Q_a},$$

so that the exponential parametrization of the projective representation U of the Lie group G is given by

$$\tilde{U}(\underline{\theta}) = e^{i\alpha(\underline{\theta})}e^{-i\theta^a Q_a}.$$

If we now want to get .....

$$e^{i\alpha(\underline{\theta})-i\theta^a k_a}\tilde{U}(\underline{\theta}) = e^{-i\theta^a \hat{Q}_a},$$

where $\hat{Q}_a = Q_a + k_a\mathbb{I}$ are the shifted generators of the representation U of the Lie group G: these operates in the same way since they differ only by a phase factor, but they have different central charges.

Now we found a true representation

- $\hat{\tilde{U}}(\underline{\tilde{j}}(\underline{\theta})) = \hat{\tilde{U}}(\underline{\theta})^{-1}$
- $\hat{\tilde{U}}(\underline{\tilde{m}}(\underline{\theta}, \boldsymbol{\theta'})) = \hat{\tilde{U}}(\underline{\theta})\hat{\tilde{U}}(\boldsymbol{\theta'})$

The term $\exp(-i\theta^a k_a)$ is an important **phase factor**, since it is the one which allows us to get a true representation from a projective representation by shifting the generators Q_a of the representation U of the Lie group G by a suitable central shift k_a. This is the reason why we introduced the concept of 2-coboundary Lie algebra, since it allows us to understand when a projective representation of a Lie group can be redefined to get a true representation of the Lie group.

This is a **multi-valued function** on the underlying group manifold G, since the phase factor usually is not an integer which comes back to the original value after a loop around the group manifold G (it is not guaranteed that the passage is single-valued): the exponential parametrization of the projective representation U of the Lie group G is a multi-valued function on the underlying group manifold G. Usually then we introduce a branch cut in the complex plane to make it single-valued, in order to treat multi valued function as single valued function on the more complex Riemann surface (such as for the square root and the logarithm).

$$\ln z = \ln|z| + i\arg z + 2\pi i n, \quad n \in \mathbb{Z}.$$

In our case the exponential parametrization of the projective representation U of the Lie group G is a multi-valued function on the underlying group manifold G, but can be made single-valued on a infinite Riemann surface which is a covering space of the underlying group manifold G: a covering group $\tilde{G}$. It is the same idea of the Riemann surface, but with more subtleties and complications, since the group structure has to be preserved.

There can be made analogies with the various covering groups we can find in physics, such as the universal covering group of the Lorentz group, which is the Spin group, and the universal covering group of the rotation group $SO(3)$, which is the special unitary group $SU(2)$. The projective representations of the Lorentz group and the rotation group $SO(3)$ can be redefined to get true representations of their respective covering groups, which are the Spin group and the special unitary group $SU(2)$. This is a consequence of the fact that the central charges of the projective representations of the Lorentz group and the rotation group $SO(3)$ can be expressed as linear combinations of the structure constants of their respective Lie algebras with coefficients given by suitable central shifts

$$\mathrm{SL}(2,\mathbb{C}) = \widetilde{\mathrm{SO}^+(1,3)}, \quad \mathrm{SU}(2) = \widetilde{\mathrm{SO}(3)}.$$

Example (The translation group $\mathrm{T}(3)$**).** The translation group $\mathrm{T}(3)$ implements the translation symmetry of the three-dimensional space:

$$\mathbf{a} \in \mathbb{E}_3, \quad T(\mathbf{x}) = \mathbf{x} + \mathbf{a},$$

where $T \in \mathrm{T}(3)$ and $\mathbb{E}_3$ is the three-dimensional Euclidean space.

The proper translation group T(3) can be represented by the unitary operators $U(\mathbf{a})$ for a translation $\mathbf{a} \in \mathbb{E}_3$ which act on the Hilbert space of states of a quantum system as

$$(U(\mathbf{a})\Psi)(\mathbf{x}) = \Psi(\mathbf{x} + \mathbf{a}),$$

where Ψ is a state of the quantum system and we want to prove that $\{U(\mathbf{a})\}_{\mathbf{a}\in\mathbb{E}_3}$ yields a unitary representation of the proper translation group T(3).

Proof. We can start by computing the composition of two translations $\mathbf{a}$ and $\mathbf{b}$: [...]

There are no multiplicative phases in the right hand side. The operator family $U(\mathbf{a})$ constitutes in this way a unitary representation of the proper translation group T(3). □

Let us now discuss the infinitesimal generators of the translation group T(3). The infinitesimal generators $\hbar^{-1}\mathbf{p}$ of the unitary representation U of the proper translation group T(3) are defined through the expansion of the unitary operators $U(\mathbf{a})$ for small translations $\delta\mathbf{a}$:

$$U(\delta\mathbf{a}) = 1 + \frac{i}{\hbar}\mathbf{p}\cdot\delta\mathbf{a} + \mathrm{O}(\delta a^2),$$

with $\mathbf{p} = (p_1, p_2, p_3)$ being the momentum operator of the quantum system

$$\mathbf{p}\Psi(\mathbf{x}) = -i\hbar\nabla\Psi(\mathbf{x}).$$

Thus we can study the expansion of the unitary operators $U(\mathbf{a})$ acting on a state Ψ for small translations $\delta\mathbf{a}$:

$$\begin{aligned}(U(\delta\mathbf{a})\Psi)(\mathbf{x}) &= \Psi(\mathbf{x} + \delta\mathbf{a}) \\ &= \Psi(\mathbf{x}) + \delta\mathbf{a}\cdot\nabla\Psi(\mathbf{x}) + \mathrm{O}(\delta a^2) \\ &= \dots\end{aligned}$$

$\mathbf{p}$ is therefore the familiar quantum linear momentum operator as anticipated by the notation used. The result obtained is in line with the standard interpretation of momentum as the infinitesimal generator of configuration space translations.

As well–known from quantum theory, its components

$$p_k = \mathbf{p}\cdot\mathbf{e}_k, \quad k = 1, 2, 3,$$

satisfy the commutation relations

$$[p_k, p_l] = 0, \quad k, l = 1, 2, 3,$$

free of central charges, as expected from the fact that the translation group T(3) is an abelian group, so that its Lie algebra is also abelian, with vanishing structure constants $f^c_{ab} = 0$.

The translation group T(3) is a special case of a more general class of groups called the Heisenberg group, which is a non-abelian group with non-vanishing structure constants and non-vanishing central charges. The Heisenberg group is the symmetry group of the quantum phase space, and its projective representations are related to the canonical commutation relations of quantum mechanics.

Thus the operators $U(\mathbf{a})$ can be written in terms of the momentum operator $\mathbf{p}$, the generators, as

$$U(\mathbf{a}) = e^{\frac{i}{\hbar}\mathbf{a}\cdot\mathbf{p}}.$$

Proof. limit for n.... □

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Example (The proper rotation group SO(3)/SU(2)**).**

The proper rotation group $SO(3)$ implements the rotation symmetry of the three-dimensional space:

$$R \in SO(3), \quad R : \mathbf{x} \to R\mathbf{x},$$

where R is a proper rotation and $\mathbf{x}$ is a vector in the three-dimensional space.

The proper rotation group $SO(3)$ can be represented by the unitary operators $U(R)$ for a proper rotation $R \in SO(3)$ which act on the Hilbert space of states of a quantum system as

$$(U(R)\Psi)(\mathbf{x}) = D(\mathbf{R})\Psi(R^{-1}\mathbf{x}),$$

where Ψ is a state of the quantum system, $\mathbf{x}$ is a point in the configuration space, and $D(\mathbf{R})$ is a unitary operator which acts on the internal degrees of freedom of the quantum system, such as spin.

Indeed we know that the spin of a quantum system is related to the representation of the rotation group $SO(3)$ on the internal degrees of freedom of the quantum system; this does not come as a surprise, since the spin of a quantum system is related to the way the quantum system transforms under rotations, and the rotation group $SO(3)$ is the symmetry group of rotations in three-dimensional space.

We parametrized the representation in order to act on $\mathbf{x}$ as the inverse of the rotation,[3] since this assumption is consistent with the composition of rotations, as we can see by computing the composition of two rotations R and R':

$$U(\mathbf{R}, \mathbf{R}') = U(\mathbf{R})U(\mathbf{R}'),$$

otherwise the the order would be inverted.

$D(\mathbf{R})$ are $(2s+1)\times(2s+1)$ unitary matrices which form a representation of the rotation group $SO(3)$ on the internal degrees of freedom of the quantum system, where s is the spin of the quantum system

$$D(\mathbf{R})^\dagger D(\mathbf{R}) = D(\mathbf{R})D(\mathbf{R})^\dagger = \mathbb{I},$$

and they satisfy the composition law[4]

$$D(\mathbf{R}\mathbf{R}') = D(\mathbf{R})D(\mathbf{R}').$$

Proof. Let us show first that the operator $U(R)$ is unitary. Define the operator

$$U^*(\mathbf{R})\Psi(\mathbf{x}) = D(\mathbf{R})^\dagger\Psi(R\mathbf{x}),$$

so that we can compute the composition of $U^*(\mathbf{R})$ and $U(\mathbf{R})$ which acts as the identity operator on the Hilbert space of states of the quantum system:

$$(U^*(\mathbf{R})U(\mathbf{R})\Psi)(\mathbf{x}) = D(\mathbf{R})^\dagger D(\mathbf{R})\Psi(\mathbf{R}^{-1}\mathbf{R}\mathbf{x}) = \Psi(\mathbf{x}).$$

[3] Which is always well defined from the properties of groups.

[4] Here the right order is implemented by the previously discussed choice of parametrization.

The next thing to show is that

$$\|U(\mathbf{R})\Psi\|^2 = \|\Psi\|^2,$$

which is the definition of unitarity. We can compute

$$\begin{aligned}\|U(\mathbf{R})\Psi\|^2 &= \int d^3x\,|U(\mathbf{R})\Psi(\mathbf{x})|^2 = \int d^3x\,|D(\mathbf{R})\Psi(R^{-1}\mathbf{x})|^2 \\ &= \int d^3x\,|\Psi(R^{-1}\mathbf{x})|^2 = \int d^3x'\,|\Psi(\mathbf{x}')|^2 = \|\Psi\|^2,\end{aligned}$$

where we have used the unitarity of $D(\mathbf{R})$ and $J = |\det(\mathbf{R})^{-1}| = 1$. The last step is to show that the composition law is satisfied, which can be done by computing

$$\begin{aligned}(U(\mathbf{R}\mathbf{R}')\Psi)(\mathbf{x}) &= D(\mathbf{R}\mathbf{R}')\Psi((\mathbf{R}\mathbf{R}')^{-1}\mathbf{x}) = D(\mathbf{R})D(\mathbf{R}')\Psi(\mathbf{R}'^{-1}\mathbf{R}^{-1}\mathbf{x}) \\ &= D(\mathbf{R})(U(\mathbf{R}')\Psi)(\mathbf{R}^{-1}\mathbf{x}) = (U(\mathbf{R})U(\mathbf{R}')\Psi)(\mathbf{x}),\end{aligned}$$

proving that $U(\mathbf{R}\mathbf{R}') = U(\mathbf{R})U(\mathbf{R}')$. □

We now aim to study the **exponential parametrization** of the representation U of the proper rotation group $SO(3)$. The generators J_k of the representation U of the proper rotation group $SO(3)$ are defined through the expansion of the unitary operators $U(R)$ for small rotations δR.

We can then parametrize the rotation R by a vector $\boldsymbol{\varphi} = (\varphi_1, \varphi_2, \varphi_3)$ which represents the axis of rotation and the angle of rotation, so that the exponential parametrization of the the proper rotation group $SO(3)$ is given by

$$R(\boldsymbol{\varphi}) = e^{-*\varphi}, \quad R(\boldsymbol{\varphi})^{-1} = e^{*\varphi},$$

where

$$*\varphi = \begin{pmatrix} 0 & \varphi_3 & -\varphi_2 \\ -\varphi_3 & 0 & \varphi_1 \\ \varphi_2 & -\varphi_1 & 0 \end{pmatrix},$$

is the antisymmetric matrix associated to the vector $\boldsymbol{\varphi} = (\varphi_1, \varphi_2, \varphi_3)$ which parametrizes the rotation R.

Now we can reflect this exponential parametrization on the representation U of the proper rotation group $SO(3)$ to get an exponential parametrization of the representation:

$$(U(\boldsymbol{\varphi})\Psi)(\mathbf{x}) = D(\boldsymbol{\varphi})\Psi(e^{*\varphi}\mathbf{x}),$$

where $D(\boldsymbol{\varphi})$ is the exponential parametrization of the representation of the rotation group $SO(3)$ on the internal degrees of freedom of the quantum system. We have to look at the expansion of the unitary operators $U(\boldsymbol{\varphi})$ for small rotations $\delta\boldsymbol{\varphi}$ to get the generators $\underline{j}$ of the representation U of the proper rotation group $SO(3)$:

$$U(\delta\boldsymbol{\varphi}) = 1 - i\underline{j}\cdot\delta\boldsymbol{\varphi} + \mathrm{O}(\delta\varphi^2),$$

where $\underline{j} = (J_1, J_2, J_3)$ are the generators of the representation U of the proper rotation group $SO(3)$, which are called the angular momentum operators of the quantum system. Then the expansion of the internal degrees transformation $D(\boldsymbol{\varphi})$ for small rotations $\delta\boldsymbol{\varphi}$ gives us

$$D(\delta\boldsymbol{\varphi}) = 1 - i\pm\cdot\delta\boldsymbol{\varphi} + \mathrm{O}(\delta\varphi^2),$$

where $\pm = (\Sigma_1, \Sigma_2, \Sigma_3)$ are the generators of the representation of the rotation group $SO(3)$ on the internal degrees of freedom of the quantum system, which are called the spin operators of the quantum system. Finally we have to compute the expansion of the spatial transformation $e^{*\varphi}$ for small rotations $\delta\boldsymbol{\varphi}$ to get

$$e^{*\delta\varphi} = 1 + *\delta\varphi + \mathrm{O}(\delta\varphi^2),$$

but since the rotation can be expressed as $-\delta\boldsymbol{\varphi} \times \mathbf{x}$, we can write

$$e^{*\delta\varphi}\mathbf{x} = \mathbf{x} - \delta\boldsymbol{\varphi} \times \mathbf{x} + \mathrm{O}(\delta\varphi^2).$$

Now we can put everything together to get the expansion of the unitary operators $U(\boldsymbol{\varphi})$ for small rotations $\delta\boldsymbol{\varphi}$:

$$\begin{aligned}
(U(\delta\boldsymbol{\varphi})\Psi)(\mathbf{x}) &= D(\delta\boldsymbol{\varphi})\Psi(e^{*\delta\varphi}\mathbf{x}) \\
&= (1 - i\pm \cdot \delta\boldsymbol{\varphi} + \mathrm{O}(\delta\varphi^2))\Psi(\mathbf{x} - \delta\boldsymbol{\varphi} \times \mathbf{x} + \mathrm{O}(\delta\varphi^2)) \\
&= (1 - i\pm \cdot \delta\boldsymbol{\varphi} + i(\delta\boldsymbol{\varphi} \times \mathbf{x}) \cdot (-i\nabla) + \mathrm{O}(\delta\varphi^2))\Psi(\mathbf{x}) \\
&\ldots \\
&= \Psi(\mathbf{x}) - \delta\boldsymbol{\varphi} \cdot (\mathbf{x} \times \nabla + i\pm)\Psi(\mathbf{x}) + \mathrm{O}(\delta\varphi^2),
\end{aligned}$$

[...]

Thus we found

$$\underline{j} = \hbar\pm - i\hbar\mathbf{x} \times \nabla,$$

where renaming $\hbar\pm = \mathbf{S}$ and $-i\hbar\mathbf{x} \times \nabla = \mathbf{L}$ we get the familiar decomposition of the angular momentum operator $\underline{j}$ into the spin operator $\mathbf{S}$ and the orbital angular momentum operator $\mathbf{L}$:

$$\underline{j} = \mathbf{S} + \mathbf{L}.$$

We know that the infinitesimal generators of a continuous symmetry group satisfy the same commutation relations as the Lie algebra of the symmetry group, so that the generators J_k of the representation U of the proper rotation group $SO(3)$ satisfy the commutation relations

$$[J_k, J_l] = i\hbar\epsilon_{klm}J_m, \quad k, l, m = 1, 2, 3,$$

while the generators Σ_k of the representation of the rotation group $SO(3)$ on the internal degrees of freedom of the quantum system satisfy the same commutation relations

$$[\Sigma_k, \Sigma_l] = i\epsilon_{klm}\Sigma_m, \quad k, l, m = 1, 2, 3,$$

where ϵ_{klm} is the Levi-Civita symbol. The commutation relations of the generators J_k and Σ_k of the representation, are in line with the structure constants of the Lie algebra $\mathfrak{so}(3)$ of the proper rotation group $SO(3)$, which are given by

$$f_{kl}^m = \epsilon_{klm}.$$

Since we are using the exponential parametrization $\mathbf{R}(\boldsymbol{\varphi})$ of SO(3), we can get the exponential parametrization of the representation $U(\boldsymbol{\varphi})$ in terms of the infinitesimal generators J_k:

$$U(\boldsymbol{\varphi}) = e^{-\frac{i}{\hbar}\boldsymbol{\varphi}\cdot\underline{j}}.$$

Proof. limit for n.... □

It should be made an extensive digression on the universal covering group, which is the group which allows us to get a single-valued representation from a multi-valued representation by lifting the representation to the covering group. The universal covering group of a Lie group G is a simply connected Lie group $\tilde{G}$ which covers G in such a way that the projection map from $\tilde{G}$ to G is a local diffeomorphism. The universal covering group $\tilde{G}$ has the same Lie algebra as G, but it can have different global properties, such as different topology and different fundamental group. The universal covering group $\tilde{G}$ allows us to get a single-valued representation from a multi-valued representation of G by lifting the representation to $\tilde{G}$.

However we do not have the time to make this digression, so we will just mention that the universal covering group of the proper rotation group $SO(3)$ is the special unitary group $SU(2)$, which is a simply connected Lie group that covers $SO(3)$ in such a way that the projection map from $SU(2)$ to $SO(3)$ is a local diffeomorphism. The universal covering group $SU(2)$ has the same Lie algebra as $SO(3)$, but it has different global properties, such as different topology and different fundamental group. The universal covering group $SU(2)$ allows us to get a single-valued representation from a multi-valued representation of $SO(3)$ by lifting the representation to $SU(2)$.

2.2 | Dynamical Symmetries and Conservation Laws

[... non lie symmetry content ...]

When a dynamical symmetry group G is a Lie group, with a parametrization $g(\underline{\theta})$, the quantum unitary representation $\tilde{U}(\underline{\theta})$ representing the action of G on the Hilbert space $\mathcal{H}$ respects

$$\tilde{U}(\underline{\theta})H\tilde{U}(\underline{\theta})^\dagger = H, \quad \forall\underline{\theta}.$$

This property restricts the form of the Hamiltonian H on one hand and has consequences for the infinitesimal generators Q_a of the unitary family $\tilde{U}(\underline{\theta})$ on the other: H commutes with the Q_a

$$[H, Q_a] = 0.$$

Proof. If we compute the derivative of the invariance condition with respect to θ^a at $\underline{\theta} = 0$, we get

$$0 = i\frac{\partial}{\partial\theta^a}(H - \tilde{U}(\underline{\theta})H\tilde{U}(\underline{\theta})^\dagger)\Big|_{\underline{\theta}=\underline{0}} = \left[-i\frac{\partial\tilde{U}(\underline{\theta})}{\partial\theta^a}H\tilde{U}(\underline{\theta})^\dagger + \tilde{U}(\underline{\theta})H\left(i\frac{\partial\tilde{U}(\underline{\theta})}{\partial\theta^a}\right)^\dagger\right]_{\underline{\theta}=\underline{0}}$$
$$= -i\frac{\partial\tilde{U}(\underline{0})}{\partial\theta^a}H\tilde{U}(\underline{0})^\dagger + \tilde{U}(\underline{0})H\left(i\frac{\partial\tilde{U}(\underline{0})}{\partial\theta^a}\right)^\dagger = -Q_aH + HQ_a = [H, Q_a],$$

since $\tilde{U}(\underline{0}) = \mathbb{I}$ and $Q_a = -i\frac{\partial\tilde{U}(\underline{\theta})}{\partial\theta^a}\big|_{\underline{\theta}=\underline{0}}$. This proves that when a system is invariant under a Lie group of dynamical symmetries, the generators of that group commute with the Hamiltonian. $\square$

This result is significant because it implies that the generators Q_a are conserved quantities of the system, as they commute with the Hamiltonian. They are obserables with meaningful physical significance. In quantum mechanics, this means that the eigenvalues of Q_a are **constants of motion**, and the system's dynamics can be analyzed in terms of these conserved quantities. This connection between symmetries and conservation laws is a fundamental aspect of both classical and quantum physics, encapsulated in **Noether's theorem**. Indeed, we can define the evolution operator

$$U(t, s) = e^{-\frac{i}{\hbar}(t-s)H},$$

such that any observable $\hat{A}$ evolves in time according to

$$\hat{A}(t) = U(t, 0)^\dagger\hat{A}U(t, 0).$$

Then we have

$$U(t, s)Q_aU(t, s)^\dagger = e^{-\frac{i}{\hbar}(t-s)H}Q_ae^{\frac{i}{\hbar}(t-s)H} = Q_a,$$

which shows that $Q_a(t) = Q_a$ is conserved under the time evolution generated by H.

Proof. This is a direct consequence of the commutation relation $[H, Q_a] = 0$: since Q_a commutes with H, it also commutes with any function of H, including the exponential $e^{-\frac{i}{\hbar}(t-s)H}$. Therefore, we can move Q_a through the exponential without changing its value, leading to the conclusion that Q_a is invariant under the time evolution generated by H. $\square$

We have just shown the quantum analog of Noether's theorem: for every continuous symmetry of the action (or equivalently, for every Lie group of dynamical symmetries), there is a corresponding conserved quantity. This profound connection between symmetries and conservation laws is a cornerstone of modern physics, providing deep insights into the fundamental structure of physical theories.

The symmetry Lie algebra $\mathfrak{g}$

$$[t_a, t_b] = f^c{}_{ab} t_c,$$

always contains a Cartan subalgebra $\mathfrak{h}$, that is a maximal subspace of $\mathfrak{g}$ the Lie brackets of all whose elements vanish: all the structure constants $f^c{}_{ab}$ vanish. Denote by h_i a set of generators of $\mathfrak{h}$. These are certain linearly independent linear combinations of the generators T_a of $\mathfrak{g}$

$$h_i = h_i^a t_a,$$

with the property that

$$[h_i, h_j] = 0.$$

Thus we are lead to introduce a set of quantum generators C_i of the Cartan subalgebra $\mathfrak{h}$ in such a way that they commute with the Hamiltonian H and with each other

$$C_i = h_i^a Q_a.$$

These generators C_i, defined in this way, respects

$$[C_i, C_j] = 0.$$

Proof.

$$0 = [h_i, h_j] = [h_i^a t_a, h_j^b t_b] \dots$$

thus we can write

$$\begin{aligned} [C_i, C_j] &= [h_i^a Q_a, h_j^b Q_b] = h_i^a\, h_j^b [Q_a, Q_b] \\ &= i h_i^a\, h_j^b f^c{}_{ab} Q_c = 0, \end{aligned}$$

since the structure constants $f^c{}_{ab}$ vanish for all a, b when t_a and t_b are in the Cartan subalgebra $\mathfrak{h}$. □

Thus

$$[C_i, C_j] = 0, \quad [C_i, H] = 0.$$

There is a one-to-one correspondence between the eigenvalues of the C_i and the generators of the cartan subalgebra $\mathfrak{h}$.

Consider now

$$\eta_{ab} = \mathrm{Tr}(t_a t_b),$$

if $\mathfrak{g}$ is a semisimple Lie algebra, the matrix η_{ab} is non-degenerate and can be used to raise and lower indices. It will be invertible and we can define η^{ab} such that $\eta^{ab}\eta_{bc} = \delta^a_c$. We obtained a metric on the Lie algebra $\mathfrak{g}$ and we can use.

Now we are going to prove that the structure constants $f_{abc} = f^d{}_{bc}\eta_{da}$ of a semisimple Lie algebra are completely antisymmetric in the indices a, b, c and not only the last two.

Proof. Since

$$\mathrm{Tr}([A, B]) = \mathrm{Tr}(AB - BA) = 0,$$

we can write

$$\begin{aligned} 0 &= \mathrm{Tr}([t_a, t_b t_c]) = \mathrm{Tr}([t_a, t_b]t_c + t_b[t_a, t_c]) = \mathrm{Tr}\big(f^d{}_{ab} t_d t_c + f^d{}_{ac} t_b t_d\big) \\ &= f^d{}_{ab}\eta_{dc} + f^d{}_{ac}\eta_{db} = f_{abc} + f_{acb} = 0, \end{aligned}$$

or equivalently

$$f^{a}_{\ b}{}^{c} + f^{b}_{\ c}{}^{a} = 0.$$

□

The antisymmetry of the structure constants f_{abc} is a crucial property that has significant implications for the representation theory of Lie algebras and the classification of semisimple Lie algebras. η_{ab} is called the **Cartan metric** of the Lie algebra $\mathfrak{g}$ and through it we can define the Quadratic Casimir operator C_2 of the Lie algebra $\mathfrak{g}$

$$C_2 = \eta^{ab} Q_a Q_b.$$

The Casimir operator C_2 is a central element of the universal enveloping algebra of $\mathfrak{g}$, meaning that it commutes with all elements of $\mathfrak{g}$. In particular, since Q_a are the generators of the symmetry group, we have

$$\begin{aligned} [Q_a, C_2] &= [Q_a, \eta^{bc} Q_b Q_c] = \eta^{bc}([Q_a, Q_b]Q_c + Q_b[Q_a, Q_c]) \\ &= i\eta^{bc}(f^d{}_{ab} Q_d Q_c + f^d{}_{ac} Q_b Q_d) = i(f^{d}{}_{a}{}^{b} Q_d Q_b + f^{d}{}_{a}{}^{b}) Q_b Q_d \\ &= i f^{d}{}_{a}{}^{b}(Q_d Q_b + Q_b Q_d) = 0, \end{aligned}$$

where we have used the antisymmetry of the structure constants f_{abc} in the last step. The Casimir operator C_2 is an important quantity in the representation theory of Lie algebras, as it takes a constant value on each irreducible representation, providing a way to classify representations and understand their properties.

$$[C_i, C_j] = 0, \quad [H, C_i] = 0, \quad [C_2, C_i] = 0, \quad [H, C_2] = 0.$$

To sum up, we have a dynmical lie symmetry group G with Lie algebra $\mathfrak{g}$ and a Cartan subalgebra $\mathfrak{h}$ of $\mathfrak{g}$. We have a set of generators C_i of the Cartan subalgebra $\mathfrak{h}$ that

$$\begin{cases} C_i = h_i^a Q_a, \\ C_2 = \eta^{ab} Q_a Q_b, \end{cases}$$

commute with the Hamiltonian H and with each other, and we have the Casimir operator C_2 that also commutes with H and with the C_i. The eigenvalues of the C_i and C_2 can be used to label the states of the system and to classify the irreducible representations of the symmetry group G.

Example (The Hydrogen Atom). The hydrogen atom has a dynamical symmetry group known as the **SO(3)** group, associated to the algebra $\mathfrak{so}(3)$ with generators

$$[t_a, t_b] = \epsilon_{abc} t_c.$$

We have an associated cartan subalgebra generated by ct_3, with $c \in \mathbb{R}$ and a cartesian metric $\eta_{ab} = \mathrm{Tr}(t_a t_b) = \delta_{ab}$. The quantum generators J_i (analogue of Q_a) of the symmetry group commute with the Hamiltonian of the hydrogen atom, and the analogue of C_i is J_3, while the analogue of the Casimir operator C_2 is $J^2 = J_1^2 + J_2^2 + J_3^2 = \delta_{ij} J_i J_j$. The eigenvalues of J_3 and J^2 can be used to label the states of the hydrogen atom and to classify the irreducible representations of the SO(3) symmetry group, which correspond to the different energy levels and angular momentum states of the electron in the hydrogen atom.

Thus

$$\begin{aligned} H\Phi_{wc\underline{e}} &= \Phi_{wc\underline{e}}w, \\ C_2\Phi_{wc\underline{e}} &= \Phi_{wc\underline{e}}c, \\ C_i\Phi_{wc\underline{e}} &= \Phi_{wc\underline{e}}e_i, \end{aligned}$$

where $\Phi_{wc\underline{e}}$ are the eigenstates of the Hamiltonian H, the Casimir operator C_2 and the generators C_i of the Cartan subalgebra, with eigenvalues w, c and e_i respectively. The quantum numbers w, c and e_i can be used to classify the states of the hydrogen atom and to understand their properties in terms of the underlying symmetries of the system.

Using w as the Bohr energy levels, we can write the energy eigenvalues of the hydrogen atom as

$$\begin{aligned} w &\to n, \quad n = 1, 2, 3, \ldots, \\ c &\to l, \quad l = 0, 1, 2, \ldots, n-1, \\ e_i &\to m, \quad m = -l, -l+1, \ldots, l-1, l. \end{aligned}$$

where in this case $\underline{e} = e$ is a single quantum number since we have only one generator C_i of the Cartan subalgebra. The energy levels of the hydrogen atom are quantized and depend on the principal quantum number n, the angular momentum quantum number l and the magnetic quantum number m, with the energy eigenvalues given by

$$c = \hbar^2 l(l+1), \quad \ldots$$

We obtained a bunch of commuting self adjoint operators H, C_i, C_2 that can be simultaneously diagonalized, with common eigenvectors $\Phi_{wc\underline{e}}$ and eigenvalues w, c, e_i. The quantum numbers w, c, e_i can be used to label the states of the system and to classify the irreducible representations of the symmetry group G. This is a powerful tool for understanding the structure of quantum systems and for predicting their behavior based on their symmetries.

We found an orthonormal basis of eigenvectors $\{\Phi_{wc\underline{e}}\}$ of the Hilbert space $\mathcal{H}$ that are common eigenvectors of the Hamiltonian H, the Casimir operator C_2 and the generators C_i of the Cartan subalgebra:

$$\begin{aligned} H\Phi_{wc\underline{e}} &= w\Phi_{wc\underline{e}}, \\ C_2\Phi_{wc\underline{e}} &= c\Phi_{wc\underline{e}}, \\ C_i\Phi_{wc\underline{e}} &= e_i\Phi_{wc\underline{e}}, \end{aligned}$$

This basis allows us to express any state of the system as a linear combination of these eigenvectors, and to analyze the dynamics and properties of the system in terms of its symmetries and conserved quantities.

3 | Vector calculus in $\mathbb{R}^3$ geometry

3.1 | Geometric Structures and Identities

3.1.1 | Euclidean and Orientational Structure

The Euclidean structure of R3 is nothing but the familiar "dot" product of vectors and is the fundamental structure characterizing $\mathbb{R}^3$. We begin with a remark on notation. Any vector $\mathbf{x} \in \mathbb{R}^3$ can be viewed as either a column or a row vector

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix}.$$

canonical orthonormal basis of $\mathbb{R}^3$ is given by

component of the vector

matrices 3x3

orientational struc

For each triple of vectors $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^3$, we can define a matrix

$$(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \begin{pmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} \mathbf{x}^T \\ \mathbf{y}^T \\ \mathbf{z}^T \end{pmatrix} = \begin{pmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{pmatrix}.$$

the reciprocally transposed matrices of R[3] having x, y, z as column and row vectors in the shown order, respectively. For fixed vectors x, y, the determinant of these matrices depends linearly on z. Therefore, there exists a vector x × y in R3 with the property that

$$\mathbf{x} \times \mathbf{y} \cdot \mathbf{z} = \det(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \det \begin{pmatrix} \mathbf{x}^T \\ \mathbf{y}^T \\ \mathbf{z}^T \end{pmatrix}.$$

...

$$\begin{aligned}\mathbf{x}\times\mathbf{y}+\mathbf{y}\times\mathbf{x}&=0,\\ \mathbf{x}\times(\mathbf{y}\times\mathbf{z})+\mathbf{y}\times(\mathbf{z}\times\mathbf{x})+\mathbf{z}\times(\mathbf{x}\times\mathbf{y})&=0,\\ (\mathbf{x}\times\mathbf{y})\cdot(\mathbf{u}\times\mathbf{v})&=\dots\\ \mathbf{x}\times(\mathbf{y}\times\mathbf{z})&=(\mathbf{x}\cdot\mathbf{z})\mathbf{y}-(\mathbf{x}\cdot\mathbf{y})\mathbf{z},\\ \mathbf{e}_i\times\mathbf{e}_j&=\sum_k\epsilon_{ijk}\mathbf{e}_k.\end{aligned}$$

Proof. We are going to prove the determinant identity... □

[...]

3.1.2 | Determinant and Trace Identities

Suppose A is a 3×3 matrix and $\mathbf{x},\mathbf{y},\mathbf{z}$ are three vectors in $\mathbb{R}^3$, then we have the following identities:

$$\begin{aligned}(A\mathbf{x})\times(A\mathbf{y})\cdot(A\mathbf{z})&=\det(A)\mathbf{x}\times\mathbf{y}\cdot\mathbf{z},\\ (A\mathbf{x})\times\mathbf{y}\cdot\mathbf{z}+\mathbf{x}\times(A\mathbf{y})\cdot\mathbf{z}+\mathbf{x}\times\mathbf{y}\cdot(A\mathbf{z})&=\operatorname{Tr}(A)\mathbf{x}\times\mathbf{y}\cdot\mathbf{z},\end{aligned}$$

The first identity can be interpreted as the invariance of the determinant under linear transformations, while the second identity can be interpreted as the infinitesimal version of the first identity, which is related to the trace of the matrix A. The first one is called the **determinant identity**, while the second one is called the **trace identity**.

Proof. where the first one can be seen as

$$(A\mathbf{x})\times(A\mathbf{y})\cdot(A\mathbf{z})=\det\begin{pmatrix}A\mathbf{x} & A\mathbf{y} & A\mathbf{z}\end{pmatrix}=\det\left(A\begin{pmatrix}\mathbf{x} & \mathbf{y} & \mathbf{z}\end{pmatrix}\right)=\det(A)\det\begin{pmatrix}\mathbf{x} & \mathbf{y} & \mathbf{z}\end{pmatrix}.$$

If $A=1+\lambda A$ and we consider an infinitesimal transformation, then we can differentiate the first identity with the leibniz rule to get the lhs of the second identity

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}(((1+\lambda A)\mathbf{x})\times((1+\lambda A)\mathbf{y})\cdot((1+\lambda A)\mathbf{z}))\bigg|_{\lambda=0}=(A\mathbf{x})\times\mathbf{y}\cdot\mathbf{z}+\mathbf{x}\times(A\mathbf{y})\cdot\mathbf{z}+\mathbf{x}\times\mathbf{y}\cdot(A\mathbf{z}),$$

and now we can differentiate the rhs of the first identity to get the rhs of the second identity

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}((1+\lambda\operatorname{Tr}(A))\mathbf{x}\times\mathbf{y}\cdot\mathbf{z})\bigg|_{\lambda=0}=\operatorname{Tr}(A)\mathbf{x}\times\mathbf{y}\cdot\mathbf{z},$$

since $\mathbf{x}\times\mathbf{y}\cdot\mathbf{z}$ is independent of λ and if $A=(\mathbf{a}_1,\mathbf{a}_2,\mathbf{a}_3)$

$$(1+\lambda A)=\begin{pmatrix}\mathbf{e}_1+\lambda\mathbf{a}_1 & \mathbf{e}_2+\lambda\mathbf{a}_2 & \mathbf{e}_3+\lambda\mathbf{a}_3\end{pmatrix},$$

so that differentiating the determinant of this matrix

$$\det(1+\lambda A)=(\mathbf{e}_1+\lambda\mathbf{a}_1)\times(\mathbf{e}_2+\lambda\mathbf{a}_2)\cdot(\mathbf{e}_3+\lambda\mathbf{a}_3),$$

with respect to λ at $\lambda=0$ gives us the sum of the diagonal entries of A, which is exactly the trace of A:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}\lambda}\det(1+\lambda A)\bigg|_{\lambda=0}&=\mathbf{a}_1\cdot\mathbf{e}_2\times\mathbf{e}_3+\mathbf{e}_1\times\mathbf{a}_2\cdot\mathbf{e}_3+\mathbf{e}_1\times\mathbf{e}_2\cdot\mathbf{a}_3\\ &=\mathbf{a}_1\cdot\mathbf{e}_2\times\mathbf{e}_3+\mathbf{a}_2\cdot\mathbf{e}_3\times\mathbf{e}_1+\mathbf{a}_3\cdot\mathbf{e}_1\times\mathbf{e}_2\\ &=\mathbf{a}_1\cdot\mathbf{e}_1+\mathbf{a}_2\cdot\mathbf{e}_2+\mathbf{a}_3\cdot\mathbf{e}_3=a_{11}+a_{22}+a_{33}=\operatorname{Tr}(A).\end{aligned}$$

□

This identities are very useful when encountering the determinant and trace of a matrix in the context of Lie groups and Lie algebras, as we will see in the next chapters.

3.2 | Algebraic Operations and Exterior Algebra

3.2.1 | Direct Product

For two vectors $\mathbf{x} = (x_1, x_2, x_3)$ and $\mathbf{y} = (y_1, y_2, y_3)$ in $\mathbb{R}^3$, we can define their direct product as a 3×3 matrix

$$\mathbf{x} \otimes \mathbf{y} = \begin{pmatrix} x_1\, y_1 & x_1\, y_2 & x_1\, y_3 \\ x_2\, y_1 & x_2\, y_2 & x_2\, y_3 \\ x_3\, y_1 & x_3\, y_2 & x_3\, y_3 \end{pmatrix} \in \mathbb{R}(3) = \mathrm{Mat}(3, \mathbb{R}).$$

Definition 3.1 (Direct Product)**.** The direct product of two vectors $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^3$ is defined as the 3×3 matrix obtained by multiplying each component of $\mathbf{x}$ with each component of $\mathbf{y}$, resulting in a matrix where the entry in the i-th row and j-th column is given by $x_i y_j$. Mathematically, it can be expressed as:

$$\otimes : \mathbb{R}^3 \times \mathbb{R}^3 \mapsto \mathbb{R}^{3\times 3},$$
$$(\mathbf{x}, \mathbf{y}) \mapsto \mathbf{x} \otimes \mathbf{y} = \begin{pmatrix} x_1\, y_1 & x_1\, y_2 & x_1\, y_3 \\ x_2\, y_1 & x_2\, y_2 & x_2\, y_3 \\ x_3\, y_1 & x_3\, y_2 & x_3\, y_3 \end{pmatrix}.$$

This operation is bilinear and has the following properties:

$$\begin{cases} (\mathbf{x} \otimes \mathbf{y})\mathbf{z} = (\mathbf{y} \cdot \mathbf{z})\mathbf{x}, \\ (\mathbf{x} \otimes \mathbf{y})^T = \mathbf{y} \otimes \mathbf{x}, \\ \mathrm{Tr}(\mathbf{x} \otimes \mathbf{y}) = \mathbf{x} \cdot \mathbf{y}, \\ \det(\mathbf{x} \otimes \mathbf{y}) = 0. \end{cases}$$

An eigenvector belong to the eigenvalue 0, so the direct product of two vectors is a singular matrix.

This operation is also known as the outer product of two vectors, and it is different from the inner product (dot product) and the cross product. The direct product of two vectors can be used to construct higher-dimensional tensors and has applications in various fields such as physics, engineering, and machine learning.

More identities can be derived from the definition of the direct product; suppose $A \in \mathbb{R}(3)$ is a 3×3 matrix and $\mathbf{x}, \mathbf{y}$ are two vectors in $\mathbb{R}^3$, then we have the following identities:

$$(A\mathbf{x}) \otimes (A\mathbf{y}) = A(\mathbf{x} \otimes \mathbf{y})A^T,$$
$$A = \sum_{i,j} A_{ij} \mathbf{e}_i \otimes \mathbf{e}_j.$$

The first identity can be interpreted as the invariance of the direct product under linear transformations, while the second identity is the expansion of a matrix in terms of the canonical basis of $\mathbb{R}^3$ and the direct product.

3.2.2 | The Star Operator

For a fixed vector $\mathbf{x} \in \mathbb{R}^3$, we can define a linear operator mapping any vector $\mathbf{y} \in \mathbb{R}^3$ to the cross product of $\mathbf{x}$ and $\mathbf{y}$:

$$\star : \mathbb{R}^3 \mapsto \mathbb{R}(3),$$

$$\mathbf{x} \mapsto \star\mathbf{x} = \begin{pmatrix} 0 & x_3 & -x_2 \\ -x_3 & 0 & x_1 \\ x_2 & -x_1 & 0 \end{pmatrix}.$$

$$(\star\mathbf{x})\mathbf{y} = -\mathbf{x} \times \mathbf{y}.$$

$$({}^{\star}\mathbf{x})_{ij} = \sum_k \epsilon_{ijk} x_k.$$

The star operator is a linear map that takes a vector in $\mathbb{R}^3$ and produces a skew-symmetric matrix. This matrix can be used to compute the cross product of the original vector with any other vector in $\mathbb{R}^3$.

Given a matrix $A \in \mathbb{R}(3)$, we can find its elements as

$$A_{ij} = \mathbf{e}_i \cdot A\mathbf{e}_j,$$

valid for any matrix A and the canonical basis vectors $\mathbf{e}_i$. Therefore, we can express the star operator in terms of the direct product as

$$\begin{aligned}(\star\mathbf{x})_{ij} = \mathbf{e}_i \cdot (\star\mathbf{x})\mathbf{e}_j &= -\mathbf{e}_i \cdot (\mathbf{x} \times \mathbf{e}_j) \\ &= +\mathbf{x} \cdot (\mathbf{e}_i \times \mathbf{e}_j) \ldots\end{aligned}$$

We can find its transpose as

$$(\star\mathbf{x})^T = - \star \mathbf{x},$$

and its determinant and trace as

$$\det(\star\mathbf{x}) = 0, \quad \mathrm{Tr}(\star\mathbf{x}) = 0,$$

since the star operator has an eigenvalue of 0: this comes from the fact that the cross product of a vector with itself is always zero, which means that the star operator has a non-trivial kernel and therefore must have an eigenvalue of 0. The trace of the star operator is also zero because it is a skew-symmetric matrix (i.e. $(\star\mathbf{x})^T = -\star\mathbf{x}$), and the trace of any skew-symmetric matrix is always zero.

We can also compute the action of the star operator on the action of a matrix A on a vector $\mathbf{x}$:

$$\star(A\mathbf{x}) = \det A A^{-1,T}(\star\mathbf{x})A^{-1},$$

and this is very interesting: imagine that A is a rotation matrix

$$A = R \in \mathrm{SO}(3), \quad \det R = 1, \quad R^{-1} = R^T,$$

then the star operator transforms under the rotation as

$$\star(R\mathbf{x}) = R(\star\mathbf{x})R^T,$$

which means that the star operator transforms covariantly under rotations, and this is a very important property in physics, especially in the study of angular momentum and spin.

Proof. Suppose $A \in \mathbb{R}(3)$ and $\mathbf{y}, \mathbf{z} \in \mathbb{R}^3$. Then,

$$\begin{aligned}(\star(A\mathbf{x})\mathbf{y}) \cdot \mathbf{z} &= -((A\mathbf{x}) \times \mathbf{y}) \cdot \mathbf{z} \\ &= -(A\mathbf{x}) \times AA^{-1}\mathbf{y} \cdot AA^{-1}\mathbf{z} \\ &= -\det A(\mathbf{x} \times A^{-1}\mathbf{y} \cdot A^{-1}\mathbf{z}) \\ &= \det A(A^{-1,T}(\mathbf{x} \times A^{-1}\mathbf{y}) \cdot \mathbf{z}) \\ &= \det A((A^{-1,T}(\star\mathbf{x})A^{-1})\mathbf{y}) \cdot \mathbf{z},\end{aligned}$$

but since $\mathbf{z}$ is arbitrary, we can conclude that

$$\star(A\mathbf{x})\mathbf{y} = (\det A)A^{-1,T}(\star\mathbf{x})A^{-1}\mathbf{y},$$

which means that

$$\star(A\mathbf{x}) = \det AA^{-1,T}(\star\mathbf{x})A^{-1}.$$

□

Ther is another important identity involving the star operator, which is basically the inverse of the first identity: suppose $A \in \mathbb{R}(3)_{AS}$, which is the space of skew-symmetric matrices, then we have

$$\star : \mathbb{R}(3)_{AS} \mapsto \mathbb{R}^3,$$
$$A \mapsto \star A = \begin{pmatrix} A_{32} \\ A_{13} \\ A_{21} \end{pmatrix},$$

so that we can find the action of the star operator on a skew-symmetric matrix A as

$$A\mathbf{x} = -(\star A) \times \mathbf{x}.$$

Then we can find the elements of the skew-symmetric matrix A in terms of the components of the vector $\star A$ as

$$(\star A)_i = \frac{1}{2}\sum_{j,k} \epsilon_{ijk}A_{jk},$$

which means that the star operator is an isomorphism between the space of skew-symmetric matrices and the space of vectors in $\mathbb{R}^3$. This is a very important result, as it allows us to identify the Lie algebra of the rotation group SO(3) with the space of skew-symmetric matrices, and therefore with the space of vectors in $\mathbb{R}^3$. This isomorphism is also known as the **Lie algebra isomorphism** between $\mathfrak{so}(3)$ and $\mathbb{R}^3$.

3.2.3 | The Wedge Product

Suppose $\mathbf{x}, \mathbf{y} \in \mathbb{R}^3$ are two vectors, then we can define their wedge product as a bivector in the exterior algebra of $\mathbb{R}^3$:

Definition 3.2 (Wedge Product)**.** The wedge product of two vectors $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^3$ is defined as a bivector in the exterior algebra of $\mathbb{R}^3$, denoted by $\mathbf{x} \wedge \mathbf{y}$. It can be expressed as:

$$\mathbf{x} \wedge \mathbf{y} = (x_1y_2 - x_2y_1)\mathbf{e}_1 \wedge \mathbf{e}_2 + (x_1y_3 - x_3y_1)\mathbf{e}_1 \wedge \mathbf{e}_3 + (x_2y_3 - x_3y_2)\mathbf{e}_2 \wedge \mathbf{e}_3,$$

where $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is the canonical basis of $\mathbb{R}^3$. The wedge product is an antisymmetric bilinear operation that takes two vectors and produces a bivector, which can be thought of as an oriented area element in $\mathbb{R}^3$.

$$\wedge : \mathbb{R}^3 \times \mathbb{R}^3 \mapsto \Lambda^2(\mathbb{R}^3),$$
$$(\mathbf{x}, \mathbf{y}) \mapsto \mathbf{x} \wedge \mathbf{y}.$$

We can also express the wedge product in terms of the direct product as

$$\mathbf{x} \wedge \mathbf{y} = \mathbf{x} \otimes \mathbf{y} - \mathbf{y} \otimes \mathbf{x},$$

and its elements can be expressed as

$$(\mathbf{x} \wedge \mathbf{y})_{ij} = x_iy_j - y_ix_j.$$

and the matrix representation of the wedge product can be expressed as

$$\mathbf{x} \wedge \mathbf{y} = \begin{pmatrix} 0 & x_1y_2 - x_2y_1 & x_1y_3 - x_3y_1 \\ -(x_1y_2 - x_2y_1) & 0 & x_2y_3 - x_3y_2 \\ -(x_1y_3 - x_3y_1) & -(x_2y_3 - x_3y_2) & 0 \end{pmatrix}.$$

It can be expressed in terms of the cross product as

$$\mathbf{x} \wedge \mathbf{y} = \star(\mathbf{x} \times \mathbf{y}),$$

which means that the wedge product of two vectors can be obtained by applying the star operator to their cross product. This relationship between the wedge product and the cross product is a fundamental result in vector calculus and has important implications in physics, particularly in the study of electromagnetism and fluid dynamics: the cross product can be used to compute the curl of a vector field, while the wedge product can be used to compute the exterior derivative of a differential form. The wedge product is also a key operation in the theory of differential forms and has applications in geometry and topology.

Proof. Suppose $\mathbf{z} \in \mathbb{R}^3$ is a vector, and let us start from the action of the star operator on the cross product of two vectors $\mathbf{x}$ and $\mathbf{y}$ applied to $\mathbf{z}$:

$$\begin{aligned} \star(\mathbf{x} \times \mathbf{y})\mathbf{z} &= -(\mathbf{x} \times \mathbf{y}) \times \mathbf{z} = -\mathbf{z} \times (\mathbf{x} \times \mathbf{y}) \\ &= \dots \end{aligned}$$

where we used the Jacobi identity between the first and second lines,... $\square$

We can compute the determinant and trace of the wedge product as

$$\det(\mathbf{x} \wedge \mathbf{y}) = 0, \quad \mathrm{Tr}(\mathbf{x} \wedge \mathbf{y}) = 0,$$

since the wedge product is a skew-symmetric matrix (since also the cross product is skew-symmetric), and any skew-symmetric matrix has an eigenvalue of 0 and a trace of 0. The wedge product can be thought of as an oriented area element in $\mathbb{R}^3$, and its determinant being zero reflects the fact that it does not have full rank, while its trace being zero reflects the fact that it is a skew-symmetric matrix.

Using the canonical ON basis of $\mathbb{R}^3$, we can express the wedge product of the basis: there are 9 wedge products of the basis vectors, but only 3 of them are linearly independent, other 3 of them are zero, and the remaining 3 of them are linearly dependent on the first 3:

$$\begin{cases} \mathbf{e}_i \wedge \mathbf{e}_i = 0, \\ \mathbf{e}_i \wedge \mathbf{e}_j = -\mathbf{e}_j \wedge \mathbf{e}_i. \end{cases}$$

Considering a matrix $A \in \mathbb{R}(3)_{AS}$ such that $A^T = -A$, we know that A can be expressed as a linear combination of the wedge products of the basis vectors:

$$A = \frac{1}{2} \sum_{i,j} A_{ij} \mathbf{e}_i \wedge \mathbf{e}_j,$$

which means that the space of skew-symmetric matrices can be identified with the space of bivectors in the exterior algebra of $\mathbb{R}^3$. This is a very important result, as it allows us to identify the Lie algebra of the rotation group SO(3) with the space of skew-symmetric matrices, and therefore with the space of bivectors in the exterior algebra of $\mathbb{R}^3$. This isomorphism is also known as the **Lie algebra isomorphism** between $\mathfrak{so}(3)$ and $\Lambda^2(\mathbb{R}^3)$.

Proof. We can write

$$\begin{aligned} A = \sum_{ij} A_{ij}\mathbf{e}_i \otimes \mathbf{e}_j &= \frac{1}{2}\sum_{ij}(A_{ij}\mathbf{e}_i \otimes \mathbf{e}_j + A_{ij}\mathbf{e}_j \otimes \mathbf{e}_i) \\ &= \frac{1}{2}\sum_{ij} A_{ij}(\mathbf{e}_i \otimes \mathbf{e}_j - \mathbf{e}_j \otimes \mathbf{e}_i) \\ &= \frac{1}{2}\sum_{ij} A_{ij}\mathbf{e}_i \wedge \mathbf{e}_j, \end{aligned}$$

where we used the fact that A is skew-symmetric, so that $A_{ij} = -A_{ji}$, and therefore we can symmetrize the sum over i and j to get the wedge product of the basis vectors. □

3.3 | Complexification and Spectral Theory

When considering a rotation matrix $R \in \mathrm{SO}(3)$, you have to imagine that it is a complex matrix which happens to have real entries:

$$R \in \mathrm{SO}(3) \subset \mathbb{C}(3).$$

Indeed its eigenvalues and vectors in general are complex. The Pauli matrices formalism is a way to represent these complex eigenvalues and eigenvectors in a more intuitive way, by using the Pauli matrices as a basis for the space of complex matrices.

Thus we consider the complex extension of $\mathbb{R}(3) \subset \mathbb{C}(3)$ of real 3×3 matrices living in the complex space.

in the real case

$$\mathbf{x} \cdot \mathbf{y} \in \mathbb{R}^3, \rightarrow \begin{cases} \mathbf{x} \cdot \mathbf{x} \geq 0, \\ \mathbf{x} \cdot \mathbf{x} = 0 \iff \mathbf{x} = 0. \end{cases}$$

For the complex case, we have to define a new inner product which is positive definite:

$$\mathbf{x} \cdot \mathbf{y} \in \mathbb{C}^3, \rightarrow \begin{cases} \mathbf{x} \cdot \mathbf{x} \geq 0, \\ \mathbf{x} \cdot \mathbf{x} = 0 \text{ even if } \mathbf{x} \neq 0. \end{cases}$$

This is the case for the standard inner product in $\mathbb{C}^3$, thus we can define the hermitian inner product as

$$\mathbf{x}^* \cdot \mathbf{y} = \sum_{i=1}^{3} x_i^* y_i, \rightarrow \begin{cases} \mathbf{x}^* \cdot \mathbf{x} \geq 0, \\ \mathbf{x}^* \cdot \mathbf{x} = 0 \iff \mathbf{x} = 0. \end{cases}$$

The orthonormal basis $\mathbf{v}_i$ of $\mathbb{C}^3$ is defined as

$$\mathbf{v}_i^* \cdot \mathbf{v}_j = \delta_{ij}.$$

Consideing this basis, we can represent any vector $\mathbf{x} \in \mathbb{C}^3$ as

$$\mathbf{x} = \sum_i (v_i^* \cdot \mathbf{x})\mathbf{v}_i = \sum_i x_i \mathbf{v}_i.$$

Proof. We can write

$$\mathbf{x} = \sum_i (v_i^* \cdot \mathbf{x})\mathbf{v}_i,$$
$$\mathbf{x} = \sum_i c_{ij}\mathbf{v}_j,$$

and

$$\mathbf{v}_i^* \cdot \mathbf{x} = \sum_j c_{ij}\mathbf{v}_i^* \cdot \mathbf{v}_j = c_{ii} = x_i,$$

thus

$$\mathbf{x} = \sum_i (v_i^* \cdot \mathbf{x})\mathbf{v}_i,$$

proving the equality.

□

$$\mathbf{x} = \sum_i (\mathbf{v}_i \otimes \mathbf{v}_i^*)\mathbf{x},$$

using the fact that $\mathbf{v}_i^* \cdot \mathbf{v}_j = \delta_{ij}$, ...

Consider $A \in \mathbb{C}(3)$ and suppose that A is diagonalizable, then there exists an orthonormal basis of eigenvectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ of $\mathbb{C}^3$ such that

$$A\mathbf{v}_i = \lambda_i \mathbf{v}_i, \quad i = 1, 2, 3, \quad \lambda_i \in \mathbb{C}.$$

This means that we can express the matrix A in terms of its eigenvalues and eigenvectors as

$$A = \sum_i \lambda_i \mathbf{v}_i \otimes \mathbf{v}_i^*,$$

which is the **spectral decomposition** of the matrix A.

Theorem 3.1 (Spectral theorem)**.** *Let A be a diagonalizable matrix in $\mathbb{C}(3)$. Then there exists an orthonormal basis of eigenvectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ of $\mathbb{C}^3$ such that*

$$A = \sum_i \lambda_i \mathbf{v}_i \otimes \mathbf{v}_i^*,$$

where λ_i are the eigenvalues of A corresponding to the eigenvectors $\mathbf{v}_i$. Then the matrix can be diagonalized as

$$A = V\Lambda V^{-1}, \quad A = \sum_i \lambda_i \mathbf{v}_i \otimes \mathbf{v}_i^*,$$

where V is the matrix whose columns are the eigenvectors $\mathbf{v}_i$ and Λ is the diagonal matrix whose diagonal entries are the eigenvalues λ_i.

This decomposition allows us to understand the structure of the matrix A in terms of its eigenvalues and eigenvectors, and it is a fundamental result in linear algebra with applications in various fields such as physics, engineering, and machine learning. The spectral decomposition is particularly useful when dealing with normal matrices, which are matrices that commute with their conjugate transpose, as they can be diagonalized by a unitary transformation.

Proof. Since A is diagonalizable, we can write A as

$$\begin{aligned} A = A\mathbb{I}_3 = A \sum_i \mathbf{v}_i \otimes \mathbf{v}_i^* = \sum_i A\mathbf{v}_i \otimes \mathbf{v}_i^* \\ = \sum_i \lambda_i \mathbf{v}_i \otimes \mathbf{v}_i^* = A. \end{aligned}$$

$\square$

4 | Applications for Physics

4.1 | Pauli Matrices Formalism

In this section, we will explore the Pauli matrices and their applications in quantum mechanics from a group theory perspective. The Pauli matrices are a set of three 2x2 complex Hermitian and unitary matrices that are widely used in the study of spin-1/2 particles and quantum information theory.

Our discussion will be centered around the following groups and their representations:

- SU(2): The special unitary group of degree 2, which is the double cover of the rotation group SO(3). The Pauli matrices serve as generators for the Lie algebra $\mathfrak{su}(2)$ associated with this group.
- SO(3): The special orthogonal group in three dimensions, which describes the symmetries of physical space. The representations of SO(3) can be constructed using the Pauli matrices.
- $\mathrm{SL}(2, \mathbb{C})$: The special linear group of degree 2 over the complex numbers, which is related to the Lorentz group in relativistic quantum mechanics. The Pauli matrices play a crucial role in the representation theory of this group as well.
- $SO_+(1,3)$: The proper orthochronous Lorentz group, which describes the symmetries of spacetime in special relativity. The Pauli matrices are used to construct the spinor representations of this group.

4.1.1 | Pauli Matrix Functions

In this section, we will define the Pauli matrices and explore their properties. The Pauli matrix function is defined as follows:

Definition 4.1 (Pauli Matrix Function)**.** A Pauli matrix function is an element of $\sigma \in \mathrm{Hom}(\mathbb{C}^3, \mathbb{C}(2))$, i.e. a linear map

$$\sigma : \mathbb{C}^3 \to \mathbb{C}(2),$$

satisfying the following properties: for all $\mathbf{x}, \mathbf{y} \in \mathbb{C}^3$, we have

- $\sigma(\mathbf{x})^\dagger = \sigma(\mathbf{x}^*)$, where $\dagger$ denotes the Hermitian conjugate and $*$ denotes complex conjugation.
- $\sigma(\mathbf{x})\sigma(\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}\mathbb{I}_3 + i\sigma(\mathbf{x} \times \mathbf{y})$, where $\cdot$ denotes the dot product and $\times$ denotes the cross product in $\mathbb{C}^3$.

(Anti)commutation Relations

The Pauli matrices satisfy specific commutation and anticommutation relations that are fundamental to their algebraic structure. These relations can be expressed as follows:

- Commutation relations: For any two Pauli matrices σ_i and σ_j, we have

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k,$$

  where ϵ_{ijk} is the Levi-Civita symbol.

- Anticommutation relations: For any two Pauli matrices σ_i and σ_j, we have

$$\{\sigma_i, \sigma_j\} = 2\delta_{ij}\mathbb{I}_2,$$

  where δ_{ij} is the Kronecker delta and $\mathbb{I}_2$ is the 2x2 identity matrix.

TODO: check and correct

These define the **Clifford algebra** associated with the Pauli matrices, which is essential for understanding their role in quantum mechanics and representation theory: the Pauli matrices generate the Clifford algebra, which is isomorphic to the algebra of 2x2 complex matrices. This algebraic structure allows us to construct representations of the groups mentioned above and to analyze the symmetries of quantum systems.

We have trace and determinant properties of the Pauli matrices as well, which are important for their applications in quantum mechanics: starting from the trace properties, we can derive the following identities for any $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{C}^3$:

$$\begin{aligned}
\mathrm{Tr}(\sigma(\mathbf{x})) &= 0, \\
\mathrm{Tr}(\sigma(\mathbf{x})\sigma(\mathbf{y})) &= 2\mathbf{x}\cdot\mathbf{y}, \\
\mathrm{Tr}(\sigma(\mathbf{x})\sigma(\mathbf{y})\sigma(\mathbf{z})) &= 2i\mathbf{x}\times\mathbf{y}\cdot\mathbf{z},
\end{aligned}$$

Proof. we have that, from the definition of the Pauli matrix function, we can write $\sigma(\mathbf{x}) = x_i\sigma_i$, where σ_i are the standard Pauli matrices. Then, we can compute the trace properties as follows:...

[only the firt two were proven] □

For the determinant properties, we can derive the following identities for any $\mathbf{x} \in \mathbb{C}^3$:

$$\det(\sigma(\mathbf{x})) = -\mathbf{x}\cdot\mathbf{x},$$

Proof. Start from $\mathbf{x} \in \mathbb{R}^3$ we can write, if $\mathbf{x} = \mathbf{y}$

$$\sigma(\mathbf{x})^2 = \mathbf{x}\cdot\mathbf{x}\mathbb{I}_2,$$

and by computing the determinant of both sides, we get

$$(\det(\sigma(\mathbf{x})))^2 = (\det(\mathbf{x}\cdot\mathbf{x}\mathbb{I}_2))^2 = (\mathbf{x}^2)^2,$$

which implies

$$\det(\sigma(\mathbf{x})) = \eta(\mathbf{x})\mathbf{x}\cdot\mathbf{x}, \quad \eta(\mathbf{x}) = \pm 1.$$

Indeed

$$\eta(\mathbf{x}) = \frac{\det(\sigma(\mathbf{x}))}{\mathbf{x}\cdot\mathbf{x}},$$

is a continuous function of $\mathbf{x}$ that takes values in

$$\mathbb{R}^3 \setminus \{0\} \to \mathbb{C},$$

and since $\mathbb{R}^3 \setminus \{0\}$ is connected, we have that $\eta(\mathbf{x})$ is constant,

$$\det(\sigma(\mathbf{x}))\det(\sigma(\mathbf{y})) = \dots$$

$\square$

4.2 | Rotations Group O(3)

[...]

Theorem 4.1 (Diagonalization theorem for O(3))**.** *Let* $R \in \mathrm{O}(3) \subset \mathbb{C}(3)$*. Then there is a sign* $\epsilon \in \{-1, +1\}$ *and a phase* $q \in [0, \pi]$ *and a basis* Q_α*, such that* $\alpha = -1, 0, 1$*. Then the following holds:*

1. $v_\alpha \cdot v_\beta = \delta_{-\alpha,\beta}$,
2. $v_\alpha^* = v_{-\alpha}$,
3. $v_0 \times v_\alpha = i\alpha v_\alpha$,
4. ...
5. ...

Proof. let $w \in \mathbb{C}^3$ invariant under R. Then we cn define its ortonormal vector $w^\perp \in \mathbb{C}^3$ which is also invariant under R. Then ...

Thus R has always one real eigenvalue and two complex conjugate eigenvalues (maybe also real). The real one $\lambda_0 = \epsilon$ while the complex ones are $\lambda_{\pm 1} = e^{\pm iq}$. Then we can define

$$\lambda_\alpha = \epsilon e^{-i\alpha q}, \qquad \alpha = -1, 0, 1.$$

Thus

$$Rv_\alpha = v_\alpha \Lambda_\alpha, \quad Rv_\alpha^* = v_\alpha^* \Lambda_{-\alpha}^* = e^{i\chi_\alpha} v_{-\alpha} \Lambda_{-\alpha}.$$

since given (?) we have that $v_\alpha^* = e^{-i\chi_\alpha} v_{-\alpha}$. [...]

$$v_1 = \frac{1}{\sqrt{2}}(v_1 + v_{-1}), \qquad v_2 = \frac{1}{\sqrt{2}i}(v_1 - v_{-1}), \qquad v_3 = v_0,$$

and we can prove this basis to be ortonormal and to satisfy the properties of the theorem. Hence $v_i \times v_j = \sum_{k=-1}^{1} \epsilon_{ijk} v_k$, proving (3) and (4). ☐

...

This shows that defining these matrices in complex space give us the ability to diagonalize them, which is not possible in real space (except for special cases). This is the reason why we need to complexify the representation of O(3) to get the irreducible representations of it.

Proposition 4.2 (...)**.** *Let* $R \in \mathrm{O}(3) \subset \mathbb{C}(3)$*. Then there is a sign* $\epsilon \in \{-1, +1\}$ *and a phase* $\phi \in [0, \pi]$ *and a unit vector* $\mathbf{n} \in \mathbb{R}^3$*,* $n^2 = 1$*, such that*

$$R = \epsilon \left[\mathbf{n} \otimes \mathbf{n} + \cos(\phi)\left(1 - \mathbf{n} \otimes \mathbf{n}\right) - \sin(\phi) \star \mathbf{n}\right].$$

Every matrix R *of this form is an element of* $\mathrm{O}(3)$*. The opposite is also true: every element of* $\mathrm{O}(3) \subset \mathbb{C}(3)$ *can be written in this form.*

Proof.

$$R = \epsilon \left[\mathbf{v}_0 \otimes \mathbf{v}_0 + (\cos(\phi) - i\sin(\phi))\mathbf{v}_1 \otimes \mathbf{v}_{-1}\right],$$

were now we can write

$$\epsilon\left[\mathbf{v}_0\otimes\mathbf{v}_0+\cos(\phi)\left(\mathbf{v}_1\otimes\mathbf{v}_{-1}+\mathbf{v}_{-1}\otimes\mathbf{v}_1\right)-i\sin(\phi)\left(\mathbf{v}_1\otimes\mathbf{v}_{-1}-\mathbf{v}_{-1}\otimes\mathbf{v}_1\right)\right],$$

since ... now

$$\sum_\alpha \mathbf{v}_\alpha\otimes\mathbf{v}_{-\alpha}=1, \qquad \Longrightarrow \qquad \mathbf{v}_1\otimes\mathbf{v}_{-1}+\mathbf{v}_{-1}\otimes\mathbf{v}_1=1-\mathbf{v}_0\otimes\mathbf{v}_0,$$

thus

$$\mathbf{v}_1\wedge\mathbf{v}_{-1}=\mathbf{v}_1\otimes\mathbf{v}_{-1}-\mathbf{v}_{-1}\otimes\mathbf{v}_1=\star(\mathbf{v}_1\times\mathbf{v}_{-1})=\star(-i\mathbf{v}_0)=-i\star\mathbf{v}_0.$$

So we get

$$R=\epsilon\left[\mathbf{v}_0\otimes\mathbf{v}_0+\cos(\phi)(1-\mathbf{v}_0\otimes\mathbf{v}_0)-i\sin(\phi)(-i\star\mathbf{v}_0)\right]=\epsilon\left[\mathbf{v}_0\otimes\mathbf{v}_0+\cos(\phi)(1-\mathbf{v}_0\otimes\mathbf{v}_0)-\sin(\phi)\star\mathbf{v}_0\right].$$

By setting $\mathbf{n}=\mathbf{v}_0$ we get the desired result, and we can do this since $\mathbf{v}_0\in\mathbb{R}^3$ and $\mathbf{v}_0^2=1$. □

We can name the previous expression as the *angle-axis parametrization* of O(3). The angle ϕ is the angle of rotation, while the axis of rotation is given by the unit vector $\mathbf{n}$. The sign ϵ tells us whether we are dealing with a proper or an improper rotation. In particular, if $\epsilon=+1$ we are dealing with a proper rotation, while if $\epsilon=-1$ we are dealing with an improper rotation (a reflection). Thus we can write any matrices in O(3) as

$$R_\epsilon(\phi,\,\mathbf{n})=\epsilon\left[\mathbf{n}\otimes\mathbf{n}+\cos(\phi)(1-\mathbf{n}\otimes\mathbf{n})-\sin(\phi)\star\mathbf{n}\right].$$

Example (Exercise). Being $A\in\mathbb{R}(3)$ and $\mathbf{v}_i$ with $i=1,2,3$ an ortonormal basis of $\mathbb{R}^3$ such that

$$A\mathbf{v}_i=\sum_{j=1}^3\hat{A}_{ji}\mathbf{v}_j, \quad (\hat{A}_{ji})\in\mathbb{R}(3).$$

Thus $\hat{A}$ is a representation matrix of A in the basis $\mathbf{v}_i$.

Now, being $R\in\mathrm{O}(3)$ and $\mathbf{v}_i$ respecting

1. $\mathbf{v}_3=\mathbf{n}$,
2. $\mathbf{v}_i\cdot\mathbf{v}_j=\delta_{ij}$ for all i,j,
3. $\mathbf{v}_i\times\mathbf{v}_j=\sum_{k=1}^3\epsilon_{ijk}\mathbf{v}_k$.

Then we can write

$$R_\epsilon(\phi,\,\mathbf{n})=\epsilon\left[\mathbf{n}\otimes\mathbf{n}+\cos(\phi)(1-\mathbf{n}\otimes\mathbf{n})-\sin(\phi)\star\mathbf{n}\right].$$

Then its action on the basis $\mathbf{v}_i$ is given by

$$R_\epsilon(\phi,\,\mathbf{n})\mathbf{v}_i=\sum_{j=1}^3\hat{R}_\epsilon(\phi,\,\mathbf{n})_{ji}\mathbf{v}_j,$$

where

$$\hat{R}_\epsilon(\phi,\,\mathbf{n})=\epsilon\begin{pmatrix}\cos(\phi) & -\sin(\phi) & 0\\ \sin(\phi) & \cos(\phi) & 0\\ 0 & 0 & 1\end{pmatrix}.$$

This can be obtained by explicitly calculating the action of $R_\epsilon(\phi,\, \mathbf{n})$ on the basis $\mathbf{v}_i$. For example, we have

$$\begin{aligned} R_\epsilon(\phi,\, \mathbf{n})\mathbf{v}_i &= \epsilon \left[\mathbf{n}\otimes\mathbf{n} + \cos(\phi)(1-\mathbf{n}\otimes\mathbf{n}) - \sin(\phi)\star\mathbf{n}\right]\mathbf{v}_i \\ &= \dots \end{aligned}$$

$\hat{A}$ depends on the choice of the basis $\mathbf{v}_i$, however its trace and determinant do not depend on the choice of the basis. In particular, we have

$$\begin{cases} \mathrm{Tr}\left(\hat{A}\right) = \mathrm{Tr}(A), \\ \det\left(\hat{A}\right) = \det(A). \end{cases}$$

Hence we now that and .

$$\begin{cases} \det(R_\epsilon(\phi,\mathbf{n})) = \det\left(\hat{R}_\epsilon(\phi,\mathbf{n})\right) = \epsilon^3 = \epsilon, \\ \mathrm{Tr}(R_\epsilon(\phi,\mathbf{n})) = \mathrm{Tr}\left(\hat{R}_\epsilon(\phi,\mathbf{n})\right) = (2\cos\phi + 1)\epsilon. \end{cases}$$

This is very useful since it allows us to determine the angle ϕ and the sign ϵ (which are equal for either the matrix and its representation) from the trace and determinant of R: we have all the element to write the angle-axis parametrization $\hat{R}$ of R.

This result implies that we can write the action of any element of O(3) on $\mathbb{R}^3$ as a rotation around an axis, in the following form

$$A\mathbf{x} = \sum_i \left(\sum_j \hat{A}_{ij}\mathbf{x}_j\right)\mathbf{v}_i.$$

Let us write

$$R_{+1}(\phi,\, \mathbf{n}) = \left[\mathbf{v}_3\otimes\mathbf{v}_3 + \cos(\phi)(1-\mathbf{v}_3\otimes\mathbf{v}_3) - \sin(\phi)\star\mathbf{v}_3\right],$$

then adopting the following cylindrical coordinates:

$$\begin{cases} \mathbf{x}_1 = \rho\cos\theta, \\ \mathbf{x}_2 = \rho\sin\theta, \\ \mathbf{x}_3 = z, \epsilon = +1, \end{cases}$$

we can write the action of the orthogonal matrix $R_{+1}(\phi,\, \mathbf{n})$ on the vector $\mathbf{x}$ as

$$\begin{aligned} (R_{+1}(\phi,\, \mathbf{n})\mathbf{x})_1 &= \rho\cos(\phi+0), \\ (R_{+1}(\phi,\, \mathbf{n})\mathbf{x})_2 &= \rho\sin(\phi+0), \\ (R_{+1}(\phi,\, \mathbf{n})\mathbf{x})_3 &= z, \end{aligned}$$

showing that the action of $R_{+1}(\phi,\, \mathbf{n})$ on any vector $\mathbf{x}\in\mathbb{R}^3$ is a rotation of angle ϕ around the axis $\mathbf{n}$.

Thus we have shown that if $R\in\mathrm{O}(3)$ then there are $\epsilon=\pm1$ and $\phi\in[0,\,\pi]$ and $\mathbf{n}\in\mathbb{R}^3$ with $\mathbf{n}^2=1$ such that

$$R_\epsilon(\phi,\, \mathbf{n}) = \epsilon\left[\mathbf{n}\otimes\mathbf{n} + \cos(\phi)(1-\mathbf{n}\otimes\mathbf{n}) - \sin(\phi)\star\mathbf{n}\right].$$

with ϕ and $\mathbf{n}$ given by the trace and determinant of R through the following relations:

$$\begin{cases} \epsilon = \det(R), \\ \phi = \arccos\left(\dfrac{\mathrm{Tr}(R)}{2\epsilon} - \dfrac{1}{2}\right). \end{cases}$$

We can also write the angle-axis parametrization of $R \in \mathrm{SO}(3)$, the special orthogonal group, with only proper rotations, as $\epsilon = 1$, but the relation is exactely the same:

$$R_{+1}(\phi,\, \mathbf{n}) = \mathbf{n} \otimes \mathbf{n} + \cos(\phi)(1 - \mathbf{n} \otimes \mathbf{n}) - \sin(\phi) \star \mathbf{n}.$$

A special case is the space reflection with respect to the origin, which is given by $\epsilon = -1$ and $\phi = 0$ and in physical space is called *parity transformation*; it is given by

$$P = R_{-1}(0, \mathbf{n}) = -1, \quad P\mathbf{x} = -\mathbf{x}.$$

With this transformation we can relate the orthogonal group O(3) to the special orthogonal group SO(3) as follows:

$$\mathrm{O}(3) = \mathrm{SO}(3) \cup P\mathrm{SO}(3).$$

At a certain point it becomes importabt to know if this parametrization is unique, i.e. if given R there is only one way to write it as $R_\epsilon(\phi,\, \mathbf{n})$. If the parameters ϵ, ϕ, and $\mathbf{n}$ are uniquely determined by R, then the parametrization is unique. But since the first two parameters ϵ and ϕ are determined by the trace and determinant of R, which are invariant under change of basis, we only need to check if the axis of rotation $\mathbf{n}$ is uniquely determined by R. This is not the case since:

- If we have a rotation of angle $\phi = \pi$ around an axis $\mathbf{n}$, then we can also write it as a rotation of angle π around the axis $-\mathbf{n}$: it is unique up to sign change;

$$R = \dots$$
$$\boldsymbol{\phi} = \pi\mathbf{n} = \pi(-\mathbf{n}).$$

- If we consider $\phi = 0$ then the parametrization is undetermined since any axis $\mathbf{n}$ can be chosen, since the rotation of angle 0 around any axis is the identity transformation: it is totally undetermined;

$$R = \dots \boldsymbol{\phi} = 0\mathbf{n} = 0\mathbf{n}' = \mathbf{0}.$$

It can be proven, however, that if $0 < \phi < \pi$, strictly comprised, then the parametrization is unique, since in this case the axis of rotation $\mathbf{n}$ is uniquely determined by R.

Then if we have defined the phase ϕ as strictly comprised between 0 and π, then in the phase space, the angle vector $\boldsymbol{\phi}$ has to have a norm equal or less than pi:

$$|\boldsymbol{\phi}| \leq \pi.$$

This tells us that the topology of the phase space of SO(3) is a ball of radius π with antipodal[1] points (on the surface) identified, since the points on the surface of the ball, which have norm equal to π, are identified with their antipodal points (the one with opposite sign), since they represent the same rotation.

Now the Lie algebra the special orthogonal group SO(3) is

$$\sigma(3) = \mathfrak{so}(3) = \{A \in \mathbb{R}(3) : A^T = -A\},$$

then we can state the following proposition:

[1] Opposite with respect to the origin.

Proposition 4.3. *For every $X \in \mathfrak{so}(3)$ there exists a unique vector $\mathbf{x} \in \mathbb{R}^3$ such that*

$$X = - \star \mathbf{x}, \quad \star\mathbf{x} = \begin{pmatrix} 0 & \mathbf{x}_3 & -\mathbf{x}_2 \\ -\mathbf{x}_3 & 0 & \mathbf{x}_1 \\ \mathbf{x}_2 & -\mathbf{x}_1 & 0 \end{pmatrix}.$$

Indeed we have seen that with the angle-axis parametrization we can write any element of SO(3) *with three parameters. We denote the map $\mathbf{x} \mapsto - \star \mathbf{x}$ with*

$$X = \ell(\mathbf{x}),$$

with the following properties:

1. $\ell(\mathbf{x})\ell(\mathbf{y}) = \ell(\mathbf{x} \times \mathbf{y})$,
2. $[\ell(\mathbf{x}),\, \ell(\mathbf{y})]\mathbf{z} = \star\mathbf{x} \times \mathbf{y}\mathbf{z} - \star\mathbf{y} \times \mathbf{x}\mathbf{z} \dots$

Indeed we know that the space of antisymmetric matrices is three-dimensional, and the map $\mathbf{x} \mapsto \star\mathbf{x}$ is a linear map from $\mathbb{R}^3$ to $\mathfrak{so}(3)$. Since both spaces have the same dimension, it is enough to show that this map is injective, i.e. that if $\star\mathbf{x} = 0$ then $\mathbf{x} = 0$. But this is true since the only vector that has zero cross product with all the other vectors is the zero vector itself. Hence the map is injective and thus it is an isomorphism, proving the proposition.

[...] then with the properties of the map ℓ we can show that the Lie bracket of $\mathfrak{so}(3)$ is given by

$$[\ell_i,\, \ell_j] = \sum_{k=1}^{3} \epsilon_{ijk} \ell_k.$$

Another useful property comes from the exponential map, since it is naturally connected to the angle-axis parametrization. Indeed we have $\Phi \in \mathbb{R}$, $\mathbf{n} \in S^2$ (the unit sphere in $\mathbb{R}^3$), then the exponential map is given by

$$\exp(\ell(\phi\mathbf{n})) = R_{+1}(\Phi, \mathbf{n}).$$

Proof. To show this, we can write the exponential map as a power series:

$$\exp(\ell(\phi\mathbf{n})) = \sum_{k=0}^{\infty} \frac{1}{k!} \ell(\phi\mathbf{n})^k = \sum_{k=0}^{\infty} \frac{1}{k!} (-\phi \star \mathbf{n})^k,$$

where now if we divide the odd and even powers of k we get

$$\exp(-\phi \star \mathbf{n}) = \sum_{k=0}^{\infty} \frac{1}{(2k)!} (-\phi)^k (\star\mathbf{n})^k = 1 + \sum_{k=1}^{\infty} \frac{\phi^{2k}}{(2k)!} (\star\mathbf{n})^{2k} - \sum_{k=0}^{\infty} \frac{\phi^{2k+1}}{(2k+1)!} (\star\mathbf{n})^{2k+1}.$$

Now let us compute $(\star\mathbf{n})^2$:

$$(\star\mathbf{n})^2\mathbf{x} = \dots = -1 + \mathbf{n} \otimes \mathbf{n},$$

which is a projector on the plane orthogonal to $\mathbf{n}$. Hence, since all the projectors are idempotent, we have

$$\left[(\star\mathbf{n})^2\right]^2 = (\star\mathbf{n})^2,$$

then, with the idea to compute the expansion of the exponential map, we can write

$$(\star\mathbf{n})^{2k+1}\mathbf{x} = \dots = (-1)^k \star \mathbf{n}\mathbf{x}, \quad (\star\mathbf{n})^{2k}\mathbf{x} = \dots = (-1)^k (1 - \mathbf{n} \otimes \mathbf{n})\mathbf{x}.$$

Hence the expansion takes the following form:

$$\begin{aligned}\exp(-\phi \star \mathbf{n}) &= 1 + \sum_{k=1}^{\infty} \frac{\phi^{2k}}{(2k)!}(-1)^k(1 - \mathbf{n} \otimes \mathbf{n}) - \sum_{k=0}^{\infty} \frac{\phi^{2k+1}}{(2k+1)!}(-1)^k \star \mathbf{n} \\ &= 1 + (\cos(\phi) - 1)(1 - \mathbf{n} \otimes \mathbf{n}) - \sin(\phi) \star \mathbf{n} = R_{+1}(\Phi, \mathbf{n}),\end{aligned}$$

which is exactely the angle-axis parametrization of a proper rotation of angle ϕ around the axis **n**, proving the desired result. □

===

4.3 | The Lorentz group $\mathrm{O}(1,3)$

In the Minkowski space $\mathbb{R}^4$ we can define the Lorentz group as the group of linear transformations that preserve the Minkowski metric $\eta = \mathrm{diag}(1,-1,-1,-1)$. We adopt the 1+3 notation for vectors in $\mathbb{R}^4$ or matrices in $\mathbb{R}(4)$, which is the most common in physics.

TODO: substitute vec with tilde under the symbol

$$\vec{u} = \begin{pmatrix} a \\ \mathbf{u} \end{pmatrix}, \quad a \in \mathbb{R},\, \mathbf{u} \in \mathbb{R}^3$$

$$\vec{u}^t = \begin{pmatrix} a & \mathbf{u}^t \end{pmatrix}$$

For the matrices

$$\vec{A} = \begin{pmatrix} x & \mathbf{u}^t \\ \mathbf{v} & P \end{pmatrix}, \quad x \in \mathbb{R},\, \mathbf{u},\, \mathbf{v} \in \mathbb{R}^3,\, P \in \mathbb{R}(3).$$

When applying an operator $\vec{A} \in \mathbb{R}(4)$ on a four-vector $\vec{u} \in \mathbb{R}^4$ we have

$$\vec{A}\vec{u} = \begin{pmatrix} x & \mathbf{u}^t \\ \mathbf{v} & P \end{pmatrix} \begin{pmatrix} a \\ \mathbf{u} \end{pmatrix} = \begin{pmatrix} xa + \mathbf{u}^t\mathbf{u} \\ a\mathbf{v} + P\mathbf{u} \end{pmatrix}.$$

TODO: Correct last expression

When multiplying two matrices $\vec{A}$ and $\vec{B}$ we have

$$\vec{A}\vec{B} = \begin{pmatrix} x & \mathbf{u}^t \\ \mathbf{v} & P \end{pmatrix} \begin{pmatrix} y & \mathbf{w}^t \\ \mathbf{z} & Q \end{pmatrix} = \begin{pmatrix} xy + \mathbf{u}^t\mathbf{z} & x\mathbf{w}^t + \mathbf{u}^t Q \\ y\mathbf{v} + P\mathbf{z} & \mathbf{v}\mathbf{w}^t + PQ \end{pmatrix}.$$

The minkowski metric can be written as

$$\vec{\eta} = \begin{pmatrix} 1 & \mathbf{0}^t \\ \mathbf{0} & -\mathbf{I}_3 \end{pmatrix}.$$

Let us define the full Lorentz group as

$$\mathrm{O}(1,3) = \{\vec{L} \in \mathbb{R}(4) : \vec{L} \text{ invertible},\, \vec{L}^T = \vec{L}^{-1},\, \vec{L}^T = \vec{\eta}L^t\vec{\eta}\}.$$

$$\vec{\eta}\vec{L}^t\vec{\eta} = \vec{L}^{-1} \quad \Longrightarrow \quad \vec{L}^t\vec{\eta}\vec{L} = \vec{\eta}.$$

Thus given $\vec{x} = \begin{pmatrix} a \\ \mathbf{x} \end{pmatrix}$ and $\vec{y} = \begin{pmatrix} b \\ \mathbf{y} \end{pmatrix}$ we have

$$\vec{x} \cdot \vec{y} = \vec{x}^t\vec{\eta}\vec{y} = ab - \mathbf{x} \cdot \mathbf{y}.$$

Applying the transformation $\vec{L}$ on the vectors $\vec{x}$ and $\vec{y}$ we have

$$(\vec{L}\vec{x}) \cdot (\vec{L}\vec{y}) = (\vec{L}\vec{x})^t\vec{\eta}(\vec{L}\vec{y}) = \vec{x}^t\vec{L}^t\vec{\eta}\vec{L}\vec{y} = \vec{x}^t\vec{\eta}\vec{y} = \vec{x} \cdot \vec{y},$$

and we could use this property to define the Lorentz group as the group of linear transformations that preserve the Minkowski metric, i.e. the group of linear transformations that preserve the Minkowski inner product:

$$\mathrm{O}(1,3) = \{\vec{L} \in \mathbb{R}(4) : \vec{L} \text{ invertible},\, (\vec{L}\vec{x}) \cdot (\vec{L}\vec{y}) = \vec{x} \cdot \vec{y} \text{ for all } \vec{x},\, \vec{y} \in \mathbb{R}^4\}.$$

The product of two four-vectors $\vec{x}$ and $\vec{y}$ is a Lorentz scalar, since it is invariant under Lorentz transformations. The norm of a four-vector $\vec{x}$ is also a Lorentz scalar, since it is given by $\vec{x} \cdot \vec{x}$, which is invariant under Lorentz transformations. This means that those measurements will be the same for all inertial observers, since they are invariant under Lorentz transformations. This is the reason why we call them scalars, since they are invariant under changes of reference frame.

Theorem 4.4 (Structure thorem)**.** *Suppose $\vec{L} \in \mathrm{O}(1,3)$. Then there is a sign $\eta \in \{-1, 1\}$, $\rho \in [0, \infty)$, $\mathbf{u} \in \mathbb{R}^3$ with unit norm $|\mathbf{u}| = 1$ and finally a transformation $\mathbf{R} \in \mathrm{O}(3)$ such that*

TODO: *change notation: matrix in 3D is bold as vector but with capital letter.*

$$\vec{L} = \begin{pmatrix} \eta \cosh(\rho) & -\eta \sinh(\rho)(\mathbf{u}^t \mathbf{R}) \\ -\sinh(\rho)\mathbf{u} & [\cosh(\rho)\mathbf{u} \otimes \mathbf{u} + (1 - \mathbf{u} \otimes \mathbf{u})]\, \mathbf{R} \end{pmatrix}$$

[...]

Proof. Let us take an element $\vec{L} \in \mathrm{O}(1,3)$ such that $\vec{L}^t \vec{\eta} \vec{L} = \vec{\eta}$. Then we can write this expression in (1+3) notation as

$$\vec{L} = \begin{pmatrix} a & \mathbf{b}^t \\ \mathbf{c} & \mathbf{D} \end{pmatrix}, \quad a \in \mathbb{R},\ \mathbf{b},\ \mathbf{c} \in \mathbb{R}^3,\ \mathbf{D} \in \mathbb{R}(3).$$

The metric in this notation is given by

$$\vec{\eta} = \begin{pmatrix} 1 & \mathbf{0}^t \\ \mathbf{0} & -\mathbf{I}_3 \end{pmatrix}.$$

Then we can write the condition $\vec{L}^t \vec{\eta} \vec{L} = \vec{\eta}$ as

$$\begin{pmatrix} a & \mathbf{c}^t \\ \mathbf{b} & \mathbf{D}^t \end{pmatrix} \begin{pmatrix} 1 & \mathbf{0}^t \\ \mathbf{0} & -\mathbf{I}_3 \end{pmatrix} \begin{pmatrix} a & \mathbf{b}^t \\ \mathbf{c} & \mathbf{D} \end{pmatrix} = \begin{pmatrix} 1 & \mathbf{0}^t \\ \mathbf{0} & -\mathbf{I}_3 \end{pmatrix}.$$

By performing the matrix multiplication we get

$$\begin{pmatrix} a & \mathbf{c}^t \\ \mathbf{b} & \mathbf{D}^t \end{pmatrix} \begin{pmatrix} a & \mathbf{b}^t \\ -\mathbf{c} & -\mathbf{D} \end{pmatrix} = \begin{pmatrix} a^2 - \mathbf{c}^2 & a\mathbf{b}^t - \mathbf{c}^t\mathbf{D} \\ a\mathbf{b} - \mathbf{D}^t\mathbf{c} & \mathbf{D}^t\mathbf{D} - \mathbf{b} \otimes \mathbf{b} \end{pmatrix} = \begin{pmatrix} 1 & \mathbf{0}^t \\ \mathbf{0} & -\mathbf{I}_3 \end{pmatrix},$$

which is equivalent to the following system of equations: the two diagonal blocks are just the transposition of each other, so we have only three independent equations give by

TODO: Check the computation.

$$\begin{cases} a^2 - \mathbf{c}^2 = 1, \\ a\mathbf{b} - \mathbf{D}^t\mathbf{c} = \mathbf{0}, \\ \mathbf{D}^t\mathbf{D} - \mathbf{b} \otimes \mathbf{b} = \mathbf{I}_3. \end{cases}$$

From this system of equations we can derive the following properties:

- From the first equation we have $a^2 - \mathbf{c}^2 = 1$, which implies that $a^2 \geq 1$ and thus $a \geq 1$ or $a \leq -1$. We can write $a = \eta \cosh(\rho)$ with $\eta = \pm 1 = \mathrm{sgn}(a)$ and $\rho \geq 0$ (since $a = \mathrm{sgn}(a)|a|$).
- From the same equation we get constraints on $\mathbf{c}$ since we have $\mathbf{c}^2 = a^2 - 1$, which implies

  $$\mathbf{c}^2 = a^2 - 1 = \eta^2 \cosh^2(\rho) - 1 = \cosh^2(\rho) - 1 = \sinh^2(\rho),$$

  thus we can write $\mathbf{c} = -\sinh(\rho)\mathbf{u}$ with $\mathbf{u} \in \mathbb{R}^3$ and $|\mathbf{u}| = 1$ ($\mathbf{u}$ is a unit vector expressing the direction of $\mathbf{c}$: $-\mathbf{u} = \mathrm{vers}(\mathbf{c})$). Thus we have

  $$\mathbf{c} = -\sinh(\rho)\mathbf{u}.$$

- From the second equation we have $a\mathbf{b} = \mathbf{D}^t\mathbf{c}$, which implies that
$$\mathbf{b} = \frac{1}{a}\mathbf{D}^t\mathbf{c} = -\frac{\sinh(\rho)}{\eta\cosh(\rho)}\mathbf{D}^t\mathbf{u}.$$
Now since η is a sign, we can move it up and recognize the hyperbolic tangent function, thus we can write
$$\mathbf{b} = -\eta\tanh(\rho)\mathbf{D}^t\mathbf{u}.$$
We will understand the form of $\mathbf{D}$ and then we will be able to write $\mathbf{b}$ in a more explicit form.

- From the third equation we have $\mathbf{D}^t\mathbf{D} = \mathbf{I}_3 + \mathbf{b}\otimes\mathbf{b}$, which implies that
$$\mathbf{D}^t\mathbf{D} - \tanh^2(\rho)(\mathbf{D}^t\mathbf{u})\otimes(\mathbf{D}^t\mathbf{u}) = \mathbf{I}_3,$$
which since $(\mathbf{A}\mathbf{x})\otimes(\mathbf{B}\mathbf{y}) = \mathbf{A}(\mathbf{x}\otimes\mathbf{y})\mathbf{B}$ we can write as
$$\mathbf{D}^t\left[1 - \tanh^2(\rho)\mathbf{u}\otimes\mathbf{u}\right]\mathbf{D} = \mathbf{I}_3.$$
Now since the identity is invertible, the other side has to be invertible as well, with a determinant given by the product of the three determinants of the terms, and since the determinant of the identity is 1, we have that each of the three determinants has to be different from zero, in particular the determinant of the middle term has to be different from zero, which implies that $\tanh^2(\rho) \neq 1$, which is true since $\rho \geq 0$ and thus $\tanh^2(\rho) < 1$. Since we have
$$\mathbf{D}^{-1} = \mathbf{D}^t\left[1 - \tanh^2(\rho)\mathbf{u}\otimes\mathbf{u}\right],$$
we have that
$$\mathbf{D}\mathbf{D}^t = \left[1 - \tanh^2(\rho)\mathbf{u}\otimes\mathbf{u}\right]^{-1}.$$
Now we have to compute the inverse of the middle term, which can be time consuming, but we can do it by knowing that $\mathbf{u}$ is a unit vector: $\mathbf{u}\otimes\mathbf{u}$ is a projector on the direction of $\mathbf{u}$, and $1 - \mathbf{u}\otimes\mathbf{u}$ is a projector on the plane orthogonal to $\mathbf{u}$. Hence we can write
$$\left[1 - \tanh^2(\rho)\mathbf{u}\otimes\mathbf{u}\right] = \left[(1 - \mathbf{u}\otimes\mathbf{u}) + \left(1 - \tanh^2(\rho)\right)\mathbf{u}\otimes\mathbf{u}\right],$$
so that we have written the term as a linear combination of two projectors, which are idempotent, and thus we can easily compute the inverse as
$$\mathbf{A} = \alpha(1 - \mathbf{u}\otimes\mathbf{u}) + \beta\mathbf{u}\otimes\mathbf{u} \quad\Longrightarrow\quad \mathbf{A}^{-1} = \frac{1}{\alpha}(1 - \mathbf{u}\otimes\mathbf{u}) + \frac{1}{\beta}\mathbf{u}\otimes\mathbf{u},$$
since we know $\alpha, \beta \neq 0$. Thus the inverse of the middle term is given by
$$\left[1 - \tanh^2(\rho)\mathbf{u}\otimes\mathbf{u}\right]^{-1} = (1 - \mathbf{u}\otimes\mathbf{u}) + \frac{1}{1 - \tanh^2(\rho)}\mathbf{u}\otimes\mathbf{u} = (1 - \mathbf{u}\otimes\mathbf{u}) + \cosh^2(\rho)\mathbf{u}\otimes\mathbf{u},$$
where we have used the identity $1 - \tanh^2(\rho) = \frac{1}{\cosh^2(\rho)}$. Hence we have
$$\mathbf{D}\mathbf{D}^t = (1 - \mathbf{u}\otimes\mathbf{u}) + \cosh^2(\rho)\mathbf{u}\otimes\mathbf{u}.$$
It is not difficult to guess a solution for $\mathbf{D}$ that satisfies this equation: imagine a matrix which has the same structure but different coefficients, which since the proectors are idempotent, when multiplied by its transposition will give the same structure but with the coefficients squared, thus we can write
$$\mathbf{D}_o = \left[\cosh(\rho)\mathbf{u}\otimes\mathbf{u} + (1 - \mathbf{u}\otimes\mathbf{u})\right] = \mathbf{D}_o^t,$$

and if we now imagine a rotation matrix $\mathbf{R} \in \mathrm{O}(3)$ and we write $\mathbf{D} = \mathbf{D}_o\mathbf{R}$, then we have $\mathbf{D}_o\mathbf{R}(D_o\mathbf{R})^t = \mathbf{D}_o\mathbf{R}\mathbf{R}^t\mathbf{D}_o^t = \mathbf{D}_o\mathbf{D}_o^t$. Hence we have found a solution for $\mathbf{D}$ that satisfies the third equation, which is given by

$$\mathbf{D} = [\cosh(\rho)\mathbf{u} \otimes \mathbf{u} + (1 - \mathbf{u} \otimes \mathbf{u})]\,\mathbf{R}.$$

- Now we can write $\mathbf{b}$ in a more explicit form by substituting the expression for $\mathbf{D}$ that we have just found, thus we have

$$\mathbf{b} = -\eta\tanh(\rho)\mathbf{D}^t\mathbf{u} = -\eta\tanh(\rho)\mathbf{R}^t\,[\cosh(\rho)\mathbf{u} \otimes \mathbf{u} + (1 - \mathbf{u} \otimes \mathbf{u})]\,\mathbf{u} = -\eta\sinh(\rho)(\mathbf{R}^t\mathbf{u}),$$

where we have used the fact that the term in square brackets is a projector on the direction of $\mathbf{u}$, thus when applied to the vector $\mathbf{u}$ gives back the same vector.

The only thing we are left to do is to check that the found parameters are uniquely determined by the matrix $\vec{L}$. The parameters η and ρ are uniquely determined since the first is the sign of a while the second is given by the relation $\rho = \mathrm{arcosh}(|a|)$. The parameter $\mathbf{u}$ is uniquely determined since it is given by the relation $\mathbf{u} = -\frac{1}{\sinh(\rho)}\mathbf{c}$, and finally the parameter $\mathbf{R}$ is uniquely determined. ...

Now if we substitute the found parameters in the expressions found before, they will be all verified and thus we can write the matrix $\vec{L}$ as

$$\vec{L} = \begin{pmatrix} \eta\cosh(\rho) & -\eta\sinh(\rho)(\mathbf{u}^t\mathbf{R}) \\ -\sinh(\rho)\mathbf{u} & [\cosh(\rho)\mathbf{u} \otimes \mathbf{u} + (1 - \mathbf{u} \otimes \mathbf{u})]\,\mathbf{R} \end{pmatrix},$$

which is the desired result, proving the structure theorem for the Lorentz group. □

4.3.1 | Subsets of the Lorentz Group

Definition 4.2 (Lorentz boost)**.** $\rho \in [0, \infty)$, $\mathbf{u} \in \mathbb{R}^3$ with $|\mathbf{u}| = 1$, we can define an element of the Lorentz group called *Lorentz boost* (SR) as

$$\vec{B}(\rho, \mathbf{u}) = \begin{pmatrix} \cosh(\rho) & -\sinh(\rho)\mathbf{u}^t \\ -\sinh(\rho)\mathbf{u} & [\cosh(\rho)\mathbf{u} \otimes \mathbf{u} + (1 - \mathbf{u} \otimes \mathbf{u})] \end{pmatrix}.$$

Definition 4.3 (Spatial rotation)**.** $\mathbf{R} = \mathbf{R}_{+1}(\phi, \mathbf{n})$, such that $\phi \in [0, \pi]$ and $\mathbf{n} \in S^2$, we can define an element of the Lorentz group called *spatial rotation* as

$$\vec{R}(\phi, \mathbf{n}) = \begin{pmatrix} 1 & \mathbf{0}^t \\ \mathbf{0} & \mathbf{R}_{+1}(\phi, \mathbf{n}) \end{pmatrix}.$$

Definition 4.4 (Time reversal)**.** Given a sign $\eta = \{-1, +1\}$, we can define an element of the Lorentz group called *time reversal* as

$$\vec{T}(\eta) = \begin{pmatrix} \eta & \mathbf{0}^t \\ \mathbf{0} & \mathbf{I}_3 \end{pmatrix},$$

for a sign $\eta = -1$ we get the time reversal transformation, while for $\eta = +1$ we get the identity transformation.

Definition 4.5 (Parity transformation)**.** Given a sign $\epsilon = \{-1, +1\}$, we can define an element of the Lorentz group called *parity transformation* as

$$\vec{P}(\epsilon) = \begin{pmatrix} 1 & \mathbf{0}^t \\ \mathbf{0} & \epsilon\mathbf{I}_3 \end{pmatrix},$$

where $\epsilon = -1$ gives the parity transformation, while $\epsilon = +1$ gives the identity transformation.

Thus we have found four subsets of the Lorentz group, which are given by the Lorentz boosts, the spatial rotations, the time reversal and the parity transformation. There is a theorem, not difficult to prove, that states that any element of the Lorentz group can be written as a product of an element of each of these subsets:

Proposition 4.5 (Factorization theorem for $\mathrm{O}(1,3)$)**.** *For every $\vec{L} \in \mathrm{O}(1,3)$ there exist two signs η, $\epsilon \in \{-1,+1\}$, a rapidity $\rho \in [0,\infty)$, two unit vectors $\mathbf{u}$, $\mathbf{n} \in S^2$ and an angle $\phi \in [0,\pi]$ such that*

$$\vec{L} = \vec{T}(\eta)\vec{B}(\rho,\mathbf{u})\vec{P}(\epsilon)\vec{R}(\phi,\mathbf{n}).$$

This is a very useful decomposition of the Lorentz group, since it allows us to understand the structure of the group and to find its representations.

Furthermore the parameters η, ϵ, ρ, $\mathbf{u}$, $\mathbf{n}$, ϕ are uniquely determined by the matrix $\vec{L}$, with exceptions on the unit vectors: $\mathbf{u}$ asks for $\rho \neq 0$ and $\mathbf{n}$ asks for $0 < \phi < \pi$ strictly, to be uniquely determined.

Proof. future work □

Part II

Advanced Lie Algebras, Weights and Roots

Introduction

1 Lie Groups and Lie Algebras

[...]

1.1 | Lie Groups as Manifolds

Definition 1.1. A Lie group G is a smooth differentiable manifold which is also a group, where the following two maps based on the group law are smooth maps between manifolds:

$$G \times G \to G, \quad (g_1, g_2) \mapsto g_1 g_2,$$
$$G \to G, \quad g \mapsto g^{-1}.$$

If the underlying manifold of a Lie group G is a complex manifold, and the above maps are holomorphic, then G is also called a complex Lie group.

Standard examples are matrix groups ("classical Lie groups"). We will see these in more details later.

Definition 1.2. A Lie group homomorphism / isomorphism is a group homomorphism / isomorphism $f : G \to H$ which is also a smooth map between manifolds.

In the context of Lie groups, we will always use "homomorphism / isomorphisms" in the above sense unless otherwise stated.

Example (example 1.2.3). [...]

[...]

A Lie group G is called compact / (simply-)connected, if G is compact / (simply-)connected as a topological space. Thus let us recall the definitions of compactness and connectedness in the context of topological spaces, and then we will see some examples of compact and non-compact Lie groups, as well as simply connected and non-simply connected Lie groups.

Definition 1.3 (Topological Compactness)**.** A topological space G is *compact* if every open cover of G has a finite subcover. In other words, if we have a collection of open sets $\{U_\alpha\}_{\alpha \in A}$ such that $G \subseteq \bigcup_{\alpha \in A} U_\alpha$, then there exists a finite subset $A' = (\alpha_1, \dots, \alpha_k)_{k<\infty} \subseteq A$ such that $G \subseteq \bigcup_{\alpha \in A'} U_\alpha$.

Example (Compactness and Lie groups).

- From the Heine Borel theorem we know that a subset of $\mathbb{R}$ is compact if and only if it is closed (there are no limiting points that belong to the set) and bounded. We can use this to show that the set of all subgroups of $\mathbb{R}$ is uncountable. Thus for matrix groups, for which $G \subset \mathbb{R}^n$ for some $n \in \mathbb{N}$, we have

  $$G \text{ compact} \iff G \text{ closed and bounded},$$

  where bounded simply means bounded magnitude of the entries of the matrices.
- The unitary group $G = \mathrm{U}(n)$ satisfies the following properties: given a unitary matrix $U \in \mathrm{U}(n)$, we have $U^\dagger U = \mathbb{I}$,and thus

  $$\sum_i U^*_{ij} U_{ik} = (U^\dagger U)_{jk} = \delta_{jk},$$

  and expanding the sum we understand that for $i = j$ we have no entries greater than 1:

  $$|U_{ij}| \leq 1.$$

  Furthermore if we have some matrix group defined by a set of polynomial equations, then the set of solutions is closed. Therefore $\mathrm{U}(n)$ is a closed and bounded subset of $\mathbb{R}^{n^2}$ and thus compact.

- $GL(n, \mathbb{R})$ or $GL(n, \mathbb{C})$, $SL(n, \mathbb{R})$, $SO(1, n-1)$ are not compact, since they are not bounded.

Definition 1.4 (Topological Connectedness)**.** A topological space G is *connected* if for any two ponts g_1, $g_2 \in G$ there exists a continuous path $\gamma : [0, 1] \to G$ such that $\gamma(0) = g_1$ and $\gamma(1) = g_2$ (it is not necessary to impose the limits on the domanin of γ, since the statement with $g_1, g_2 \in \gamma$ would have been sufficient, but it is convenient to do so). This means that we can connect any two points in G with a continuous path, and thus there are no isolated points or disconnected components in G.

This curve is a continuous map, thus any function cannot vary discontinuously along it. In particular, if we have a continuous homomorphism $f : G \to H$ and G is connected, then the image of f must be connected as well. This is because the image of a continuous map is always connected.

If we start from a point where $\det(f(g)) = 1$ for some $g \in G$, then we can infere that $\det(f(g)) = 1$ for all $g \in \gamma \subset G$. This is because the determinant is a continuous function, and if it is unitary at one point, it must be unitary along the entire path connecting that point to any other point in G.

Example (Connectedness and Lie groups).

- The orthogonal group $\mathrm{O}(n)$ is not connected, since it has two disconnected components: the component of the identity, which consists of all orthogonal matrices with determinant 1 (the special orthogonal group $\mathrm{SO}(n)$), and the component consisting of all orthogonal matrices with determinant -1. We cannot have a continuous curve connecting $Q_1 \in \mathrm{O}(n)$ and $Q_2 \in \mathrm{O}(n)$ if they belong to different components ($\det(Q_1) = 1$ and $\det(Q_2) = -1$), since the determinant would have to change continuously from 1 to -1, which is not possible.
- The unitary group $\mathrm{U}(n)$ is connected, since any unitary matrix can be continuously deformed to the identity matrix by a path of unitary matrices. This is because the determinant of a unitary matrix is a complex number of magnitude 1

$$u \in \mathrm{U}(n) \implies |\det(U)| = 1,$$

  and we can continuously change the phase of the determinant from 1 to any other value on the unit circle without leaving the set of unitary matrices. Thus we can connect any two unitary matrices with a continuous path in $\mathrm{U}(n)$.
- The special orthogonal group $\mathrm{SO}(n)$ and the special unitary group $\mathrm{SU}(n)$ are also connected, since they are defined as the connected components of $\mathrm{O}(n)$ and $\mathrm{U}(n)$ respectively that contain the identity element. Thus any two elements in $\mathrm{SO}(n)$ or $\mathrm{SU}(n)$ can be connected by a continuous path within the same component.

Definition 1.5 (Simply connected Lie groups)**.** If a group G is connected and any closed curve (or loop) $\gamma \subset G$ can be continuously contracted (smoothly deformed) to a point, then G is called *simply connected*.

It is easier to think about topological spaces as being made of rubber, and the continuous deformation as being a process of stretching and bending the space without tearing it. In this way, a simply connected space is one that has no holes or loops that cannot be contracted to a point. As a visual example, the case of a thorus, a can connect any two points with a continuous path, but there are loops around the hole of the thorus that cannot be contracted to a point. On the other hand, a sphere is simply connected, because any loop on the sphere can be contracted to a point.

Example (Simply connectedness in Lie groups).

- The unitary group $U(n)$ is not simply connected, since it contains $U(1) \subset U(n)$ which is topologically a circle
$$\mathrm{U}(1) \cong \mathrm{SO}(2) \cong S^1,$$
and thus has loops that cannot be contracted to a point. On the other hand, the special unitary group $\mathrm{SU}(n)$ is simply connected, because it does not contain any loops that cannot be contracted to a point. Indeed, the determinant is a homomorphism in $\mathrm{U}(n)$ that maps the group to $\mathrm{U}(1)$, and it is possible to define a curve which composition with the determinant gives a "*winding number*" different from zero, which means that the curve cannot be contracted to a point.
- Another example is the special orthogonal group $\mathrm{SO}(3)$ in three dimensional space: For $n \geq 3$, $\mathrm{SO}(n)$ is not simply connected. However, for $n = 2$, $\mathrm{SO}(2)$ is simply connected. If we think about $\mathrm{SU}(2)$, which is the double cover of $\mathrm{SO}(3)$, we can see that it is simply connected: the inehrited loops from $\mathrm{SO}(3)$ can be contracted to a point in $\mathrm{SU}(2)$ because of the double cover structure. In fact, $\mathrm{SU}(2)$ is topologically equivalent to the 3-sphere S^3, which is simply connected. Thus spin-$\frac{1}{2}$ objects, which are described by representations of $\mathrm{SU}(2)$, can be thought of as living on a simply connected space, while the rotations in three-dimensional space, described by $\mathrm{SO}(3)$, have a non-simply connected structure due to the presence of loops that cannot be contracted to a point.

Remark (Fundamental groups and connectedness). *Talking about fundamental groups, we are implying that we are working with a connected manifold, since the fundamental group is defined for connected spaces. If the space is not connected, we can consider the fundamental group of each connected component separately.*

1.2 | Lie Algebra from Tangent Vectors

As hopefully familiar from GR and/or differential geometry, tangent vectors play a crucial role in the study of manifolds. In this section we will see that on a Lie group G, the tangent vectors at the identity element $e \in G$ carry an algebra structure, the so-called Lie algebra $\mathfrak{g}$. The big advantage of the Lie algebra over the group is that the algebra is first and foremost a vector space, and vector spaces are much more "rigid" and thus easier to study than the Lie group. It will play an important role for our understanding and classification of Lie groups and their representations

Definition 1.6 (Tangent vectors)**.** Let $(\mathcal{M}, \{u_\alpha\})$ and a point $p \in \mathcal{M}$ in the manifold. Let us imagine a smooth curve $\gamma : (a, b) \to \mathcal{M}$ such that $\gamma(t_0) = p$ for some $t_0 \in (a, b)$. We can define the tangent vector to the curve at the point p, $\dot{\gamma}_p$ as as a differential operator acting on smooth functions $\phi \in \mathbb{C}^\infty(\mathcal{M})$ as follows:

$$\dot{\gamma}_p(\phi) := \frac{\mathrm{d}}{\mathrm{d}t}(\phi \circ \gamma)(t)\bigg|_{t=t_0}.$$

A set of tangent vectors at a point p spans a n-dimensional vector space ($n = \dim(\mathcal{M})$), called the tangent space at p, denoted by $T_p\mathcal{M}$. The dimension of the tangent space is thus equal to the dimension of the manifold.

Let us now consider a local chart $p \in U$ with coordinate functions$x \equiv x^i : U \to \mathbb{R}^n$. We can define a set of tangent vectors at p as a natural basis for the tangent space $T_p\mathcal{M}$ as follows:

$$\dot{\gamma}_p(\phi) = \frac{\mathrm{d}}{\mathrm{d}t}(\phi \circ x^{-1} \circ x \circ \gamma)(t)\bigg|_{t=t_0} = \frac{\mathrm{d}}{\mathrm{d}t}(\phi \circ x^{-1})(x(\gamma(t)))\bigg|_{t=t_0}.$$

Applying the chain rule, we can write this as:

$$\dot{\gamma}^i_p(\phi) = \frac{\mathrm{d}x^j}{\mathrm{d}t}\frac{\partial}{\partial x^j}(\phi \circ x^{-1})(x(\gamma(t)))\bigg|_{t=t_0} = \frac{\mathrm{d}x^j}{\mathrm{d}t}\bigg|_{t=t_0}\frac{\partial}{\partial x^j}(\phi \circ x^{-1})(x)\bigg|_{x=x(p)}.$$

TODO: correct the expression below, it is not correct

Thus we find that

$$\dot{\gamma}_p(\phi) = \sum_i \gamma^i(t_0)\frac{\partial}{\partial x^i}(\phi \circ x^{-1})(x(p)) = \sum_i \gamma^i(t_0)\frac{\partial}{\partial x^i}(\phi(p)).$$

In a local coordinate patch thus the tangent vector can be expressed as a linear combination of the partial derivatives with respect to the coordinates, evaluated at the point p. The coefficients of this linear combination are given by the components of the curve γ at $t = t_0$. Thus the tangent space has dimension equal to the number of coordinates, which is equal to the dimension of the manifold.

Corollary 1.1. *If we have two manifolds $\mathcal{M}_1$ and $\mathcal{M}_2$ of dimensions m_1 and m_2 respectively, then the tangent space of their product is the direct sum of their tangent spaces:*

$$T_p(\mathcal{M}_1 \times \mathcal{M}_2) \cong T_p\mathcal{M}_1 \oplus T_p\mathcal{M}_2 = T_p\mathcal{M},$$

where $p = (p_1, p_2)$ is a point in the product manifold $\mathcal{M}_1 \times \mathcal{M}_2$.

Definition 1.7 (Tangent bundle)**.** If one take the union of all the tangent spaces at all points of the manifold, we obtain the **tangent bundle** $T\mathcal{M} = \bigcup_{p \in \mathcal{M}} T_p\mathcal{M}$. The tangent bundle is itself a manifold of dimension $2n$, where n is the dimension of the original manifold $\mathcal{M}$ (we have to keep the original maniforld in the definition of the tangent bundle, since the tangent spaces are defined at each point of the manifold):

$$T\mathcal{M} = \{(p, \mathbf{v}) | p \in \mathcal{M}, \mathbf{v} \in T_p\mathcal{M}\},$$

and furthermore we have a natural projection map $\pi : T\mathcal{M} \to \mathcal{M}$ defined by $\pi(p, \mathbf{v}) = p$. The tangent bundle is a vector bundle, since each fiber (the tangent space at each point) is a vector space.

Definition 1.8 (Vector fields)**.** A **vector field** on a manifold $\mathcal{M}$ is a smooth section of the tangent bundle, that is, a smooth map $V : \mathcal{M} \to T\mathcal{M}$ such that $\pi \circ V = \mathrm{id}_{\mathcal{M}}$. In other words, for each point $p \in \mathcal{M}$, the vector field assigns a tangent vector $V(p) \in T_p\mathcal{M}$. It is also called a *section* of the tangent bundle $T\mathcal{M}$.

Thus we associate to each point of the manifold a tangent space, and a vector field is a smooth assignment of a tangent vector to each point of the manifold. We are cutting the fibers of the bundle, and we are left with a single vector at each point of the manifold.

Since locally this vector field is a map from the manifold to the tangent space, if and only if the map is smooth, then the vector field is smooth and can be written as $\Gamma(T\mathcal{M})$.

$$\dot{\gamma}_p(\phi) \in T_p\mathcal{M},$$

$$\dot{\gamma}_p(\phi) = \frac{\mathrm{d}}{\mathrm{d}t}(\phi \circ \gamma)(t_0) = \sum_i \gamma^i(t_0) \frac{\partial}{\partial x^i}(\phi(p)),$$

where $\partial_i = \frac{\partial}{\partial x^i}$ is the basis for the tangent space at the point p. Thus we can express the tangent vector as a linear combination of the basis vectors of the tangent space, with coefficients given by the components of the curve γ at $t = t_0$. This shows that the tangent space at each point of the manifold is a vector space, and that the dimension of the tangent space is equal to the dimension of the manifold.

The vector field is a smooth assignment of a tangent vector to each point of the manifold

$$V : \mathcal{M} \to T\mathcal{M}, \quad p \mapsto (p, V_p) \in T_p\mathcal{M},$$

and it can be thought of as a smooth map from the manifold to the tangent bundle. The vector field can be expressed in local coordinates as a linear combination of the basis vectors of the tangent space, with coefficients that are smooth functions on the manifold:

$$x^i(p) \in \mathbb{R}^n, \quad x^i(p) \mapsto (x^i(p), V^i(p)) \in T_p\mathcal{M} \equiv \mathbb{R}^n \times \mathbb{R}^n,$$

This allows us to perform calculus on the manifold using the vector fields, and to define concepts such as divergence, curl, and gradient in a coordinate-independent way. If this mapping is smooth for every choice of chart, then the vector field is smooth and can be written as $\Gamma(T\mathcal{M})$, and it forms a Lie algebra under the Lie bracket operation defined by the commutator of vector fields. Thus we managed to generalize the notion of vector fields from Euclidean space to arbitrary manifolds, and we can use this to define the Lie algebra of a Lie group as the tangent space at the identity element of the group.

We can also apply a vector field $V \in \Gamma(T\mathcal{M})$ on a function $\phi \in \mathbb{C}^\infty(\mathcal{M})$ to obtain a new function $V(\phi) : \mathcal{M} \to \mathbb{R}$ defined by

$$p \mapsto V(\phi)(p) := V_p(\phi) \in \mathbb{R}, \quad V(\phi) \in \mathbb{C}^\infty(\mathcal{M}),$$

where V_p is the tangent vector assigned to the point p by the vector field V. This allows us to define the action of a vector field on functions, and to perform calculus on the manifold using the vector fields.

If we now take two vector fields $V, W \in \Gamma(T\mathcal{M})$, we can apply the previous construction to the function $V(\phi)$ to obtain a new function $W(V(\phi)) : \mathcal{M} \to \mathbb{R}$ defined by

$$W(V(\phi))(p) := W_p(V(\phi)) \in \mathbb{R}, \quad W(V(\phi)) \in \mathbb{C}^\infty(\mathcal{M}),$$

where W_p is the tangent vector assigned to the point p by the vector field W. This can be computed as follows:

$$W_p(V(\phi)) = \sum_i W_p^i \partial_i (V(\phi))(p) = \sum_{ij} W_p^i \partial_i (V^j(p) \partial_j (\phi \circ x^{-1}))(x(p)),$$

where we have used the local coordinate expression for the vector fields; we can continue to compute this expression by applying the product rule for differentiation:

$$W_p(V(\phi)) = \sum_{ij} W_p^i \partial_i (V^j(p)) \partial_j (\phi \circ x^{-1})(x(p)) + \sum_{ij} W_p^i V_p^j \partial_i \partial_j (\phi \circ x^{-1})(x(p)).$$

Note that a second derivative term has appeared in the expression, which we do not now how to interpret in terms of the original vector fields V and W: this new term is not a vector anymore, it is not an object behaving as a vector field $X(\phi)$ with $X \in \Gamma(T\mathcal{M})$. We cannot interpret this result as another vector field operating on the same function ϕ, since the second derivative term does not have the same structure as the original vector fields.

Definition 1.9 (Lie bracket of vector fields)**.** The Lie bracket of two vector fields (or commutator) $V, W \in \Gamma(T\mathcal{M})$ is defined as the new vector field $[V, W] \in \Gamma(T\mathcal{M})$ such that

$$[V, W](\phi) := V(W(\phi)) - W(V(\phi)),$$

for all smooth functions $\phi \in \mathbb{C}^\infty(\mathcal{M})$. This new vector field is defined by the difference of the two second derivative terms that appeared in the previous expression, and it can be shown that it satisfies the properties of a Lie bracket, as we will see.

With this new object defined, we have an operation that takes two vector fields and produces a new vector field, with in charts can be written as

$$[V, W]^j = \sum_i \left(V^i \partial_i W^j - W^i \partial_i V^j \right) \partial_j.$$

Proposition 1.2. *For any three vector fields $X, Y, Z \in \Gamma(T\mathcal{M})$, the Lie bracket satisfies the following properties:*

- ***Antisymmetry****:* $[X, Y] = -[Y, X]$.
- ***Bilinearity****:* $[aX + bY, Z] = a[X, Z] + b[Y, Z]$ *and* $[Z, aX + bY] = a[Z, X] + b[Z, Y]$ *for all scalars* $a, b \in \mathbb{R}$.
- ***Jacobi identity****:* $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$.

Corollary 1.3. *With the previously defined Lie bracket, the set of vector fields $\Gamma(T\mathcal{M})$ can be endowed with the structure of a non-associative algebra over $\mathbb{R}$. We said non-associative because the Lie bracket does not satisfy the associative property, but it satisfies the Jacobi identity instead:*

$$[[X, Y], Z] \neq [X, [Y, Z]].$$

The Jacobi identity is a weaker condition than associativity, and it ensures that the Lie bracket defines a consistent algebraic structure on the space of vector fields.

It is now time to connect this construction to the Lie algebra of a Lie group.

Definition 1.10. Given a smooth map $f : \mathcal{M} \to \mathcal{N}$ between two manifolds, we can define an induced map (which will be identified with a *vector space homomorphism* for every point in the manifold) on the tangent spaces, called the **pushforward** of f, denoted by $(f_*)_p : T_p\mathcal{M} \to T_{f(p)}\mathcal{N}$. The pushforward is defined by

$$(f_*)_p(\dot{\gamma}_p) = (f \dot{\circ} \gamma)_{f(p)} \quad \forall p \in \mathcal{M},$$

then let $h \in \mathbb{C}^\infty(\mathcal{N})$, and $v = \dot{\gamma}_p \in T_p\mathcal{M}$, then we have

$$\implies [(f_*)_p(v)]\,(h) = v(h \circ f).$$

In coordinates x for $\mathcal{M}$ and y for $\mathcal{N}$, we can write the pushforward as

$$[(f_*)_p]^i_k = \partial_k(y^i \circ f \circ x^{-1})(x(p)) = (\mathrm{d}f)^i_k(x(p)),$$

where $\mathrm{d}f$ is the Jacobian matrix of the map f at the point p: $f^i(x^k)$. This is just the jacobian matrix of a vector-valued function, which is a linear map that takes a tangent vector at p and maps it to a tangent vector at $f(p)$ by applying the derivative of the function f to the components of the tangent vector.

This concept is known in the field of general relativity as the **Lie dragging** of a vector field along a curve, and it is used to describe how vector fields change as they are transported along curves in a manifold. The pushforward allows us to compare vector fields at different points in the manifold, and to understand how they transform under smooth maps between manifolds.

If the function f is a diffeomorphism (smooth and invertible), then the pushforward is an isomorphism between the tangent spaces at p and $f(p)$, and it allows us to transfer the structure of the tangent space from one point to another. This is particularly useful when we want to define the Lie algebra of a Lie group, as we will see successively.

Proposition 1.4. *If $f : \mathcal{M} \to \mathcal{N}$ is a diffeomorphism, then the pushforward $(f_*)_p$ is an isomorphism between vector fields on $\mathcal{M}$ and vector fields on $\mathcal{N}$:*

$$f_* : \Gamma(T\mathcal{M}) \to \Gamma(T\mathcal{N}), \quad V \mapsto f_*(V),$$

$$(f_*(V))_{f(p)} = (f_*)_p(V_p) \in T_{f(p)}\mathcal{N},$$

and since consequently the jacobian matrix of f is invertible, we can also define the inverse pushforward $(f_*^{-1})_{f(p)} : T_{f(p)}\mathcal{N} \to T_p\mathcal{M}$ as the inverse of the pushforward, and it is also an isomorphism between vector fields on $\mathcal{N}$ and vector fields on $\mathcal{M}$.

We can now imagine to concatenate functions; let $f_1 \in \mathbb{C}^\infty(\mathcal{M}, \mathcal{N})$ and $f_2 \in \mathbb{C}^\infty(\mathcal{N}, \mathcal{Q})$ be two diffeomorphisms, then their composition $f_2 \circ f_1 : \mathcal{M} \to \mathcal{Q}$ is also a diffeomorphism, and we can define the pushforward of the composition as follows:

$$\forall p \in \mathcal{M}, \quad [(f_2 \circ f_1)_*]_p = [(f_2)_*]_{f_1(p)} \circ [(f_1)_*]_p : T_p\mathcal{M} \to T_{f_2(f_1(p))}\mathcal{Q},$$

which can be read as the pushforward of the composition is the composition of the pushforwards. This property is known as the **functoriality** of the pushforward, and it ensures that the push-forward behaves well under composition of functions, which is a fundamental property for any mathematical construction that aims to capture the structure of smooth maps between manifolds.

Now we are ready to define the Lie algebra of a Lie group as the tangent space at the identity element of the group, and to show how the Lie bracket of vector fields induces a Lie bracket on this tangent space, which will give us the structure of a Lie algebra on the tangent space at the identity element. This is a crucial step in understanding the relationship between Lie groups and their associated Lie algebras, and it will allow us to study the properties of Lie groups using the tools of Lie algebras.

Definition 1.11. Let G be a Lie group, and let e be the identity element of G. Then any $a \in G$ defines a "*left-translation*" which maps any element $g \in G$ to ag:

$$L_a : G \to G, \quad g \mapsto L_a g = ag.$$

This left-translation is a diffeomorphism, since it is smooth and has a smooth inverse given by $(L_a)^{-1} = L_{a^{-1}}$. Left translation are natural diffeomorphisms of the Lie group, and they allow us to transfer the structure of the tangent space at the identity element to any other point in the group. In particular, we can use left translations to define a vector field on the Lie group by translating a tangent vector at the identity element to every other point in the group.

Definition 1.12. Let us consider a class of vector fields $V \in \Gamma(TG)$ on a Lie group G that are invariant under left translations; they are called **left-invariant vector fields**, and they satisfy the following property:

$$\forall a \in G, \quad (L_a)_* V = V,$$

which means that the vector field is unchanged under left translations. In other words, for any point $g \in G$, the value of the vector field

$$\forall a, g \in G, \quad [(L_a)_* V]_a g = [(L_a)_*]_g (V_g) = V_{ag},$$

which reads as the value of the vector field at the point ag is equal to the pushforward of the vector field at the point g under the left translation by a. This property ensures that the vector field is consistent with the group structure of G.

Lemma 1.5. *Given a tangent vector $v \in T_e G$ (any point would have been fine, so we chose the identity element for convenience), there exists a unique left-invariant vector field $V^{(v)} \in \Gamma(TG)$ by translating v to every point in the group using left translations, which satisfies the initial conditions*

$$V_e^{(v)} = v.$$

Since it is a left-invariant vector field, for every point $g \in G$ we can use left translation to define the value of the vector field at that point.

On a lie group we have a one to one correspondence between tangent vectors at a point (the identity element for convenience) and left-invariant vector fields, which allows us to transfer the structure of the tangent space at the identity element to the entire group

$$v \in T_e G, \quad v \mapsto V^{(v)} \in \Gamma(TG), \quad (V^{(v)})_g = (L_g)_*(v) \in T_g G.$$

The other ingredient we need to define the Lie algebra of a Lie group is the closeness of the Lie bracket of vector fields, which we have already defined.

Proposition 1.6. *The set of left-invariant vector fields on a Lie group G is closed under the Lie bracket operation $[\cdot,\cdot]$, which means that if $V, W \in \Gamma(TG)$ are left-invariant vector fields, then*

$$[V, W] \in \Gamma(TG) \text{ is left-invariant.}$$

In order to prove this proposition, we need to show another final lemma:

Lemma 1.7. *Let $f : \mathcal{M} \to \mathcal{N}$ suriective and $\phi \in \mathbb{C}^\infty(\mathcal{N})$ (so that $(\phi \circ f \in \mathbb{C}^\infty(\mathcal{M}))$) and $V \in \Gamma(T\mathcal{M})$ be a vector field on $\mathcal{M}$ such that $(f_*)(V)$ is a well defined vector field on $\mathcal{N}$, then we have*

$$V(\phi \circ f) = (f_*(V))(\phi) \circ f \iff V(\phi \circ f) = (f_*(V))_{f(p)}(\phi) = ((f_*)(V)(\phi) \circ f)(p).$$

This means that ...

Proof. Let $\phi \in \mathbb{C}^\infty(G)$. Then, for any $a, g \in G$ we have

$$\begin{aligned}[(L_a)_*[V,W]]_{ag}(\phi) = \dots \\ = [V,W]_g(\phi \circ L_a) = V_g(W(\phi \circ L_a)) - W_g(V(\phi \circ L_a)),\end{aligned}$$

and since we have the previous lemma, we can write for a generic vector field U on G

$$[(L_a)_*U]_a g = U_{ag} \implies U(\phi \circ L_a)(g) = [(L_a)_*U]_{ag}(\phi) = U_{ag}(\phi) = U(\phi)(L_a(g)) = (U(\phi) \circ L_a)(g).$$

Thus we can write

$$\begin{aligned}((L_a)_*[V,W])_{ag}(\phi) &= V_g(W(\phi) \circ L_a) - W_g(V(\phi) \circ L_a) \\ &= (((L_a)_*)_g V)_g(W(\phi)) - (((L_a)_*)_g W)_g(V(\phi)) \\ &= [(L_a)_*V]_{ag}(W(\phi)) - [(L_a)_*W]_{ag}(V(\phi)) \\ &= V_{ag}(W(\phi)) - W_{ag}(V(\phi)) = [V,W]_{ag}(\phi),\end{aligned}$$

since both V and W are left-invariant vector fields, and thus we have shown that $[(L_a)_*[V,W]]_{ag}(\phi) = [V,W]_{ag}(\phi)$ for all $a, g \in G$, which means that $[V,W]$ is left-invariant. □

Thus we have shown that the set of left-invariant vector fields on a Lie group G is closed under the Lie bracket operation, which means that we can define a Lie algebra structure on the tangent space at the identity element of the group by using the Lie bracket of left-invariant vector fields. This is a crucial step in understanding the relationship between Lie groups and their associated Lie algebras.

Definition 1.13. Let G be a Lie group, and let e be the identity element of G. The **Lie algebra** of G (or *associated to* G), denoted by $\mathfrak{g}$, is defined as the $\mathbb{R}$-vector space $\mathfrak{g} = T_eG$ with the bilinear product given by the Lie bracket of left-invariant vector fields:

$$[\cdot,\cdot]_{\mathfrak{g}} : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}, \quad (v, w) \mapsto [V^{(v)}, W^{(w)}]_{\mathfrak{g}} := [V^{(v)}, V^{(w)}]_e,$$

where $(\mathfrak{g}, [\cdot,\cdot]_{\mathfrak{g}})$ satisfies antisymmetry, bilinearity, and the Jacobi identity, and thus it is abstractly defining a Lie algebra.

Definition 1.14. A vector space basis $\{T_i\}_{i=1,\dots,n}$ with $n = \dim\mathfrak{g}$ of the Lie algebra $\mathfrak{g}$ is called a **basis of generators** of the Lie algebra. Since the Lie brackets of elements of the Lie algebra can be expressed as linear combinations of the basis elements (since the algebra is closed under the Lie bracket), we can write

$$[T_i,\, T_j] = \sum_{k=1}^{n} c_{ij}^k T_k,$$

where c_{ij}^k are the **structure constants** of the Lie algebra $\mathfrak{g}$ with the basis of generators $\{T_i\}$.

We can highlight some properties of the structure constants c_{ij}^k that follow from the properties of the Lie bracket:

- They are antisymmetric in the first two (lower) indices: $c_{ij}^k = -c_{ji}^k$, which follows from the antisymmetry of the Lie bracket.
- They satisfy the Jacobi identity: $\sum_{l=1}^{n}(c_{ij}^l c_{lk}^m + c_{jk}^l c_{li}^m + c_{ki}^l c_{lj}^m) = 0$ for all i, j, k, m, which follows from the Jacobi identity of the Lie bracket.

These comes directly from the properties of the Lie bracket and thus their application onto the generators of the Lie algebra. Furthermore an abstract Lie algebra can be defined by specifying a choice of basis $\{T_i\}$ along with a set of structure constants c_{ij}^k that satisfy the above properties, without any reference to a specific Lie group.

As an easy example, we can consider the tensor ϵ_{ij}^k and check that it respects the above properties: along with a good choice of basis, it can be used to define the structure constants of the Lie algebra $\mathfrak{su}(2)$ associated to the Lie group $SU(2)$.

We used real coefficients for the Lie algebra, but we could have used complex coefficients as well, and thus we would have defined a complex Lie algebra. The choice of real or complex coefficients depends on the context and the applications we have in mind, and it does not affect the general properties of the Lie algebra. The extension and generalization is straightforward, and it is just a matter of changing the field over which the vector space is defined.

1.3 | Lie Algebra of Matrix Groups

If we consider matrices $G \subseteq \mathrm{GL}(V)$, with V $(n = \mathrm{dim}V)$ a vector space over $\mathbb{R}$ or $\mathbb{C}$ (with a proper basis chosen), then $\mathrm{GL}(V) \simeq \mathrm{GL}(n, \mathbb{R})$ or $\mathrm{GL}(n, \mathbb{C})$, respectively (for simplicity let us stick to the real case, the complex case is just a straightforward generalization).

Taking than any element $g \in G \subseteq \mathrm{GL}(n, \mathbb{R})$ can be thought of as an $n \times n$ matrix

$$g = (g^{AB})_{AB}, \quad g^{AB} : G \to \mathbb{R}, \quad g \to g^{AB},$$

for any fixed pair of indices A, B. Thus we can think of the elements of the group as being labeled by a set of real numbers, which are the entries of the matrix.

Definition 1.15. Let $G \subset \mathrm{GL}(n, \mathbb{R})$ be a matrix group. Let $v \in T_k G$ and $(U, x \equiv x^k)$, such that a test function $\phi \in \mathbb{C}^\infty(G)$ with $k = 1, \ldots, \mathrm{dim}G \neq n^2$ and a point $h \in U$, then

$$\begin{aligned} v(\phi) &= \sum_k v^k \partial_k \phi\big|_{x(h)} \\ &= \sum_{k,A,B} v^k \frac{\partial g^{AB}}{\partial x^k}\bigg|_{x(h)} \frac{\partial \phi}{\partial g^{AB}} \\ &= \sum_{A,B} v^{AB} \frac{\partial \phi}{\partial g^{AB}}, \end{aligned}$$

where we can see how the summation over k defines a sort of *tangent matrix*

$$v^{AB} := \sum_k v^k \frac{\partial g^{AB}}{\partial x^k}\bigg|_{x(h)}, \quad \frac{\mathrm{d}}{\mathrm{d}t}(\gamma^{AB}(t))\bigg|_{t=t_0} = v^{AB},$$

which is the tangent vector to the curve $\gamma^{AB}(t)$ in the space of matrices, and it can be thought of as an element of the Lie algebra associated to the matrix group G. Thus we can identify the tangent space at the identity element of the group with a subspace of the space of matrices, and this subspace is precisely the Lie algebra of the group.

Definition 1.16. Let $G \subset \mathrm{GL}(n, \mathbb{R}) \subset \mathbb{R}^{n \times n}$ be a matrix group, where the groups from left to right are included injectively. Then there exists a $g \in G$ such that $g \mapsto (g^{AB}) \in \mathbb{R}^{n^2}$ is a smooth and injective map, and thus we can identify the elements of the group with points in $\mathbb{R}^{n^2}$.

Let now $v \in T_h G$ then $(g^{AB})_*(v) \in T_{h^{AB}} \mathbb{R}^{n \times n} \simeq \mathbb{R}^{n^2}$ is injective. We will denote $(g^{AB})_*(v) = v^{AB}$ is the tangent matrix:

$$\{v^{AB}\} \subseteq T_{h^{AB}} \mathbb{R}^{n \times n} \simeq \mathbb{R}^{n^2}.$$

For $v = \dot{\gamma}(t_0)$ with $\gamma : \mathbb{R} \to G$ a curve with $\gamma(t_0) = h$, the tangent vector v corresponds to a tangent matrix v^{AB} in $\mathbb{R}^{n^2}$. We have $\forall \phi \in \mathbb{C}^\infty(G)$,

$$[(g^{AB})_*(v)](\phi) = \frac{\mathrm{d}}{\mathrm{d}t}(\phi \circ g^{AB} \circ \gamma)(t_0) = \frac{\mathrm{d}}{\mathrm{d}t}(g^{AB} \circ \gamma)(t_0) \frac{\partial \phi}{\partial g^{AB}},$$

which implies

$$v^{AB} = \frac{\mathrm{d}}{\mathrm{d}t}(g^{AB} \circ \gamma)(t_0).$$

[...]

Example (General Linear group). Let us consider $G = \mathrm{GL}(n,\mathbb{R}) = \{M \in \mathbb{R}^{n\times n} : \det(M) \neq 0\}$, we are interested in finding the Lie algebra of the grop as the tangent space at the identity element $e = \mathbb{I}_n$ of the group. We can identify the elements of the group with points in $\mathbb{R}^{n^2}$ by mapping each matrix M to its entries M^{AB}. Locally, around the identity element, the tangent space can be identified with the space of all $n \times n$ matrices:

$$T_e G \simeq \mathfrak{g} = T_e \mathbb{R}^{n\times n} \simeq \mathbb{R}^{n^2},$$

and since the dimension of the algebra, the group, the manifold etc. coincides, we know it to be of dimension n^2. When passing to the complex counterpart, we have $G = \mathrm{GL}(n,\mathbb{C})$ and the Lie algebra is $\mathfrak{g} = T_e G \simeq \mathbb{C}^{n^2}$, which is of dimension $2n^2$ as a real vector space.

Example (Special Linear group). If we consider the special linear group $G = \mathrm{SL}(n,\mathbb{R}) = \{M \in \mathrm{GL}(n,\mathbb{R}) : \det(M) = 1\}$, we can identify the elements of the group with points in $\mathbb{R}^{n^2}$ by mapping each matrix M to its entries M^{AB}. Let $M(t) : \mathbb{R} \to G$ be a curve in the group with $M(0) = e = \mathbb{I}_n$. Then the tangent vector at the identity is given by

$$v = \frac{\mathrm{d}}{\mathrm{d}t} M(t)\bigg|_{t=0}.$$

Since $M(t)$ is a matrix in $\mathrm{SL}(n,\mathbb{R})$, we have $\det(M(t)) = 1$ for all t. Taking the derivative of this condition with respect to t and evaluating at $t = 0$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \det(M(t))\bigg|_{t=0} = 0.$$

Using the formula for the derivative of the determinant

$$\frac{\mathrm{d}}{\mathrm{d}t} \det(M(t))\bigg|_{t=0} = \mathrm{Tr}\left[\det(M(t))(M^{-1}(t))^T \dot{M}(t)\right]_{t=0} = \mathrm{Tr}\big(\dot{M}(0)\big) = \mathrm{Tr}(v),$$

this implies

$$\mathrm{tr}(v) = 0,$$

which means that the tangent vector v (derivative of an element) at the identity element must be a traceless matrix. Therefore, the Lie algebra of the special linear group is the set of all traceless $n \times n$ matrices, which can be identified with a subspace of $\mathbb{R}^{n^2}$:

$$\mathfrak{sl}(n,\mathbb{R}) = T_e G \simeq \{X \in \mathbb{R}^{n\times n} : \mathrm{tr}(X) = 0\}.$$

The dimension of this Lie algebra is $n^2 - 1$, since the condition of being traceless reduces the number of independent parameters by one (but is the same as the original group, since there we had the condition on the determinant). In the complex case, we have $G = \mathrm{SL}(n,\mathbb{C})$ and the Lie algebra is $\mathfrak{sl}(n,\mathbb{C}) = T_e G \simeq \{X \in \mathbb{C}^{n\times n} : \mathrm{tr}(X) = 0\}$, which is of dimension $2(n^2 - 1)$ as a real vector space.

Example (Orthogonal group). Consider the orthogonal group $G = \mathrm{O}(n,\mathbb{R}) = \{M \in \mathbb{R}^{n\times n} : M^T M = \mathbb{I}_n\}$. To find its Lie algebra, we look at the tangent space at the identity element $e = \mathbb{I}_n$. A curve $M(t) : \mathbb{R} \to G$ with $M(0) = \mathbb{I}_n$ satisfies $M(t)^T M(t) = \mathbb{I}_n$ for all t. Differentiating this condition with respect to t and evaluating at $t = 0$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t}(M(t)^T M(t))\bigg|_{t=0} = 0.$$

Using the product rule,

$$\dot{M}(0)^T \mathbb{I}_n + \mathbb{I}_n^T \dot{M}(0) = 0,$$

which simplifies to

$$\dot{M}(0)^T + \dot{M}(0) = 0.$$

This means that the tangent vector $v = \dot{M}(0)$ must be skew-symmetric. Therefore, the Lie algebra of the orthogonal group is the set of all skew-symmetric $n \times n$ matrices:

$$\mathfrak{o}(n, \mathbb{R}) = T_e G \simeq \{X \in \mathbb{R}^{n \times n} : X^T + X = 0\}.$$

The dimension of this Lie algebra is $n(n-1)/2$, as there are $n(n-1)/2$ independent parameters for a skew-symmetric matrix.

Example (Special Orthogonal group). For the special orthogonal group $G = \mathrm{SO}(n, \mathbb{R}) = \{M \in \mathrm{O}(n, \mathbb{R}) : \det(M) = 1\}$, the Lie algebra is the same as that of the orthogonal group, since the condition of having determinant equal to one does not impose any additional constraints on the tangent vectors at the identity element. Therefore, we have

$$\mathfrak{so}(n, \mathbb{R}) = T_e G \simeq \{X \in \mathbb{R}^{n \times n} : X^T + X = 0\},$$

which is of dimension $n(n-1)/2$.

We can study this group as the intersection of the special linear group and the orthogonal group, since it is defined by the conditions of being both orthogonal and having determinant equal to one. Thus we can write

$$\mathrm{SO}(n, \mathbb{R}) = \mathrm{SL}(n, \mathbb{R}) \cap \mathrm{O}(n, \mathbb{R}),$$

and consequently we can write the Lie algebra as the intersection of the Lie algebras of the special linear group and the orthogonal group:

$$\mathfrak{so}(n, \mathbb{R}) = \mathfrak{sl}(n, \mathbb{R}) \cap \mathfrak{o}(n, \mathbb{R}) = \{X \in \mathbb{R}^{n \times n} : X^T + X = 0, \mathrm{tr}(X) = 0\}.$$

But since the condition of being skew-symmetric already implies that the trace is zero, we can simplify this to

$$\mathfrak{so}(n, \mathbb{R}) = \{X \in \mathbb{R}^{n \times n} : X^T + X = 0\} = \mathfrak{o}(n, \mathbb{R}),$$

which is the same as the Lie algebra of the orthogonal group.

Example (Unitary group). For the unitary group $G = \mathrm{U}(n, \mathbb{C}) = \{U \in \mathbb{C}^{n \times n} : U^\dagger U = \mathbb{I}_n\}$, we can find its Lie algebra by looking at the tangent space at the identity element $e = \mathbb{I}_n$. A curve $U(t) : \mathbb{R} \to G$ with $U(0) = \mathbb{I}_n$ satisfies $U(t)^\dagger U(t) = \mathbb{I}_n$ for all t. Differentiating this condition with respect to t and evaluating at $t = 0$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t}(U(t)^\dagger U(t))\bigg|_{t=0} = 0.$$

Using the product rule,

$$\dot{U}(0)^\dagger \mathbb{I}_n + \mathbb{I}_n^\dagger \dot{U}(0) = 0,$$

which simplifies to

$$\dot{U}(0)^\dagger + \dot{U}(0) = 0.$$

This means that the tangent vector $v = \dot{U}(0)$ must be anti-Hermitian. Therefore, the Lie algebra of the unitary group is the set of all anti-Hermitian $n \times n$ matrices:

$$\mathfrak{u}(n, \mathbb{C}) = T_e G \simeq \{X \in \mathbb{C}^{n \times n} : X^\dagger + X = 0\}.$$

The real dimension of this Lie algebra is n^2, as there are n^2 independent parameters for an anti-Hermitian matrix: the upper half of the matrix has $\frac{n(n-1)}{2}$ independent parameters, and the diagonal elements (all purely imaginary since they have to be changed in sign when daggered) have n independent parameters, for a total of $2\frac{n(n-1)}{2} + n = n^2$ independent parameters.

Example (Special Unitary group). For the special unitary group $G = \mathrm{SU}(n, \mathbb{C}) = \{U \in \mathrm{U}(n, \mathbb{C}) : \det(U) = 1\}$, the Lie algebra is the set of all anti-Hermitian $n \times n$ matrices with trace zero:

$$\mathfrak{su}(n, \mathbb{C}) = T_e G \simeq \{X \in \mathbb{C}^{n\times n} : X^\dagger + X = 0, \mathrm{tr}(X) = 0\}.$$

The real dimension of this Lie algebra is $n^2 - 1$, as the condition of being traceless reduces the number of independent parameters by one.

We are computing the **real dimension** of the lie algebra even in the complex case, since we are considering the Lie algebra as a real vector space. If we were to consider it as a complex vector space, we would have to consider that when the coefficients are complex, the dimension would be halved, since each complex parameter corresponds to two real parameters (the real and imaginary parts). Thus, when we have an odd number of independent parameters, the complex dimension would be a half-integer, which is not possible for a vector space. Therefore, we consider the Lie algebra as a real vector space, even in the complex case, to ensure that the dimension is always an integer.

We want to impose only **holomorphic functions** on the complex Lie groups, since we want to preserve the complex structure of the group and its Lie algebra. If we were to allow non-holomorphic functions, we would lose the complex structure and we would not be able to define a complex Lie algebra. The holomorphic functions (for example the determinant consition, which is just a polynomial equation in complex variables) ensure that the Lie algebra is closed under the Lie bracket and that it has a well-defined complex structure, which is essential for many applications in mathematics and physics.

Lemma 1.8. *Let G be a matrix Lie group, and let $v \in T_h G$ be a tangent vector in h and consider $a \in G$; then the pushforward of v under the left translation by a is given by*

$$[[(L_a)_*]_h(v)]^{AB} = \sum_{C=1}^{n} a^{AC} v^{CB} = (a \cdot v)^{AB},$$

where the last equality makes sense only in a matrix group, since we are using the matrix multiplication to define the action of the group on the tangent vector. This shows how the pushforward of a tangent vector under left translation can be expressed in terms of the matrix multiplication of the group element and the tangent vector, which is a key property that allows us to understand how the Lie algebra is related to the structure of the Lie group.

Proof. We can use a test function $\phi \in \mathbb{C}^\infty(G)$ to compute the pushforward of v under left translation by a:

$$[(L_a)_*]_h(v)(\phi) = v(\phi \circ L_a) = \sum_k v^k \partial_k(\phi \circ L_a)\big|_{x(h)} \equiv v(\tilde{\phi}),$$

where we defined $\tilde{\phi} := \phi(ag)$. Now we can express the function $\tilde{\phi}$ in terms of the matrix entries g^{AB} if we compute

$$[[(L_a)_*]_h(v)]^{AB} = v^{AB} \frac{\partial \phi}{\partial g^{PQ}}\bigg|_{g=ah} \frac{\partial}{\partial g^{AB}} (ag)^{PQ}\bigg|_{g=h},$$

which can be rewritten as

$$= v^{AB} \frac{\partial \phi}{\partial g^{PQ}}\bigg|_{g=ah} a^{PR}\delta^{RA}\delta^{QB} = a^{PR}v^{RQ}\frac{\partial \phi}{\partial g^{PQ}} = (a \cdot v)^{AB}\frac{\partial \phi}{\partial g^{AB}},$$

[...] □

Proposition 1.9. *Let G be the matrix group and $v, w \in T_eG \implies \mathfrak{g}$, then we can associate to v and w the left-invariant vector fields $V^{(v)}$ and $W^{(w)}$, and we can compute the Lie bracket of these vector fields at the identity element:*

$$[V^{(v)}, W^{(w)}]_e = [v, w]_{\mathfrak{g}},$$

where the Lie bracket on the right-hand side is defined by the commutator of the corresponding tangent matrices:

$$[v, w]_{\mathfrak{g}} = (vw)^{AB} - (wv)^{AB} = v^{AC}w^{CB} - w^{AC}v^{CB},$$

which is the standard Lie bracket for matrix Lie algebras and makes sense in the context of matrix groups, since the tangent vectors can be identified with matrices and the Lie bracket can be defined as the commutator of these matrices.

We are not going to prove this proposition, since it is a straightforward computation that follows from the definition of the Lie bracket of vector fields and the properties of the pushforward under left translations. The key point is that the left-invariant vector fields are defined by translating the tangent vectors at the identity element to every other point in the group using left translations, and thus their Lie bracket can be computed by using the properties of the pushforward under left translations and the definition of the Lie bracket of vector fields.

1.4 | Exponential Map

We are now going to define the exponential map for a Lie group, which is a crucial tool for understanding the relationship between the Lie algebra and the Lie group. The exponential map allows us to relate elements of the Lie algebra (tangent vectors at the identity element) to elements of the Lie group, and it provides a way to construct curves in the Lie group from tangent vectors in the Lie algebra. It is the tool which let us take a finite transformation from an infinitesimal one, and thus it is fundamental for applications in physics, where we often deal with infinitesimal transformations and we want to understand how they relate to finite transformations.

Definition 1.17 (Integral curve)**.** Let $V \in \Gamma(T\mathcal{M})$ be a vector field and let $\gamma : (a, b) \to \mathcal{M}$ be an integral curve for V, i.e.,a smooth curve whose tangent vector at each point $\gamma(t)$ is equal to the value of the vector field at that point $V_{\gamma(t)}$, i.e.

$$\dot{\gamma}(t) = V_{\gamma(t)} \quad \forall t \in (a, b).$$

If we use a chart (U, x) around a point $p \in \mathcal{M}$ such that $\gamma(t_0) = p$ for some $t_0 \in (a, b)$, then we can write the integral curve in coordinates as

$$\dot{\gamma}^i(t) = \frac{\mathrm{d}}{\mathrm{d}t}\gamma^i(t) = V^i(\gamma(t)) \quad \forall t \in (a, b), \quad \text{(ODE)},$$

which is a system of ordinary differential equations (ODEs) that can be solved to find the integral curve $\gamma(t)$ given the vector field V and the initial condition $\gamma(t_0) = p$.

The existence and uniqueness of solutions to this ODE is guaranteed by the standard theory of ODEs, and thus we can always find an integral curve for any given vector field and initial condition.

Definition 1.18 (One-parameter subgroup)**.** Let a smooth curve $\sigma : \mathbb{R} \to G$ in a Lie group G is called a **one-parameter subgroup** of G if it satisfies the group homomorphism property

$$\sigma(s + t) = \sigma(s)\sigma(t) \quad \forall s, t \in \mathbb{R},$$

with $\sigma(0) = e$ is the identity element of the Lie group G.

This last definition means that a one-parameter subgroup is a curve in the Lie group that respects the group structure, i.e., the value of the curve at the sum of two parameters is equal to the product (group operation) of the values of the curve at each parameter. This property is crucial for understanding how one-parameter subgroups relate to the Lie algebra and how they can be used to construct finite transformations from infinitesimal ones.

We will adopt the notation in which

$$\sigma \equiv \{\sigma(t) \mid t \in \mathbb{R}\},$$

as a subgroup of G generated by the one-parameter subgroup σ. This notation emphasizes the fact that the one-parameter subgroup generates a subgroup of G that is isomorphic to $\mathbb{R}$ under addition, and it allows us to think of the one-parameter subgroup as a continuous family of group elements parameterized by a real variable t.

Proposition 1.10. *Let G be a Lie group and let $V \in \Gamma(TG)$ be a left-invariant vector field on G. Then the integral curve $\sigma : \mathbb{R} \to G$ of V with initial condition $\sigma(0) = e$ is a one-parameter subgroup of G.*

The idea here is that by taking the integral curve of a left-invariant vector field starting at the identity element, we obtain a curve that respects the group structure and thus defines a one-parameter subgroup. The left-invariance of the vector field ensures that the flow generated by the vector field is compatible with the group operation, which is essential for the resulting curve to be a one-parameter subgroup.

Proposition 1.11. *Let σ be a one-parameter subgroup of a Lie group G. Then the tangent vector $V_{\sigma(t)} = \frac{d}{dt}\sigma(t)$ to the curve $\sigma(t)$ at any point t can be computed as the pushforward of the tangent vector at the identity element V_e under the left translation by $\sigma(t)$:*

$$V_{\sigma(t)} = [(L_{\sigma(t)})_*]_e(\dot{\sigma}(0)),$$

which implies that $V_{\sigma(t)}$ is a left-invariant vector field along the curve $\sigma(t)$, and thus the one-parameter subgroup σ is generated by the left-invariant vector field V that corresponds to the tangent vector at the identity element.

Thus every one-parameter subgroup of a Lie group is generated by a left-invariant vector field, and conversely, every left-invariant vector field generates a one-parameter subgroup. This establishes a deep connection between the Lie algebra (the space of left-invariant vector fields) and the structure of the Lie group (the one-parameter subgroups), which is fundamental for understanding how the Lie algebra encodes information about the local structure of the Lie group and how it can be used to construct finite transformations from infinitesimal ones.

It is a one-to-one correspondence between the elements of the Lie algebra and the one-parameter subgroups of the Lie group, which is a key result in the theory of Lie groups and Lie algebras.

Corollary 1.12. *Suppose to have two different one-parameter subgroups $\sigma(t)$ and $\mu(t)$ of a Lie group G with $\dot{\sigma}(0) = \dot{\mu}(0) = v \in T_eG$. Then $\sigma(t) = \mu(t)$ for all $t \in \mathbb{R}$.*

This corollary can be read as saying that the one-parameter subgroup generated by a given tangent vector at the identity element is unique. This emphasizes the one-to-one correspondence between the elements of the Lie algebra and the one-parameter subgroups of the Lie group, and it ensures that each element of the Lie algebra generates a unique one-parameter subgroup.

Definition 1.19. Let $X \in \mathfrak{g}$ be an element of the Lie algebra of a Lie group G. We denote as σ_X the unique one-parameter subgroup of G constructed as follows:

1. $X \in \mathfrak{g} \iff V^{(X)} \in \Gamma(TG)$ is left-invariant vector field on G such that $V_e^{(X)} = X$.
2. $\sigma_X : \mathbb{R} \to G$ is the unique integral curve of $V^{(X)}$ with initial condition $\sigma_X(0) = e$ (passing through the identity element) with $\dot{\sigma}_X(0) = V_e^{(X)} = X$; it is a one-parameter subgroup of G.

Since along the curve $\sigma_X(t)$ we have a possible mapping from the Lie algebra to the Lie group, we can define the exponential map as follows.

Definition 1.20 (Exponential map)**.** The exponential map

$$\exp : \mathfrak{g} \to G, \quad X \mapsto \sigma_X(1),$$

is called the exponential map of the Lie group G. It maps an element X of the Lie algebra $\mathfrak{g}$ to the value of the one-parameter subgroup σ_X at $t = 1$, which is an element of the Lie group G.

Proposition 1.13. *If $X \in \mathfrak{g}$, $\lambda \in \mathbb{R}$ then we have*

$$\exp(\lambda X) = \sigma_X(\lambda).$$

Since this is a one-parameter subgroup, we have

$$\exp((\lambda + \mu)X) = \sigma_X(\lambda + \mu) = \sigma_X(\lambda)\sigma_X(\mu) = \exp(\lambda X)\exp(\mu X), \quad \forall \lambda, \mu \in \mathbb{R},$$

and we can also derive straightforwardly that $\exp(0) = e$ *and* $\exp(-X) = \sigma_X(-1) = \sigma_X(1)^{-1} = \exp(X)^{-1}$. *This shows that the exponential map has properties that are consistent with the group structure, and it allows us to relate the Lie algebra to the Lie group in a way that respects the algebraic structure of both.*

This definition allows us to plug in the map any vector from the Lie algebra and obtain an element of the Lie group, which is a powerful tool for understanding how the Lie algebra encodes information about the local structure of the Lie group and how it can be used to construct finite transformations from infinitesimal ones.

Proof. For $\lambda \neq 0$ we can define a new curve $\tilde{\sigma} : \mathbb{R} \to G$ as

$$\tilde{\sigma}(t) := \sigma_X(\lambda t),$$

which is still a one-parameter subgroup of G since it satisfies the group homomorphism property. If we now define another one-parameter subgroup from $\sigma_{\lambda X}(0) = \lambda X$, then we have also

$$\dot{\tilde{\sigma}}(0) = \frac{\mathrm{d}}{\mathrm{d}t}\sigma_X(\lambda t)\bigg|_{t=0} = \lambda\dot{\sigma}_X(0) = \lambda X,$$

which means that $\tilde{\sigma}$ and $\sigma_{\lambda X}$ are two one-parameter subgroups of G with the same tangent vector at the identity element (initial conditions), and thus by the previous corollary they must coincide, i.e., $\tilde{\sigma}(t) = \sigma_{\lambda X}(t)$ for all $t \in \mathbb{R}$. If this is the case, then we can use the definition of the exponential map with respect to the one-parameter subgroup $\sigma_{\lambda X}$ to write

$$\exp(\lambda X) = \sigma_{\lambda X}(1) = \tilde{\sigma}(1) = \sigma_X(\lambda),$$

which is the desired result. □

We are now going to express some basic and important properties of the exponential map.

Proposition 1.14. *The exponential map has the following properties:*

1. *The image of the exponential map is a neighborhood of the identity element e in G, which means that*

$$\exp(\mathfrak{g}) \subseteq G_0 = \{g \in G | \exists \gamma : [a, b] \to G \textit{ continuous, with } \gamma(t_0) = e, \gamma(t_1) = g\},$$

   where G_0 is the connected component of the identity element in G. For example $\mathrm{SO}(n) = [O(n)]_e$.

2. *The Lie algebra $\mathfrak{g}$ is a $\mathbb{R}$-vector space $\simeq \mathbb{R}^n$, then we can say that the exponential map is a smooth map from $\mathbb{R}^n$ to G, and thus it is a* ***local diffeomorphism*** *around the identity element e. This means that there exists a neighborhood U of $0 \in \mathfrak{g}$ such that* $\exp : U \to \exp(U)$ *is a diffeomorphism onto its image, which is a neighborhood of the identity element in G. With these restrictions the exponential map is thus injective.*

3. *The exponential map* $\exp : \mathfrak{g} \to G$ *is a smooth map, but generally not surjective, since there may be elements of the Lie group that cannot be reached by exponentiating elements of the Lie algebra. But if* G_0 *is a compact (connected component) then it is surjective. For example, if* $G = \mathrm{SL}(2, \mathbb{R})$*, which is not compact, then the exponential map is not surjective. On the other hand, if* $G = \mathrm{SU}(n)$*, which is compact, then the exponential map is surjective. There may be cases of surjectivity even for non-compact groups, but this is not guaranteed in general (for example, the exponential map for* $\mathrm{GL}(n, \mathbb{C})$ *and* $\mathrm{SO}(1, n)$ *is surjective even if the group is not compact).*

4. *Any* $g \in G_0$ *can be written as*

$$g = \exp(X_1)\exp(X_2)\cdots\exp(X_k),$$

for some $X_1, X_2, \ldots, X_k \in \mathfrak{g}$ *and some* $k \in \mathbb{N}$*. This means that any element of the connected component of the identity can be expressed as a finite product of exponentials of elements of the Lie algebra, which do not contradicts the previous point about surjectivity, since we are not claiming that every element of* G *can be written as a single exponential, but rather that it can be written as a finite product of exponentials.*

5. ***Baker-Campbell-Hausdorff (BCH) formula****: For any* $X, Y \in \mathfrak{g}$*, we have then*

$$Z = X + Y + \frac{1}{2}[X, Y]_{\mathfrak{g}} + \frac{1}{12}[X, [X, Y]_{\mathfrak{g}}]_{\mathfrak{g}} - \frac{1}{12}[Y, [X, Y]_{\mathfrak{g}}]_{\mathfrak{g}} + \cdots ,$$

where the series continues with higher-order commutators. This formula let us connect ... Furtheore, if this series converges, we have

$$\exp(Z) = \exp(X)\exp(Y) = \exp\left(X + Y + \frac{1}{2}[X, Y]_{\mathfrak{g}} + \frac{1}{12}[X, [X, Y]_{\mathfrak{g}}]_{\mathfrak{g}} - \frac{1}{12}[Y, [X, Y]_{\mathfrak{g}}]_{\mathfrak{g}} + \cdots\right),$$

has a solution for Z *in terms of* X *and* Y*. This formula is fundamental for understanding how the group operation in* G *can be expressed in terms of the Lie algebra* $\mathfrak{g}$*, and it provides a way to compute the product of two exponentials in terms of the Lie bracket and higher-order commutators. If the group is compact or the map is surjective, then we can use the BCH formula to express any element of the group as an exponential of an element of the Lie algebra (since the series would always converge in this case). For the case in which* $[X, Y] = 0$ *then the BCH formula simplifies to*

$$\exp(X)\exp(Y) = \exp(X + Y),$$

which is a special case that occurs when the Lie algebra is abelian (i.e., when all elements commute with each other).

6. *Let* $F : G_1 \to G_2$ *be a smooth map between two Lie groups (Lie group homomorphism), then*

$$\exp_2(F_*(X)) = F(\exp_1(X)), \quad \forall X \in \mathfrak{g}_1,$$

where $\exp_1$ *and* $\exp_2$ *are the exponential maps of* G_1 *and* G_2 *respectively, and* F_* *is the pushforward of* F*. This property shows that the exponential map is compatible with Lie group homomorphisms, and it allows us to relate the exponential maps of different Lie groups through the pushforward of a homomorphism between them.*

1.4.1 | Will see

[...]

Proposition 1.15 (Exponential on matrix group)**.** *Let G be a matrix group, then $\forall X \in \mathfrak{g}$ we have*

$$[\exp(X)]^{AB} = (e^X)^{AB},$$

where $X = X^{AB}$ is the tangent matrix and

$$e^X = \sum_{n=0}^{\infty} \frac{X^n}{n!},$$

and this is exactly the exponential map defined in the previous section.

Proof. Let σ_X be the integral curve of the left-invariant vector field associated with X, $V^{(X)}$; we have from the definition of integral curve that

$$\dot{\sigma}_X(t) = V^{(X)}_{\sigma_X(t)} = [(L_{\sigma_X(t)})_*]_e(X).$$

Now we can use the exponential map to write

$$(\frac{\mathrm{d}}{\mathrm{d}t}\exp(tX))^{AB} = [\dot{\sigma}_X(t)]^{AB} = [\sigma_X(t)]^{AC}X^{CB} = [\exp(tX)]^{AC}X^{CB}.$$

Since the one parameter subgroup σ_X satisfies the composition

$$\sigma_X(t+s) = \sigma_X(t)\sigma_X(s),$$

then if we fix one of the parameters, say s, we can treat $\sigma_X(s)$ as a constant matrix and solve the above ODE to get

$$\frac{\mathrm{d}}{\mathrm{d}t}[\sigma_X(t+s)]^{AB}\bigg|_s = [\sigma_X(s)]^{AC}X^{CB} \implies [\sigma_X(s)]^{AB} = \frac{\mathrm{d}}{\mathrm{d}\tau}\sigma(\tau)^{AB}\bigg|_{\tau=t+s} = \left([(L_{\sigma_X(t+s)})_*]_e(X)\right)^{AB},$$

which can be read as the tangent vector of σ_X at the point $\sigma_X(t+s)$ which is given by the pushforward at the identity. We have then:

$$\frac{\mathrm{d}}{\mathrm{d}t}[\sigma_X(t+s)]^{AB}\bigg|_s = \sigma_X(t+s)^{AC}X^{CB} = \sigma_X(t)^{AC}\sigma_X(s)^{CD}X^{DB},$$

since we are still making computations which makes sense only when speaking about matrices. Then

$$\frac{\mathrm{d}}{\mathrm{d}t}[\sigma_X(t+s)]^{AB}\bigg|_s = \dots$$

When we consider $t_1 = 0$ the term $\sigma_X(t_1)$ is just the identity matrix, so we have

TODO: substitute t + s with t + t2 at t1

$$(\sigma_X(t_2)X)^{AB} = (X\sigma_X(t_2))^{AB}, \quad \forall t_2 \in \mathbb{R},$$

thus we have that $\sigma_X(t)$ commutes with X for all t. Then the above ODE can be solved by the power series expansion of the exponential function, and we have

$$\implies (\frac{\mathrm{d}}{\mathrm{d}t}\exp(tX))^{AB} = (X\exp(tX))^{AB} = (\exp(tX)X)^{AB}.$$

Given this equation we know that the solution is given by the power series expansion of the exponential function, and we needed to show the commutation of $\sigma_X(t)$ with X to be able to solve the ODE (since ...). $\square$

2 | Representations of Lie Groups and Algebras

From notes [...] (2 - representation of Lie groups and Lie algebras)

We are going to exploit why when we have a group representation we also have an algebra representation, and how the exponential map is the link between the two. We are going to see that the exponential map is a homomorphism between the Lie algebra and the Lie group, and that it allows us to construct representations of the Lie group from representations of the Lie algebra.

The key is the finding how given any grouo representation of a lie group we can act on this representation with the exponential map to get a representation of the lie algebra, and vice versa.

2.0.1 | Representation of Groups: Recap

In this section we will refer to a general group as G and a vector space V on a field $\mathbb{K} = \mathbb{C}$ or $\mathbb{R}$.

Definition 2.1. A representation of a group G on a vector space V is a homomorphism $D : G \to GL(V)$, $D \in \mathrm{Hom}(G, GL(V))$ where $GL(V)$ is the group of invertible linear transformations on V. Thus $D : V \to V$ is an invertible linear transformation for each $g \in G$ and we have

$$D(g_1, g_2) = \dots$$

A few common nomenclatures:

- Representation space
- Dimension of the representation
- Infinite or finite dimensional representation
- Real or complex representation

Definition 2.2. Let us consider two representations and two vector spaces, $D_1 : G \to GL(V_1)$ and $D_2 : G \to GL(V_2)$. We can define direct sum and tensor product of representations as follows:

$$\begin{cases} aaaaa \\ bbbbb \end{cases}$$

which are still representations of G on the vector spaces $V_1 \oplus V_2$ and $V_1 \otimes V_2$ respectively: **direct sum representation** and **tensor product representation**. We are linearly extending the definition of the representation to the direct sum and tensor product of vector spaces.

Definition 2.3 (Complex conjugate representation)**.** Let us consider a representation $D : G \to GL(V)$ on a vector space V. We can define the complex conjugate representation as follows:

$$\overline{D} : G \to GL(V^*), \quad g \mapsto \overline{D}(g) = (D(g))^*,$$

where V^* is the dual vector space of V and $D(g)^*$ is the complex conjugate of the linear transformation $D(g)$.

Definition 2.4 (Dual representation)**.** Let us consider a representation $D : G \to GL(V)$ on a vector space V. We can define the dual representation as follows:

$$D^\vee : G \to GL(V^\vee), \quad D^\vee(g) = D(g)^*,$$

where $V^\vee$ is the dual vector space of V and $D(g)^*$ is the dual of the linear transformation $D(g)$.

dual space

transpose map

basis for space has a dual basis for the dual space: relation between the two is given by the delta function

application of dual on vector written as sum of components of the vector times the components of the dual vector

scalar product provides a canonical isomorphism between the vector space and its dual: we can identify a vector with its dual by using the scalar product

natural paring between the dual space and the original space: left invariant under the combined action of the representation and its dual representation on it

$$[D^\vee(g)(\omega)](D(g)(v)) = \omega(v), \quad \forall g \in G, \omega \in V^\vee, v \in V.$$

Looking at the previous equation as matrix multiplication, we can see that it holds if and only if $D^\vee(g) = D(g)^*$, which is the definition of the dual representation: the two matrices in the middle have to become the identity matrix, which is the condition for the natural pairing to be invariant under the combined action of the representation and its dual representation on it.

Definition 2.5 (Equivalent representations)**.** Two different representations of the same group are equivalent if there exists an isomorphism between the two representation spaces (vector spaces) that intertwines the two representations such that

$$\forall g \in G, \quad AD_1(g)A^{-1} = D_2(g),$$

...

It is easy to show that this is an equivalence relation, since we have

- Reflexivity: D_1 is equivalent to itself, since we can take $A = I$ the identity matrix.
- Symmetry: if D_1 is equivalent to D_2, then D_2 is equivalent to D_1, since we can take A^{-1} as the intertwining operator.
- Transitivity: if D_1 is equivalent to D_2 and D_2 is equivalent to D_3, then D_1 is equivalent to D_3, since we can take the product of the two intertwining operators as the intertwining operator between D_1 and D_3.

> **Example (Equivalent representations in physics).** For example electrons and up-quarks are different particles, but they transform in the same way under the gauge group of the Standard Model, so they are equivalent representations of the gauge group: they are both half-integer spin representations of the Lorentz group. As far as the representation under rotation group is concerned, they are equivalent representations, since they have the same spin.

Definition 2.6 (Invariant subspace)**.** Let us consider a representation $D : G \to GL(V)$ on a vector space V. A subspace $W \subset V$ is invariant under the action of the representation if $D(g)(W) \subset W$ for all $g \in G$. ... This will define a smaller representation of G on the subspace W by restricting the representation to W.

There always exists at least two invariant subspaces, the trivial subspace $\{0\}$ and the whole space V, which are invariant under the action of any representation. We are interested in finding non-trivial invariant subspaces, which will allow us to decompose the representation into smaller representations.

Definition 2.7 (Reducibility of representations)**.** Let (D, V) be a representation of a group G. We say that:

1. (D, V) is called **reducible** if there exists a non-trivial invariant subspace $W \subset V$ such that $D(g)(W) \subset W$ for all $g \in G$.
2. (D, V) is called **irreducible** if there does not exist a non-trivial invariant subspace $W \subset V$ such that $D(g)(W) \subset W$ for all $g \in G$.
3. (D, V) is called **completely reducible** if it can be decomposed into a direct sum of irreducible representations, i.e. if there exists a decomposition of the representation space V into a direct sum of invariant subspaces $V = W_1 \oplus W_2 \oplus \cdots \oplus W_n$ such that each W_i is an irreducible representation of G.
   - $D|_{L_k}$ has no non-trivial invariant subspace, so it is irreducible.
   - $L_i \cap L_j = \{0\}$ for $i \neq j$:...

...

Proposition 2.1. *A representation is completely reducible if and only if there exists irreducible representations* (D_k, V_k) *such that*

$$(D, V) = \left(\bigoplus_k D_k, \bigoplus_k V_k \right),$$

such that the irreducible representations D_k *are a sort of "building blocks" of the representation* D *in the sense that any representation can be decomposed into a direct sum of irreducible representations, and the irreducible representations are the "atoms" of the representation theory, since they cannot be decomposed further into smaller representations.*

We need a notion of inner product on the representation space to be able to talk about orthogonality of subspaces and to be able to decompose the representation into irreducible representations. We will see that for compact groups we can always find an inner product on the representation space that is invariant under the action of the group, which will allow us to decompose the representation into irreducible representations.

Definition 2.8 (Hermitian inner product and representations)**.** Let V a finite dimensional vector space over $\mathbb{C}$ endowed with a Hermitian inner product. Then a representation $D : G \to GL(V)$ is called unitary if $D(g)$ is a unitary operator for all $g \in G$:

$$D(g)^{-1} = D(g)^{\dagger}, \quad \forall g \in G,$$

where the dagger of a linear map O on a vector space with a Hermitian inner product is defined as the unique linear map $O^{\dagger}$ such that

$$\langle v|Ow\rangle = \left\langle O^{\dagger}v\middle|w\right\rangle, \quad \forall v, w \in V.$$

An obvious consequence of the definition of unitary representation is that the inner product is invariant under the action of the group, since we have

$$\langle v|v\rangle = \left\langle v\middle|D(g)^{-1}D(g)w\right\rangle = \langle D(g)v|D(g)w\rangle = \langle v'|w'\rangle,$$

where we have defined $v' = D(g)v$ and $w' = D(g)w$. This means that the inner product is invariant under the action of the group, which will allow us to decompose the representation into irreducible representations by using the orthogonality of subspaces.

For unitary representation, we can give a version of the powerful Weyl theorem, which states:

Theorem 2.2 (Weyl's unitary trick)**.** *Let V be a complex vector space with an Hermitian inner product, and let (D, V) a representation of a group G with respect to this inner product. Then any representation of G on V is equivalent to a unitary representation. ...*

The idea is to make use of orthogonal decomposition of the representation space to decompose the representation into irreducible representations, and then to use the fact that for compact groups we can always find an invariant inner product on the representation space to make the representation unitary.

We need orthogonality and compatibility of the representation with the inner product to be able to decompose the representation into irreducible representations, and we need the invariance of the inner product under the action of the group to be able to make the representation unitary.

[...]

reducible and completely reducible becomes the same

we need to define the concept of an integration measure

Proposition 2.3. *Let D be a representation of a finite group G on a finite dimensional complex vector space V. Then there exists an positive Hermitian inner product on V with respect to which D is a unitary representation.*

Proof. Pick a basis and define an inner product on V ...

Now we can define a new inner product by averaging over the group (diagonal sum with complex conjugation on first argument):

$$\langle u|v\rangle := \frac{1}{|G|}\sum_{g\in G}\langle D(g)u|D(g)v\rangle_0,$$

For finite group it is a finite linear combination of inner products, so it is still an inner product, so all the defining properties of an inner product are still satisfied.

On a Lie group the previous expression would look like

$$\frac{1}{V(G)} \int_G \mathrm{d}\mu_G \left\langle D(g)u|D(g)v\right\rangle_0 ,$$

...

Then we have

$$[\ldots]$$
$$[\ldots]$$
$$[\ldots].$$

So in the end

$$\implies D(g)^\dagger = D(g)^{-1}, \quad \forall g \in G,$$

which means that D is a unitary representation with respect to the new inner product. □

Thus for finite groups we can always find an inner product on the representation space that makes the representation unitary, and for compact groups we can always find an invariant inner product on the representation space that makes the representation unitary. This is a very powerful result, since it allows us to restrict our attention to unitary representations when studying the representation theory of finite and compact groups, which are the most common types of groups in physics.

Example (Reducible representation). Let us consider the group $G = (\mathbb{R}, +)$ on the vector space $V = \mathbb{R}^2$ with the representation defined by

$$D : G \mapsto \mathrm{GL}(V) = \mathrm{GL}(2, \mathbb{R}), \quad D(r) = \begin{pmatrix} 1 & r \\ 0 & 1 \end{pmatrix},$$

respecting obviously the group composition law, since we have

$$D(r_1 + r_2) = D(r_1)D(r_2), \quad D(-r) = D(r)^{-1}.$$

Then the action of the representation on a vector

$$\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2,$$

is given by

$$D(r) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + ry \\ y \end{pmatrix},$$

which means that we have a non trivial invariant subspace given by the y-axis, which is

$$L = \left\{ \begin{pmatrix} x \\ 0 \end{pmatrix} : x \in \mathbb{R} \right\} \subseteq \mathbb{R}^2,$$

such that

$$D(r)\big|_L = \mathbb{I} \implies D(r)l = l, \quad \forall l \in L, r \in G.$$

Suppose there exists a non trivial invariant subspace $L' \neq \{0\}$ of the vector space $V \sim \mathbb{R}^2$, with dimension $\dim(L') = 1$, then

$$L' = \{\lambda v | \lambda \in \mathbb{R}\}, \text{ for some } v \in \mathbb{R}^2, v \neq 0,$$

and thus for any value of the group parameter r we have

$$\forall r \in \mathbb{R}, \quad D(r)v = \lambda_r v, \text{ for some } \lambda_r \in \mathbb{R},$$

which means that v is an eigenvector of the representation $D(r)$ for all r, and thus we have

$$\iff \begin{pmatrix} v_1 + rv_2 \\ v_2 \end{pmatrix} = \lambda_r \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}, \iff v_2 = \lambda_r v_2, \quad v_1 + rv_2 = \lambda_r v_2, \quad \forall r \in \mathbb{R},$$

which means that either $v_2 = 0 \implies L' = L$ or $\lambda_r = 1$ for all r. This means that the representation D is reducible, since it has a non trivial invariant subspace L, but it is **not completely reducible**, since it cannot be decomposed into a direct sum of irreducible representations, since the only invariant subspace is L and it does not have a complementary invariant subspace.

Proposition 2.4. *Let D be a representation of a finite group G on a finite dimensional complex vector space V. Then there exists a positive Hermitian inner product on V with respect to which D is a unitary representation.*

Theorem 2.5. *Let V be a complex vector space with an Hermitian inner product, and let (D, V) a unitary representation of a group G with respect to this inner product. Then*

$$D \text{ is reducible} \iff D \text{ is completely reducible.}$$

This means that compact groups can be represented by unitary representations, and thus we can always decompose a representation of a compact group into irreducible representations, which are the building blocks of the representation theory of compact groups. This is a very powerful result, since it allows us to restrict our attention to unitary representations when studying the representation theory of compact groups, which are the most common types of groups in physics. This is expressed by the next two propositions:

Proposition 2.6. *Let (D, V) be a finite dimensional complex representation of a compact Lie group G. Then there exists a positive Hermitian inner product on V with respect to which D is a unitary representation*

Proposition 2.7. *Let G be a compact Lie group. Then any complex representation of G is completely reducible if and only if it is reducible.*

Dual Space

Let $V^\vee = \{\varphi : V \to \mathbb{K} \text{ linear function}\}$, then given a basis $\{b_k\}$ of V we can notice how any linear function $\varphi \in V^\vee$ is completely determined by its action on the basis vectors, since we can write any vector $v \in V$ as a linear combination of the basis vectors, and thus we have

$$\varphi(v) = \varphi(v^k b_k) = v^k \varphi(b_k),$$

which directly implies that the dual space $V^\vee$ has the same dimension as V, and thus we can define a **dual basis** $\{e^\ell\}$ of $V^\vee$ such that

$$e^\ell(b_k) = \delta^\ell_k, \implies \varphi = \varphi_\ell e^\ell,$$

then we can write the action of the dual representation on the dual space as

$$\varphi(v) = v^k \varphi(b_k) = v^k \varphi_\ell e^\ell(b_k) = v^k \varphi_\ell \delta^\ell_k = v^k \varphi_k.$$

Adopting the same bases, we can represent the vectors of V as column vectors and the linear functions of $V^\vee$ as row vectors, and thus given two vectors $v, w \in V$

$$v \simeq \vec{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}, \quad w \simeq \vec{w} = \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix},$$

then we can define the action of φ on v as

$$\varphi(v) = \vec{\varphi} \cdot \vec{v} = \vec{w}^T \cdot \vec{v} = w_k v^k,$$

where we have defined $\vec{\varphi} = (w_1, \ldots, w_n)$ as the row vector representation of the linear function φ. This is the natural pairing between the dual space and the original space, which is left invariant under the combined action of the representation and its dual representation on it.

Definition 2.9. Let $E : V \mapsto V$ be a linear map, we can define a matrix representation of E with respect to the basis $\{b_k\}$ as

$$E(b_k) = E^l{}_k b_l,$$

then the action of E on a vector v can be written as

$$E(v) = v^k E^l{}_k b_l = (E^l{}_k v^k) b_l,$$

where the term between the last parentheses is the matrix multiplication of the matrix representation of E with the column vector representation of v: $E \cdot \vec{v}$. We can thus define the dual map $E^\vee : V^\vee \mapsto V^\vee$ as

$$[E^\vee(\varphi)](v) = \varphi(E(v)),$$

which can be read as the action of the dual map on the dual space, and we are requesting that it will be compatible with the action of the original map on the original space, in the sense that the natural pairing between the dual space and the original space is invariant under the combined action of the representation and its dual representation on it. This are endomorphisms of the vector space V and its dual $V^\vee$, and we can write

$$E^\vee(e^\ell) = (E^\vee)_n{}^\ell e^n,$$

which implies

$$\begin{aligned}[E^\vee(e^\ell)](b_k) &= (E^\vee)_n{}^\ell e^n(b_k) = (E^\vee)_k{}^\ell \\ &= e^\ell(E(b_k)) = e^\ell(E^j{}_k b_j) = E^\ell{}_k,\end{aligned}$$

thus in this matrix representation we have that the dual map is represented by the transpose of the matrix representation of the original map, i.e. we have

$$E^\vee = E^T,$$

since from the position of the indices we can see that the matrix representation of the dual map is obtained by inverting rows with columns of the matrix representation of the original map.

Suppose that we have a linear map B on a vector space V and another linear map A on the dual space $V^\vee$ such that

$$\begin{cases} A : V^\vee \mapsto V^\vee, & A(\varphi) = A_n{}^\ell \varphi_\ell e^n, \\ B : V \mapsto V, & B(v) = B^l{}_k v^k b_l, \end{cases}$$

then the natural pairing between the dual space and the original space is invariant under the combined action of A and B on it if and only if

$$[A(\varphi)][B(v)] = \varphi_\ell A_n{}^\ell B^n{}_k v^k = \varphi_\ell (A^T)_k{}^\ell B^k{}_\ell v^k = \varphi(v) = \vec{\varphi}^T \cdot A^T \cdot (B\vec{v}) = (A\vec{\varphi})^T \cdot (B\vec{v}),$$

[...]

If $D : G \mapsto \mathrm{GL}(V)$ is a representation of a group G on a vector space V such that

$$D(g) \in \mathrm{GL}(V) \subseteq \mathrm{End}(V), \quad \forall g \in G,$$

then its dual representation $D^\vee : G \mapsto \mathrm{GL}(V^\vee)$ is defined by

$$D^\vee(g) = (D(g^{-1}))^T \in \mathrm{End}(V^\vee), \quad \forall g \in G,$$

we have

$$[D^\vee(g)(\varphi)](D(g)(v)) = [D(g^{-1})^T \vec{\varphi}]^T (D(g)\vec{v}) = \vec{\varphi}^T \cdot D(g^{-1}) \cdot D(g) \cdot \vec{v} = \vec{\varphi}^T \cdot \vec{v} = \varphi(v).$$

Equivalent Representations

Proposition 2.8. *Let G be a compact Lie group, and let (D, V) be a complex representation of G on a finite dimensional vector space V. Then there exists an equivalent representation $(\overline{D}, V)$ such that*

$$(\overline{D}, V) = (D^\vee, V^\vee) \iff \exists A : V \mapsto V^\vee \text{ isomorphism s. t. } A^{-1}\overline{D}(g)A = D^\vee(g), \quad \forall g \in G,$$

it is equivalent to the existence of an isomorphism between the representation space V and its dual $V^\vee$ that intertwines the representation D with its dual representation $D^\vee$.

Proof. There exists a G-invariant Hermitian product $\langle\cdot|\cdot\rangle$ on V such that (D, V) is unitary with respect to it, since G is compact. Then we can define an isomorphism $A : V \mapsto V^\vee$ by

$$\begin{cases} \langle\cdot|\cdot\rangle : V \times V \mapsto \mathbb{K} \text{ Hermitian,} \\ A : V \mapsto V^\vee, \quad A(v) = \langle v|\cdot\rangle \in V^\vee, \end{cases}$$

which is an isomorphism since the inner product is non-degenerate. Then picking an ON basis $\{b_k\}$ of V with respect to the inner product, we can write

$$\implies (D(g))^\dagger = (D(g)^*)^T = D(g)^{-1}, \quad \forall g \in G,$$

in the matrix representation with respect to the basis $\{b_k\}$ (the basis is necessary to be able to write the representation as a matrix, in any other basis the dagger may not be defined as the transpose of the c.c.), and thus we have

$$D^\vee(g) = (D(g^{-1}))^T \text{ in dual basis,}$$

in the matrices sense, then it is easy to verify that

$$D^\vee(g) = (D(g^{-1}))^T = (D(g)^{-1})^T = (D(g)^{-1})^T = ((D(g)^*)^T)^T = \overline{D}(g),$$

which means that the dual representation is equivalent to the complex conjugate representation. Even though V and $V^\vee$ are differnt vector spaces, they are isomorphic, and thus we can identify them by using the isomorphism A defined above, which means that we can identify the dual representation with the complex conjugate representation. □

In particle physics we are often interested in finding representations that are equivalent to their complex conjugate, since they correspond to particles that are their own antiparticles, such as the photon or the gluon. This is a very important property, since it allows us to identify particles with their antiparticles, which is a fundamental aspect of particle physics. For example, the charge conjugation operator in quantum field theory is an example of an operator that maps a representation to its complex conjugate, and thus it allows us to identify particles with their antiparticles. This is a very important aspect of particle physics, since it allows us to understand the properties of particles and their interactions.

2.1 | Representation of Lie Algebras

Definition 2.10. Let (D, V) be the representation of a Lie group G on a vector space V. In group theory

$$D : G \mapsto \mathrm{GL}(V) \sim \mathbb{R}^{n\times n},$$

is a smooth map from the Lie group G to the group of invertible linear transformations on $V \sim \mathbb{K}^n$ (is a technical requirement that will let us take derivatives and compute push-forwards). Any such representation is a "model" of G in $\mathrm{GL}(V)$ as

$$D(G) = \{D(g)|g \in G\} \subset \mathrm{GL}(V),$$

it needn't be "accurate" in the sense that it can be a homomorphism that is not injective, but it is still a representation of G on V, for example

$$D(g) = \mathbb{I}, \quad \forall g \in G,$$

so that it is a very "crude" model of G in $\mathrm{GL}(V)$, but it is still a representation of G on V.

Definition 2.11. Let us give some definitions that will be useful to define the representation of a Lie algebra, and to understand the relation between the representation of a Lie group and the representation of its Lie algebra.

(a) A **Lie algebra homomorphism** is a linear map $f : \mathfrak{g} \to \mathfrak{h}$ between two Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$ is a $\mathbb{R}$-vector space homomorphism that preserves the Lie bracket, i.e. it satisfies

$$f([X, Y]_{\mathfrak{g}}) = [f(X), f(Y)]_{\mathfrak{h}}, \quad \forall X, Y \in \mathfrak{g}.$$

(b) Given a Lie algebra $\mathfrak{g}$, a **Lie algebra representation** (d, V) on a $\mathbb{R}$-vector space V, is a Lie algebra homomorphism $\rho : \mathfrak{g} \to \mathfrak{gl}(V) \simeq \mathrm{End}(V)$ from the Lie algebra $\mathfrak{g}$ to the Lie algebra of endomorphisms of a vector space V (including non invertible ones). These carry a canonical Lie algebra structure on $\mathrm{End}(V)$ such that given two endomorphisms $A, B \in \mathrm{End}(V)$ and some real number $\lambda \in \mathbb{R}$ we have

$$\begin{cases} \lambda A : v \mapsto \lambda A(v), \\ (A + B) : v \mapsto A(v) + B(v), \end{cases}$$

which does define a canonical vector space structure on $\mathrm{End}(V)$, and the Lie bracket is given by the commutator of endomorphisms, i.e. we have

$$\begin{cases} [A, B]_{\mathrm{End}(V)} : v \mapsto A(B(v)) - B(A(v)), \\ [\lambda A, B]_{\mathrm{End}(V)} = \lambda [A, B]_{\mathrm{End}(V)}, \\ [A + B, C]_{\mathrm{End}(V)} = [A, C]_{\mathrm{End}(V)} + [B, C]_{\mathrm{End}(V)}, \end{cases}$$

from the induced Lie algebra structure on $\mathrm{End}(V)$. In particular, the representation of the Lie brackets on $\mathfrak{g}$ becomes the commutator of the endomorphisms on V, i.e. we have

$$d([X, Y]_{\mathfrak{g}}) = d(X)d(Y) - d(Y)d(X), \quad \forall X, Y \in \mathfrak{g}.$$

Lemma 2.9. *The exponential map*

$$\exp : \mathrm{End}(V) \mapsto \mathrm{GL}(V),$$

is defined as the power series

$$\exp(A) = e^A : V \mapsto V, \quad v \mapsto \sum_{n=0}^{\infty} \frac{A^n(v)}{n!}.$$

TODO: ref

Proof. From proposition 1.5.10 plus the choice of a basis on V we can identify $\mathrm{End}(V)$ with the space of $n \times n$ matrices, and thus we can define the exponential map as the power series defined above, which is well defined since it converges for any matrix A. □

Lemma 2.10. *Let $F : G_1 \mapsto G_2$ be a Lie group homomorphism between two Lie groups G_1 and G_2 (with Lie algebras $\mathfrak{g}_1$ and $\mathfrak{g}_2$), then the push-forward of F at the identity element $e \in G_1$ defines a Lie algebra homomorphism between the Lie algebras of G_1 and G_2, i.e. we have*

$$F_*([X, Y]_{\mathfrak{g}_1}) = [F_*(X), F_*(Y)]_{\mathfrak{g}_2}, \quad \forall X, Y \in \mathfrak{g}_1.$$

We will not prove this lemma since it is just an exercise of application of definitions and concepts of differential geometry, but the idea is that the push-forward of a Lie group homomorphism at the identity element defines a linear map between the tangent spaces of the two Lie groups at their identity elements, which are isomorphic to their Lie algebras, and thus we can identify the push-forward with a linear map between the Lie algebras, which is a Lie algebra homomorphism since it preserves the Lie bracket.

Proposition 2.11. *Let $D : G \mapsto \mathrm{GL}(V)$ be a representation of a Lie group G on a vector space V. Then the push-forward of D at the identity element $e \in G$ defines a representation of the Lie algebra $\mathfrak{g}$ of G on V, i.e. we have*

$$d = (D_*)_e : \mathfrak{g} \mapsto \mathfrak{gl}(V) \simeq \mathrm{End}(V), \quad d(X) = (D_*)_e(X), \quad \forall X \in \mathfrak{g},$$

which is a Lie algebra homomorphism since it preserves the Lie bracket.

Corollary 2.12. *Let (D, V) be a representation of the group G and (d, V) the associated representation of the Lie algebra $\mathfrak{g}$ obtained by taking the push-forward of D at the identity element. Then we have*

$$D(\exp(X)) \in \mathrm{GL}(V), \quad D(\exp(X)) = \exp(d(X)), \quad \forall X \in \mathfrak{g},$$

which means that the representation of the Lie group G can be obtained by exponentiating the representation of its Lie algebra $\mathfrak{g}$.

Proof. Use tha fact that $F : G_1 \mapsto G_2$ is a Lie group homomorphism, then we have

$$\exp_2(F_*(X)) = F(\exp_1(X)), \quad \forall X \in \mathfrak{g}_1,$$

where $\exp_1$ and $\exp_2$ are the exponential maps of the Lie groups G_1 and G_2 respectively, then we can apply this result to the representation as $F \sim D$ of the Lie group G, which is a Lie group homomorphism from G to $\mathrm{GL}(V)$, and thus we have

TODO: check if correct

$$\exp_{\mathrm{GL}(V)}(d(X)) = D(\exp_G(X)), \quad \forall X \in \mathfrak{g},$$

which means that the representation of the Lie group G can be obtained by exponentiating the representation of its Lie algebra $\mathfrak{g}$. □

This is what we need to connect finite transformations with infinitesimal transformations, since the representation of the Lie algebra $\mathfrak{g}$ can be thought of as the infinitesimal version of the representation of the Lie group G, and thus we can obtain the representation of the Lie group by exponentiating the representation of its Lie algebra. This is a very powerful result, since it allows us to study the representation theory of Lie groups by studying the representation theory of their

Lie algebras, which is often simpler and more tractable. This is particularly useful in physics, where we are often interested in studying the symmetries of physical systems, which are described by Lie groups, and thus we can study their representations by studying the representations of their Lie algebras.

Corollary 2.13. *If we have a representation $D : G \mapsto \mathrm{GL}(V)$ which is unitary, then the associated representation of the Lie algebra $\mathfrak{g}$ is given by anti-Hermitian operators, i.e.*

$$d(X)^T = -d(X), \quad \forall X \in \mathfrak{g}$$

Proof. If the representation of the Lie group $e^{d(X)}$ is unitary, then we have

$$\left(e^{d(X)}\right)^\dagger = \sum_n \left(\frac{(d(X))^n}{n!}\right)^\dagger = \sum_n \frac{(d(X)^\dagger)^n}{n!} = e^{d(X)^\dagger} = e^{-d(X)},$$

which comes directly from the definition of unitary representation, thus proving the claim. □

Remark. *In physics it is common to define the exponential map with an extra factor of i in the exponent, i.e. we have*

$$D(g) = e^{id(X)},$$

which strictly speaking makes sense only for complex Lie algebras, in which case

$$d_{phys} \coloneqq -i(D_*)_e = -id,$$

which is just an alternative convention for the representation of the Lie algebra, used by physicists since it allows to work with Hermitian operators instead of anti-Hermitian operators, which are more common in the fields of application. This is just a matter of convention, and it does not change the mathematical structure of the representation theory, but it is important to be aware of this convention when working with representations in physics.

Example.

1. If $G \subseteq \mathrm{GL}(V)$ is a matrix group to begin with, then the "defining representation" of G is just the inclusion map, i.e. we have

$$D : G \mapsto \mathrm{GL}(V), \quad D(g) = g, \quad \forall g \in G,$$

which is a representation of G on V since it is a group homomorphism from G to $\mathrm{GL}(V)$. Then the associated defining representation of the Lie algebra $d : \mathfrak{g} \mapsto \mathrm{End}(V)$ is given by

$$d(X) = (D_*)_e(X) = X, \quad \forall X \in \mathfrak{g},$$

which means that the representation of the Lie algebra is just the identity map on the Lie algebra, which is a very simple representation of the Lie algebra. It is also known as the **foundamental representation**. We implicitly computed them in the section about Lie groups, when we computed the Lie algebra of a matrix group by taking the tangent space at the identity element, which is just the Lie algebra itself.

2. The "**trivial representation**" is defined by

$$D : G \mapsto \mathrm{GL}(V), \quad D(g) = \mathbb{I}, \quad \forall g \in G,$$

which is a representation of G on V since it is a group homomorphism from G to $\mathrm{GL}(V)$. Then the associated representation of the Lie algebra is given by

$$d(X) = (D_*)_e(X) = 0, \quad \forall X \in \mathfrak{g},$$

which means that the representation of the Lie algebra is just the zero map on the Lie algebra, which is a very simple representation of the Lie algebra.

To sum up:

- $d : \mathfrak{g} \mapsto \mathrm{End}(V)$ is a lie algebra representation ...
- If $D : G \mapsto \mathrm{GL}(V)$ is a group representation, then ...

[...]

2.1.1 | The Adjoint Representation

Definition 2.12. Let G be a Lie group then the conjugation by $g \in G$ is a smooth map

$$C_g : G \mapsto \dots$$

This is also called the "adjoint action" (by g) on G iself

$$\implies (C_g)_* : T_h G \mapsto T_{C_g(h)} G$$

is a homomorphism of Lie groups, and in particular, the map

$$h = e, \dots$$

...

[... DIO MALEDICA IL POLLINE ...]

Proposition 2.14. *The adjoint action induces an endomorphism ... [...]*

Proof. 1. $\mathrm{Ad}_g = [(C_g)_*]_e$ is an invertible endomorphism for any $g \in G$ since $C_g : G \mapsto G$ is a diffeomorphism (its inverse is smooth and $(C_g)^{-1} = C_{g^{-1}}$):

$$\implies (C_g)_* \text{ is an invertible endomorphism } \forall g \in G.$$

2. $g \mapsto \mathrm{Ad}_g$ respects the group structure:

$$C_{g_1 g_2}(h) = g_1(g_2 h g_2^{-1}) g_1^{-1} = C_{g_1}(C_{g_2}(h)),$$

[...]

□

We are going to express a corollary which helps us to understand the meaning of the adjoint representation.

Corollary 2.15. *Let* $\mathrm{ad} : \mathfrak{g} \mapsto \mathrm{End}(\mathfrak{g})$*, with* $X \mapsto \mathrm{ad}_X$ *be the Lie algebra representation induced by the adjoint representation of* G*,* $\mathrm{Ad} : G \mapsto \mathrm{GL}(\mathfrak{g})$*. This algebra representation satisfies the following relations*

1. $\mathrm{ad}_X(Y) = [X, Y]$ *for any* $X, Y \in \mathfrak{g}$.
2. $\mathrm{Ad}_{\exp(X)}(Y) = \exp(\mathrm{ad}_X)(Y) = e^{\mathrm{ad}_X}(Y)$ *for any* $X, Y \in \mathfrak{g}$.

Proof. Following corollary (2.2.6) we have that **TODO:** ref

$$D(\exp(X)) = \exp(d(X)) = e^{d(X)} \quad \forall X \in \mathfrak{g},$$

which proves the second point almost trivially. For the first point, we have that $\forall X \in \mathfrak{g}$

$$X \mapsto V^{(X)} \mapsto \sigma_X(t) = \exp(tX), \quad \dot{\sigma}_X(0) = V_e^{(X)} = X,$$

and thus $\forall g \in G, Y \in \mathfrak{g}, \phi \in C^\infty(G)$,, we have that

$$\mathrm{Ad}_g(Y)(\phi) = [(C_g)_*]_e(Y)(\phi) = [(C_g)_* V^{(Y)}(\phi) \circ C_g]_e,$$

[...] thus from lemma (1.3.13) we have that **TODO:** ref

$$\begin{aligned}\mathrm{Ad}_g(Y)(\phi) = V^{(Y)}(\phi \circ C_g)(e) = V_e^{(Y)}(\phi \circ C_g) = \frac{\mathrm{d}}{\mathrm{d}t}(\phi \circ C_g \circ \exp(tY))_{t=0} \\ = \dots\end{aligned}$$

which in the end

$$\implies \dots$$

and any g spits out an endomorphism $\mathrm{Ad} : G \mapsto \mathrm{GL}(\mathfrak{g})$. $\square$

In order to compute the adjoint representation of the lie algebra we need to take the push forward and ...

$$\mathrm{ad}_X = (\mathrm{Ad})_*(X) = (\mathrm{Ad}_*)\dot{\sigma}_X(0) = \frac{\mathrm{d}}{\mathrm{d}s}(\mathrm{Ad} \circ \sigma_X(s))_{s=0}$$

$$\implies \mathrm{ad}_X(Y) = \frac{\mathrm{d}}{\mathrm{d}s}(\mathrm{Ad}_{\exp(sX)}(Y))_{s=0} = \frac{\partial}{\partial s}\left[\frac{\partial}{\partial t}\exp(sX)\exp(tY)\exp(-sX)\Big|_{t=0}\right]_{s=0},$$

and using the Baker-Campbell-Hausdorff formula we have that

$$\begin{aligned}BCH \implies \exp(sX)\exp(tY)\exp(-sX) = \exp\left(tY + s[X, Y] + \dots\right) \\ = \exp(\dots)\end{aligned}$$

and by deriving with respect to t (it is not difficult to believe that it is the same as deriving the usual exponential) we have that

$$\left(Y + \frac{s}{2}([X, Y] - [Y, X]) + s^2(\dots) + st(\dots) + \mathcal{O}(s^2 t)\right)\exp(\dots),$$

which needs to be evaluated at $t = 0$ and then derived with respect to s at $s = 0$, thus we have that

$$\xrightarrow{t=0} \left(Y + s[X, Y] + \mathcal{O}(s^2)\right) \xrightarrow{\partial/\partial s} [X, Y] + \mathcal{O}(s) \xrightarrow{s=0} [X, Y].$$

Pick basis ... applying the adjoint representation of one of the elements of the basis element for the lie algebra on another of such besis elemnts

$$\mathrm{ad}_{T_i} T_j = [T_i, T_j] = C_{ij}{}^k T_k,$$

then we know how the representation of the basis elements acts ...

$$E(b_j) = E^i{}_j b_i = E(v^j b_j) = (E^i{}_j v^j) b_i,$$

which means that the matrix elements of the representation of the basis elements are given by the structure constants of the lie algebra

$$\implies (\mathrm{ad}_{T_i})^k{}_j = C_{ij}{}^k.$$

We now have a very complete picture of the tools we have at our disposal to study the representations of lie groups and lie algebras...

Example. Let us consider the lie group SU(2) and its lie algebra $\mathfrak{su}(2) = \{$anti-Hermitian, traceless$\}$, which elements can be written as

$$X = \begin{pmatrix} ai & z \\ -z^* & -ai \end{pmatrix}, \quad a \in \mathbb{R}, z \in \mathbb{C}.$$

The standard basis for $\mathfrak{su}(2)$ is given by

$$T_1 = -\frac{i}{2}\sigma_x, \quad T_2 = -\frac{i}{2}\sigma_y, \quad T_3 = -\frac{i}{2}\sigma_z,$$

where $\sigma_x, \sigma_y, \sigma_z$ are the Pauli matrices. The structure constants of $\mathfrak{su}(2)$ are given by

$$[T_i, T_j] = \epsilon_{ij}{}^k T_k \implies \mathrm{ad}_{T_n} \equiv L_n \in \mathrm{End}(\mathfrak{su}(2)) \, : \, X \mapsto [T_n, X] = L_n(X),$$

defines the Pauli basis

$$(L_n)^k = \epsilon_{nj}{}^k,$$

TODO: check indices

with

$$(L_n^T)^k{}_j = (L_n)^j{}_k = \epsilon_{nj}{}^k = -\epsilon_{nk}{}^j = -(L_n)^j{}_k = -(L_n^T)^k{}_j,$$

which means that the matrices L_n are antisymmetric and thus they are anti-Hermitian, and thus they span the set of all antisymmetric 3×3 matrices, which is the lie algebra $\mathfrak{so}(3)$ of the lie group SO(3). We have thus shown that the adjoint representation of $\mathfrak{su}(2)$ is isomorphic to the fundamental representation of $\mathfrak{so}(3)$.

[...]

The group adjoint representation of SU(2) is the same as the defining representation of SO(3) for the same observations as before

$$\mathrm{Ad}^{\mathrm{SU}(2)} \cong \text{ defining rep. of SO(3)} \cong \mathrm{Ad}^{\mathrm{SO}(3)}.$$

This is not true in general:

1. if $\mathfrak{g} \cong \mathfrak{h}$ are isomorphic lie algebras, then certainly $\mathrm{ad}^{\mathfrak{g}} \cong \mathrm{ad}^{\mathfrak{h}}$ are isomorphic lie algebra representations, but
2. the defining representation of $\mathrm{SO}(n) \ncong \mathrm{Ad}^{\mathrm{SO(n)}}$ is not isomorphic to the adjoint representation of $\mathrm{SO}(n)$ for $n > 3$ in general.

2.2 | Reductibility of Lie Group and Algebra Representations

Definition 2.13. Let (d, V) be the representation of a lie algebra $\mathfrak{g}$. A non trivial invariant subspace $L \subset V$ is a subspace different from $\{0\}$ and V such that $d(X)(L) \in L$ for any $X \in \mathfrak{g}$, $d \in L$.

Proposition 2.16. *Let G be the connected lie group and (D, V) be a finite-dimensional representation of G. Then, the representation is (ir-)reducible if and only if the induced representation of the lie algebra $\mathfrak{g}$, $(d = (D_*)_e, V)$ is (ir-)reducible.*

... we need to prove two implications:

...

...

Assume d is reducible, ther tehre exist a non trivial invariant subspace $L \subset V$ such that $\forall X \in \mathfrak{g}$, $\forall v \in L$ we have that $d(X)(v) \in L$. Then we can write

$$D(\exp(X))(v) = \exp(d(X))(v) = e^{d(X)}(v) = \lim_{m\to\infty} \sum_{n}^{m} \frac{d(X)^n}{n!}(v) \in L = \lim_{m\to\infty} a_m,$$

which means that

$$\forall m \in \mathbb{N}, a_m \in L \implies \lim_{m\to\infty} a_m \in L \implies D(\exp(X))(v) \in L.$$

The other way is almost analogous; we start from the assumption that D is reducible, thus there exist a non trivial invariant subspace $L \subset V$ such that for a curve $\sigma(t) \in G$ we have that $D(\sigma(t))(v) \in L$ for any $v \in L$. Since any $X \in \mathfrak{g}$ is tangent to $\sigma_X(t) = \exp(tX)$ at $t = 0$, then we have that

$$\begin{aligned}\forall v \in L, \quad d(X)(v) = \frac{\mathrm{d}}{\mathrm{d}t}\left(e^{td(X)}(v)\right)_{t=0} &= \frac{\mathrm{d}}{\mathrm{d}t}\left(D(\exp(tX))(v)\right)_{t=0} \\ &= \lim_{t\to 0} \frac{1}{t}\left(D(\exp(tX))(v) - v\right) \in L \forall t,\end{aligned}$$

which means that $d(X)(v) \in L$ for any $X \in \mathfrak{g}$ and $v \in L$. □

Definition 2.14. A lie algebra $\mathfrak{g}$ ia called abelian if $[X, Y] = 0$ for any $X, Y \in \mathfrak{g}$. By corollary (2.3.3) we have that the adjoint representation of an abelian lie algebra is trivial, **TODO:** ref

$$\mathrm{ad}_X = 0 \quad \forall X \in \mathfrak{g},$$

thus any one-dimensional lie algebra is abelian, and any representation of an abelian lie algebra is reducible since any subspace is invariant under the action of the lie algebra. **TODO:** check

If G is connected then its Lie algebra is abelian if and only if G is abelian, thus any representation of an abelian lie group is reducible.

[...]

Theorem 2.17 (Schwarz's lemma)**.** *Let (D_1, V_1) and (D_2, V_2) be irreducible representations of a lie group G or a Lie algebra $\mathfrak{g}$. If we have an homomorphism of representations $A \in Hom(V_1, V_2)$:*

$V_1 \mapsto V_2$ *such that* $A \circ D_1(g) = D_2(g) \circ A$ *for any* $g \in G$ *(or* $A \circ d_1(X) = d_2(X) \circ A$ *for any* $X \in \mathfrak{g}$*), then either* $A = 0$ *or* A *is invertible, i.e. it is an isomorphism of representations. In particular, if it is invertible then the two representations are isomorphic, and if* $V_1 = V_2$ *then* A *is a scalar multiple of the identity.*

Proof. For any element $x \in \ker(A) \subset V_1$, then we have

$$\forall g \in G\,(\text{or }\mathfrak{g}) \implies A(D_1(g)(x)) = D_2(g)(A(x)) = 0 \implies D_1(g)(x) \in \ker(A),$$

thus $\ker(A)$ is an invariant subspace of V_1, and since V_1 is irreducible, then either $\ker(A) = \{0\}$ or $\ker(A) = V_1$. In the first case, A is *injective*, and in the second case $A = 0$.

For any element $y \in \operatorname{im}(A) \subset V_2$, then there exist $x \in V_1$ such that $A(x) = y$, thus we have that $A(D_1(g)(x)) \in \operatorname{im}(A)$ and

$$A(D_1(g)(x)) = D_2(g)(A(x)) = D_2(g)(y),$$

thus $\operatorname{im}(A)$ is an ariant subspace of D_2 and since D_2 is irreducible, then either $\operatorname{im}(A) = \{0\}$ or $\operatorname{im}(A) = V_2$. In the first case, $A = 0$, and in the second case A is *surjective*. Thus, if $A \neq 0$, then it is both injective and surjective, thus it is an isomorphism of representations. $\square$

[...]

[...]

[...]

2.3 | Complexification of Lie Algebras

Intro

3 bullet points

theorem 2.6.18: for integrating and obtaining a finite result from an infinitesimal one, we need the group to be compact to use the unitarity trick of reducing the representation... ones we classify all the complex reducible representations of a lie algebra, we can obtain all the irreducible representations of the lie algebra by taking the complexification of the lie algebra and then restricting the representation to the original lie algebra. **TODO:** check

theorem 2.6.19: ...

3 | Roots and Weights

3.1 | Motivating example: $\mathfrak{su}(2)$

3.1.1 | Projective representations and simply-connected groups

representations as linear maps on vectors

when we say the representation of a group is in GL or the one of an algebra is in End, we are saying that the group or algebra is represented as linear maps on vectors.

when we measure we normalize with resoect to the inner product, we need the norm of the vectors and thus the inner product...

all of the vectors on the same ray (same direction) are to be considered in the same state. it is a one dimensional subspace of the vector space. if this is what carachterizes the system, we need to really care only about the projected representation on the projective space, which is the space of rays. in qm symmetries should be realized as projective representattions of the group, rather than as linear representations. but we can always lift a projective representation to a linear one, by going to the universal cover of the group. for example, the projective representations of SO(3) are the linear representations of SU(2). this is why we can use SU(2) to describe spin, even if spin is not a representation of SO(3).

the mathematical key is the one saying that the a finite dimensional projective representation of a lie group on a finite dimensional complex vector space lifts to a unitary linear representation the covering group $\overline{G}$, which is the unique simply connected Lie group with the same lie algebra as G. hard to prove, absolutely non trivial.

projective representations $\equiv$ unitary linear representations of the covering group $\equiv$ anti hermitian linear representations of the lie algebra $\equiv$ linear representations of the complexified lie algebra

In fact, for the special case of $G = SO(3)$ and $\mathfrak{g} = \mathfrak{su}(2)$, this math-result, combined with the "axiomatic" definition of quantum mechanics using the (projective) Hilbert space, can be viewed as predicting half-integer spin particles (i.e., linear representations of $\mathfrak{su}(2)$ which are not $SO(3)$ reps) in the quantum world: they arise from projective representations of the rotation symmetry $SO(3)$, which we do not have in classical mechanics.

[details on axiomatics of qm]

3.1.2 | Ladder Operators and Spin Representation

Complexification and Hermitian basis. from properties of complexification we can take complex linear combinations of the generators of the algebra, and we can choose them in a way that they have nice properties. since the original generators are anti-hermitian, we can take linear combinations that are hermitian. so we can diagonalize them. thinking of the adjoint rep in that basis, which is an endomorphism of the algebra, we can see that the adj rep of the generators is also an endomorphism of the algebra, and thus we can diagonalize it as well, since it will remain hermitian. this property will extend to all the other complex finite dimensional representations, and thus they will have a basis in which the generators are hermitian and thus diagonalizable.

Eigenvalues of $d(J_3)$ from ladder operators. we can pick an on basis thanks to the usual unitarity trick which allows us to make the representation unitary, and thus the generators hermitian. in the following we prove that for irreducible representations the dimension of the subspace of eigenvectors of $d(J_3)$ with a given eigenvalue is one, and that the eigenvalues are equally spaced. the ladder operators are not hermitian. after definingthem through the commutation relations, we can study how the eigenvalues of $d(J_3)$ change when we apply the ladder operators (it moves up and down in the different eigenvector subspaces, the state changes but it remains an eigenvector). we can see that they change by one unit, and thus the eigenvalues are equally spaced (and thus the entire lie algebra moves among the invariant eigenvectors subspaces). this dont work in irreducible representations, since there do not exist trivial invariant subspaces. if the representation is finite, there must exist a maximum range of eigenvalues, and thus there must be a highest weight vector, which is an eigenvector of $d(J_3)$ that is annihilated by the raising operator. same for the minimum value (annihilated by the lowering operator).thus the representation space is made of a finite direct sum of eigenspaces of $d(J_3)$ with equally spaced eigenvalues, and the ladder operators move us among them.

The Casimir operators and range of J_3-eigenvalues. to further restrict the ... the casimir operator commutes with all the generators, thus it is a multiple of the identity in an irreducible representation. we can apply schurs lemma to infer that the eigenvalue of the casimir operator is a constant in an irreducible representation. if we apply the obtained formulae to the basis of eigenvectors of $d(J_3)$, we can find a relation between the eigenvalue of the casimir operator and the range of the eigenvalues of $d(J_3)$. this allows us to find the possible values of the eigenvalues of $d(J_3)$ and thus to classify all the irreducible representations. the proprierties we derive constrain the spacing of the eigenvalues to be integer or half-integer (and > 0), and thus we can label the irreducible representations by a non-negative integer or half-integer j, which is the maximum eigenvalue of $d(J_3)$ (the highest weight). the dimension of the representation is then $2j + 1$.

Dimensionality of eigenspaces. we want to argue that all the eigenspaces of $d(J_3)$ are one-dimensional. the span of the maximum eigenvectors and consecutive applications of the lowering operator is an invariant subspace, thus it must be the entire representation space, since the representation is irreducible. thus all the eigenvectors of $d(J_3)$ are obtained by applying the lowering operator to the maximum eigenvector, and thus they are all one-dimensional (we never get to the same eigenvalue twice, since the eigenvalues are equally spaced and we have a finite number of them).

Normaliation and numerical coefficients. [...] we find how to completely specify the representation by giving the value of j and the normalization of the ladder operators. we can choose a normalization such that the action of the ladder operators on the eigenvectors is given by a simple formula, which is very convenient for computations.

3.2 | Reformulation with Roots and Weights

[...]

3.3 | Killing Form and Cartan Subalgebra

For a general Lie algebra $\mathfrak{g}$, we first need the analogue of the subalgebra spanned by J_3 in $\mathfrak{sl}(2,\mathbb{C})$ which we choose to diagonalize.

Definition 3.1 (Killing form)**.** [...]

It takes two elements of the algebra and gives a number, thus it is a bilinear form on the algebra. since we are taking a trace of endomorphisms, we now there exist a basis invariant way to define it, and thus it is invariant under the adjoint action of the algebra on itself. this is a very important property, since it allows us to use the killing form to define an inner product on the algebra, which is invariant under the adjoint action. this will allow us to define a notion of orthogonality and thus to define a notion of diagonalization for the adjoint representation, which is crucial for the classification of representations.

The symmetric property comes from the fact that the trace of a product of endomorphisms is invariant under cyclic permutations, thus we can swap the two elements without changing the value of the killing form. the invariance property comes from the fact that the killing form is defined in terms of the adjoint representation, which is invariant under the adjoint action.

Given a basis for the lie algebra, given by the generators, we can write the killing form in terms of the structure constants of the algebra, with structure constants defined by the commutation relations of the generators. we know tha adjoint representation is given by the structure constants, thus we can write the killing form in terms of the structure constants. this allows us to compute the killing form explicitly for any given lie algebra, once we know its structure constants. Thus it is written in the same basis.

$$K_{mn} = \dots$$

in particular for $\mathfrak{g} = \mathfrak{su}(2)$ in the previous antihermitian basis we have $K_{mn} = -\delta_{mn}$ and for the hermitian basis for the complexification $\mathfrak{sl}(2,\mathbb{C})$ we have $K_{mn} = \delta_{mn}$.

There is an ambiguity when it comes to subalgebras: there are a priori two pairings we can define on a subalgebra, the one given by the killing form of the original algebra and the one given by the killing form of the subalgebra (which is a Lie algebra iself). in general they are not the same, thus we need to choose which one to use. whats relevant for us is when they are the same: in any case in which the subalgebra is an invariant subspace of the adjoint then we have the same killing form, since the killing form is defined in terms of the adjoint representation. thus we will always choose a subalgebra that is an invariant subspace of the adjoint, so that we can use the same killing form to define an inner product on it. thus for a semisimple lie algebra, we have a unique killing form, made by the direct sum of the killing forms of the simple components. another relevant case is when the subalgebra is real (from the complexification) then the real basis can be used to define the complex basis, therefore structure constants and the pairing matrix in the basis are the same, thus the following corollary holds:

Corollary 3.1. *Let $\mathfrak{h}$ be a real Lie algebra and $\mathfrak{g} = \mathfrak{h}_{\mathbb{C}}$ its complexification. Then the killing form of $\mathfrak{h}$ and the killing form of $\mathfrak{g}$ coincide on $\mathfrak{h}$:*

$$K^{\mathfrak{h}} = K^{\mathfrak{g}}\big|_{\mathfrak{h}}.$$

Proposition 3.2. *The Killing form on $\mathfrak{g}$ satisfies $\forall$ ...*

$$\ldots$$

symmetric pairing - invariant their sum is zero

... ☐

The notion of "invariance" becomes intuitive when we inspect the interplay between the Killing form on $\mathfrak{g}$ and the adjoint representation of the Lie group G with algebra $\mathfrak{g}$. Namely, from Corollary 2.3.3 we have for all $X, Y, Z \in \mathfrak{g}$:

$$\ldots$$

from which we can see that the infinitesimal transformation $\mathrm{ad}_{\exp(Z)}$ does not change the Killing form of any two elements of the algebra, thus it is invariant under the adjoint action of the group on the algebra. this is a very important property, since it allows us to use the killing form to define an inner product on the algebra, which is invariant under the adjoint action.

Example (...). [...]

[...]

Theorem 3.3. *A Lie algebra is semi-simple if and only if the Killing form is a non-degenerate bilinear form, i.e., if $K(X, Y) = K(Y, X) = 0$ for all $Y \in \mathfrak{g}$, then $X = 0$.*

While this theorem can be clearly stated, and is important for us in the following, its proof utilizes concepts of ideals and radicals of Lie algebras, which we have not introduced. This is a bit hard to prove so we wont.

Theorem 3.4. *Let G be a compact real Lie group and $\mathfrak{g}$ its Lie algebra. Then its Lie algebra has to be **reductive**:*

$$\mathfrak{g} \equiv \mathfrak{g}^{(ab)} \oplus \mathfrak{g}^{(ss)},$$

where $\mathfrak{g}^{(ab)}$ is an abelian Lie algebra and $\mathfrak{g}^{(ss)}$ is a semisimple Lie algebra. Furthermore, the Killing form is negative-semi-definite: ...

Conversely, if $\mathfrak{g}$ is a real Lie algebra with negative-definite Killing form (then $\mathfrak{g}$ must be semi-simple), then any connected Lie group with algebra $\mathfrak{g}$ is compact. If a real Lie algebra $\mathfrak{g}$ has positive-definite Killing form, then $\mathfrak{g} = 0$.

Thus the Killing form relates to the semisimplicity of the algebra, and thus to the compactness of the group. In particular, for a compact real Lie group, the killing form is negative-semi-definite, and thus it is negative-definite on the semisimple part of the algebra, and zero on the abelian part.

The relevance of these results to us is highlighted in the following way. Consider first a real semi-simple algebra $\mathfrak{g}$ with compact G_0. Pick an $\mathbb{R}$-basis $\{T_i\}$ such that

$$K_{ij} = K(T_i, T_j) = \mathrm{diag}\,(-1, \ldots, -1)\,.$$

which is always possible by Gram–Schmidt. One particularly nice property of this basis is that the structure constants

$$[T_i, T_j] = \mathrm{ad}_{T_i}(T_j) = c_{ij}^k T_k,$$

are constrained by the invariance of the Killing form

$$\ldots$$

thus using the invariance and the symmetry of the Killing form we can derive a constraint on the structure constants, which is a very useful property for computations: they are **completely antisymmetric** in the three indices, thus they are the structure constants of a Lie algebra with respect to an orthonormal basis. We already know that the structure constants were antisymmetric in the first two indices (due to the definition through commutators), but now we have a stronger constraint, which is that they are completely antisymmetric in all three indices.

Passing to the complexification, we can define a $\mathbb{C}$-basis of $\mathfrak{g}_{\mathbb{C}}$, $\{J_m = iT_m\}$, which by Corollary 3.3.2 is orthonormal, i.e., $K^{\mathfrak{g}_{\mathbb{C}}}(J_m, J_n) = K^{\mathfrak{g}_{\mathbb{C}}}(iT_m, iT_n) = -K^{\mathfrak{g}_{\mathbb{C}}}(T_m, T_n) = -K(T_m, T_n) = \delta_{mn}$. In this new basis, the adjoint rep ad of $\mathfrak{g}_{\mathbb{C}}$ satisfies

$$\ldots$$

which is an Hermitian matrix. We now know that in this basis we can diagonalize the adjoint representation, since we have found a basis in which the adjoint representation is Hermitian. So $\mathrm{ad}_{J_l} \in \mathrm{End}(\mathfrak{g}_{\mathbb{C}})$ for the particular generators J_l of the complexified algebra can be diagonalized.

Given the many details / proofs we have left out in arriving at this statement, we summarize the main result in the following:

Corollary 3.5. *Let $\mathfrak{g}$ be a complex semi-simple Lie algebra. Then there exists elements $X \in \mathfrak{g}$ such that $\mathrm{ad}_X \in \mathrm{End}(\mathfrak{g})$ is diagonalizable.*

This is also a sort of justification for the habit of physicists to use infinitesimal transformations with complex coefficients... you can find the real.... ?

In the following, we will mostly work on complex algebras, so $\mathfrak{g}$ will denote a complex semi-simple Lie algebra unless otherwise stated.

Cartan subalgebra

Definition 3.2. We have two statements

(i) toroidal...

(ii) cartan subalgebra...

In the example $\mathfrak{g} = \mathfrak{su}(2)_{\mathbb{C}} = \mathfrak{sl}(2; \mathbb{C})$, we alredy saw that J_3 spans a Cartan subalgebra (since the generators do not commute we can take one of them). However it is also clear that any of the other generators $J_{1/2}$ would equally span a Cartan algebra. So the Cartan is no unique, but basically "a basis change away".

This is made precise in the following

Theorem 3.6. *Let $\mathfrak{h}_1, \mathfrak{h}_2 \subset \mathfrak{g}$ be two Cartan subalgebras of a complex semi-simple Lie algebra $\mathfrak{g}$. Then there exists an element in any group G with such algebra that maps one Cartan subalgebra to the other by the adjoint action:*

$$\mathfrak{h}_1 = \mathrm{Ad}_{\mathfrak{g}}(\mathfrak{h}_2).$$

Consequently, $\dim(\mathfrak{h}_1) = \dim(\mathfrak{h}_2)$.

This is in a sense just a rotation, since the Killing form is left invariant by the adjoint action, thus it is a rotation in the space of generators. Thus all Cartan subalgebras are equivalent.

Definition 3.3. The **rank** of a complex semi-simple Lie algebra $\mathfrak{g}$ is the dimension of its Cartan subalgebra(s) $\mathfrak{h}$:

$$\mathrm{rank}(\mathfrak{g}) := \dim(\mathfrak{h}).$$

Thus $\mathfrak{su}(2)_{\mathbb{C}} = \mathfrak{sl}(2;\mathbb{C})$ has rank 1; more generally, $\mathfrak{su}(n)_{\mathbb{C}} = \mathfrak{sl}(n;\mathbb{C})$ has rank $n-1$; $\mathrm{rank}(\mathfrak{so}(2n)_{\mathbb{C}}) = \mathrm{rank}(\mathfrak{so}(2n+1)_{\mathbb{C}}) = n$.

The name "toral/toroidal" algebra comes from the following. By definition, a toral algebra $\mathfrak{h} \subset \mathfrak{g}$ is Abelian. If we pick a compact real form $\mathfrak{k} \subset \mathfrak{g}$, then $\mathfrak{h} \cap \mathfrak{k}$ is a real toral algebra, which exponentiates into a compact Abelian subgroup (these elements commute), i.e., of the form $\mathrm{U}(1)^n \equiv (\mathbb{S}^1)^n$ which is topologically a torus. A Cartan subalgebra exponentiates into a "maximal torus", which is a common name in the literature. Just as their algebras, maximal tori $\mathcal{H}_{1,2}$ are conjugate to each other:

$$\exists g \in G \,:\, g\mathcal{H}_1 g^{-1} = \mathcal{H}_2.$$

3.4 | Roots and Root Decomposition

In the following, we assume that we have fixed a Cartan algebra $\mathfrak{h} \subset \mathfrak{g}$; other choices would yield identical results given that they differ by a basis change $\mathrm{Ad}_g \in G \to \mathrm{GL}(\mathfrak{g})$. Since by definition all elements of $\mathfrak{h}$ commute, i.e.

$$\forall H, H' \in \mathfrak{h}, \quad [\mathrm{ad}_H, \mathrm{ad}_{H'}] = \mathrm{ad}_{[H,H']} = 0,$$

we can diagonalize ad_H for all $H \in \mathfrak{h}$ simultaneously, i.e., we can find a basis of $\mathfrak{g}$ which consists of (simultaneous) eigenvectors X of all ad_H, whose eigenvalue we denote by $\lambda(H) \in \mathbb{C}$:

$$\implies \mathrm{ad}_H(X) = [H, X] = \lambda(H)X.$$

Such eigenvalues satisfy for any $H_1, H_2 \in \mathfrak{h}$:

$$\begin{aligned} \mathrm{ad}_{H_1+H_2}(X) &= \mathrm{ad}_{H_1}(X) + \mathrm{ad}_{H_2}(X) = \lambda(H_1)X + \lambda(H_2)X = (\lambda(H_1) + \lambda(H_2))X, \\ \mathrm{ad}_{cH_1}(X) &= c\,\mathrm{ad}_{H_1}(X) = c\lambda(H_1)X, \quad c \in \mathbb{C}, \end{aligned}$$

So every (simultaneous) eigenvector $X \in \mathfrak{g}$ defines a linear functional $\lambda : \mathfrak{h} \to \mathbb{C}$, $H \mapsto \lambda(H)$, i.e., $\lambda \in \mathfrak{h}^\vee$ is an element of the vector space dual to the Cartan algebra.

For any $\alpha \in \mathfrak{h}^\vee$, we can then define the eigenspace

$$\mathfrak{g}_\alpha := \{X \in \mathfrak{g} \,:\, \mathrm{ad}_H(X) = [H, X] = \alpha(H)X, \forall H \in \mathfrak{h}\},$$

i.e., the set of eigenvectors whose eigenvalues agree with the linear functional α. Note that, as defined here, $\mathfrak{g}_\alpha$ is a vector space, but a priori not a Lie algebra. Clearly, a generic $\alpha \in \mathfrak{h}^\vee$ will not be the eigenvalues of ad_H for any $H \in \mathfrak{h}$, so $\mathfrak{g}_\alpha = \{0\}$ generically. Furthermore, since $\mathfrak{h}$ is the maximal Abelian subalgebra, it is easy to see that $\mathfrak{h} = \mathfrak{g}_0$. So this is a vector space but not a Lie algebra in general.

Definition 3.4 (Root system)**.** [...]

The set of all dual vector as appears ...

Lemma 3.7. *$\mathcal{R}$ is a finite set.*

Proof. This is obvious from the fact that eigenspaces $\mathfrak{g}_\alpha$ associated to different eigenvalues of a diagonalizable endomorphism are linearly independent. Since $\mathfrak{g} \supset \mathfrak{g}_\alpha$ is finite dimensional, there can only be finitely many α such that $\mathfrak{g}_\alpha \neq \{0\}$. □

Let us revisit $\mathfrak{g} = \mathfrak{sl}(2;\mathbb{C})$. We fix $\mathfrak{h} = \{\mu J_3 \,|\, \mu \in \mathbb{C}\} \subset \mathfrak{sl}(2;\mathbb{C})$ as the Cartan algebra. Given the commutation relation $[J_3,\, J_\pm] = \pm J_\pm$, we see that ad_{J_3} only has ± 1 as non-zero eigenvalues. So $R = \{\alpha_+ \equiv \alpha, \alpha_- = -\alpha\} \subset \mathfrak{h}^\vee$ with $\alpha : \mathfrak{h} \to \mathbb{C}$, $J_3 \mapsto 1$, and $\mathfrak{g}_{\alpha_\pm} = \{\mu J_\pm \,|\, \mu \in \mathbb{C}\}$. Notice how, as a vector space, $\mathfrak{sl}(2;\mathbb{C}) = \mathfrak{h} \oplus \mathfrak{g}_{\alpha_+} \oplus \mathfrak{g}_{\alpha_-}$. These features, together with a few other ones that are easy to check explicitly for $\mathfrak{sl}(2;\mathbb{C})$, are the content of the following theorem.

Theorem 3.8 (Root decomposition)**.** *We fix $\mathfrak{h} \subset \mathfrak{g}$ as the Cartan subalgebra, and R the root system of $\mathfrak{g}$ with $\mathfrak{h} = \mathfrak{g}_0$.*

(i) We can decompose the Lie algebra as a vector space as

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \mathcal{R} \Subset} \mathfrak{g}_\alpha,$$

as the cartan subalgebra plus the root spaces. This is called the root decomposition of the algebra. It is not a sum of algebras, since the root spaces are not closed under the Lie bracket, they are just vector spaces.

(ii) If we take two such elements ... then the commutator lives in the space associated to the sum of the two roots ... and the brackets must be closed...

(iii) If we have two roots that do not sum to zero ...

(iv) For any root α, one can take the killing form restricted to the root space $\mathfrak{g}_\alpha$ and the root space $\mathfrak{g}_{-\alpha}$, and this gives a non-degenerate pairing between these two spaces, thus they have the same dimension.

Proof. (i) Follows directly from $\{\mathrm{ad}_H \,|\, H \in \mathfrak{h}\}$ being simultaneously diagonalizable, so $\mathfrak{g}$ decomposes into the direct sum of the eigenspaces, which are by definition the root subspaces $\mathfrak{g}_\alpha$.

(ii) Direct computation gives for any $H \in \mathfrak{h}$:

$$\ldots$$

(iii) We can rewrite (ii) also as $\mathrm{ad}_{E_\alpha}(E_\beta) \in \mathfrak{g}_{\alpha+\beta}$. So, for $E_\gamma \in \mathfrak{g}_\gamma$ we have

$$\mathrm{ad}_{E_\alpha} \circ \mathrm{ad}_{E_\beta} = \sum_{\alpha,i} \lambda_\alpha^{(i)} E_\alpha^{(i)} \in \mathfrak{g}_{\alpha+\beta+\gamma},$$

So if $\alpha + \beta \neq 0$ then $E_\gamma \notin \mathfrak{g}_{\alpha+\beta+\gamma}$, which implies that $\lambda_\gamma^{(i)} = 0$: all of the diagonal elements of the matrix representing this endomorphism in the root-eigenbasis are zero.

Then, unless $\alpha + \beta = 0$, $\mathrm{ad}_{E_\alpha} \circ \mathrm{ad}_{E_\beta}$ maps any basis vector T which by (i) can be chosen to be an eigenvector with certain eigenvalue $\gamma \in \mathcal{R}$, to a linear combination of basis vectors that are eigenvectors with eigenvalues $\alpha + \beta + \gamma \neq \gamma$. This again implies that the diagonal entries for the matrix of $\mathrm{ad}_{E_\alpha} \circ \mathrm{ad}_{E_\beta}$ in the root-eigenbasis must be all 0, so $\mathrm{Tr}(\mathrm{ad}_{E_\alpha} \circ \mathrm{ad}_{E_\beta}) = K(E_\alpha, E_\beta) = 0$.

(iv) By (iii), the only elements $0 \neq Y \in \mathfrak{g}$ such that $K(X,Y) \neq 0$ for an element $0 \neq X \in \mathfrak{g}_\alpha \subset \mathfrak{g}$ must be in $\mathfrak{g}_{-\alpha} \subset \mathfrak{g}$. Since we assumed the Lie algebra to be semi-simple, then its Killing form K is non-degenerate on $\mathfrak{g}$: there must exist such a $0 \neq Y \in \mathfrak{g}_{-\alpha}$. So K provides a non-degenerate pairing $\mathfrak{g}_\alpha \times \mathfrak{g}_{-\alpha} \to \mathbb{C}$. Thus, if $\mathfrak{g}_\alpha$ is non-trivial, also $\mathfrak{g}_{-\alpha} \neq \{0\}$, hence $-\alpha \in \mathcal{R}$.

□

Thus if we take commutators of vectors from different groups, we get vectors in the group associated to the sum of the roots. In particular, if we take commutators of vectors in the same group, we get vectors in the group associated to twice the root. So if $2\alpha \notin \mathcal{R}$ then $\mathfrak{g}_{2\alpha} = \{0\}$ and thus $[\mathfrak{g}_\alpha, \mathfrak{g}_\alpha] = \{0\}$.

TODO: check

This is the single most important result about semi-simple Lie algebras. Everything we will learn further about their structure, including the classification and constructing their representations, builds on this decomposition.

Definition 3.5. Using the fact that $K|_{\mathfrak{h}}$ is a non-degenerate symmetric pairing, we have a vector space isomorphism $\kappa : \mathfrak{h}^\vee \xrightarrow{\simeq} \mathfrak{h}$. This gives us a non trivial notion of duality, which is invertible:

$$\begin{aligned}\kappa^{-1} \ : \mathfrak{h} &\xrightarrow{\simeq} \mathfrak{h}^\vee, \\ H &\mapsto \kappa(H, \cdot).\end{aligned}$$

1. Let $\alpha \in \mathfrak{h}^\vee$, then we can denote $H_\alpha := \kappa(\alpha)$. It satisfies $K(H_\alpha, H_\beta) = \alpha(H_\beta)$ for all $\alpha, \beta \in \mathfrak{h}^\vee$.
2. ...

For a better intuition about it, let us fix a basis $\{H_i\}$ of $\mathfrak{h}$, in which $K|_{\mathfrak{h}}$ has the matrix $\kappa_{ij} = K(H_i, H_j)$. In the dual basis $\{\theta_i\}$ of $\mathfrak{h}^\vee$ (i.e., $\theta_i(H_j) = \delta_{ij}$), we have for all $\alpha \in \mathfrak{h}^\vee$ the equation

$$H_\alpha = \kappa(\alpha_i \theta_i) = \kappa_{ij} \alpha_i H_j,$$

where $\kappa_{ij} \equiv (\kappa^{-1})_{ij}$ is the inverse matrix, which also encodes the pairing on $\mathfrak{h}^\vee$:

$$\begin{aligned}(\alpha, \beta) = K(H_\alpha, H_\beta) = K(\kappa^{ij}\alpha_i H_j, \kappa^{lm}\beta_l H_m) &= \alpha_i \beta_l \kappa^{ij} \kappa^{lm} K(H_j, H_m) \\ &= \alpha_i \beta_l \kappa^{ij} \kappa^{lm} \kappa_{jm} = \alpha_i \beta_l \kappa^{ij} \delta^l{}_j = k^{ij} \alpha_i \beta_j.\end{aligned}$$

For $\mathfrak{g} = \mathfrak{sl}(2; \mathbb{C})$, we have used for $\mathfrak{h}$ the basis $H = J_3$, with $K(J_3, J_3) = 2$ and

$$[J_3, X] = \alpha(J_3) X, \quad [J_e, J_\pm] = \pm J_\pm,$$

and the roots $\alpha^\pm : J_3 \mapsto \pm 1$ associated with eigenvectors $E_{\pm\alpha} = J_\pm$:

$$\mathfrak{g}_{\alpha^\pm} = \{\mu J_\pm \,|\, \mu \in \mathbb{C}\} = \{\mu E_{\pm\alpha} \,|\, \mu \in \mathbb{C}\}.$$

Then:

1. $H_\alpha = \kappa(\alpha) = \lambda J_3 \in \mathfrak{h}$, is determined by the condition...

   If we choose a different basis for the Cartan subalgebra, for example J_1 instead of J_3, then we would have a different root system. However, the root system is basically the same...
2. $\alpha(J_3) = 1$, they form a dual pair of basis for...

We found the roots of the root system identifying the eigenvalues of the adjoint representation of the Cartan subalgebra... We restrict the killing form to the cartan subspace and use it to define a non-degenerate pairing (given by a non degenerate matrix) between the root spaces associated to opposite roots, which allows us to find the dual vector H_α associated to each root α (it gives an isomorphism between $\mathfrak{h}^\vee$ and $\mathfrak{h}$). We can use this matrix κ to identify vectors in the Cartan subalgebra with linear functionals on it. We can use this pairing as a geometric euclidean pairing on the root space, which allows us to define angles and lengths of roots (through scalar products). We can use the root decomposition to find the commutation relations among the generators of the algebra, which are constrained by the root structure.

Let us discuss some further properties of the root decomposition, which are important for the classification of representations.

Lemma 3.9. *Let $E_+ \in \mathfrak{g}_\alpha$ and $E_- \in \mathfrak{g}_{-\alpha}$, then*

$$[E_+, E_-] = K(E_+, E_-) = H_\alpha.$$

... □

Lemma 3.10. *Let $E_+ \in \mathfrak{g}_\alpha$ and $E_- \in \mathfrak{g}_{-\alpha}$, then*

$$[E_+, E_-] = K(E_+, E_-)H_\alpha,$$

for $\alpha \in \mathcal{R}$, where $H_\alpha = \kappa(\alpha)$ is the dual vector associated to α through the Killing form.

Proof.

$$H := \dots$$

$$K\big|_{\mathfrak{h}} = \dots$$

This means that by leaving the second argument of the Killing form free, we can identify it with the dual vector H_α associated to α through the Killing form...

$$H = \kappa(\kappa^{-1}(H)) = \dots$$

□

Lemma 3.11. *Let $\alpha \in \mathcal{R} \subset \mathfrak{h}^\vee$ be a root $\mathfrak{g}$. Then $(\alpha, \alpha) = K(H_\alpha, H_\alpha) \neq 0$.*

Proof. Let us make a futher assumption: the basis of the cartan subalgebra can diagonalize ... such that ad_{H_α} is Hermitian (highly non trivial and we cannot prove it at this point, but it is true). Then we can compute

$$K(H_\alpha, H_\alpha) = \mathrm{Tr}(\mathrm{ad}_{H_\alpha} \circ \mathrm{ad}_{H_\alpha}) = \sum_{\beta \in \mathcal{R}} \beta(H_\alpha)^2 \dim(\mathfrak{g}_\beta),$$

we knew that we can diagonalize ad_{H_α}, but knowing that it is Hermitian allows us to conclude that its eigenvalues are real, thus $\beta(H_\alpha)^2 \geq 0$ for all $\beta \in \mathcal{R}$ (also the dimension of the subspaces are positive), so the sum is positive, thus $K(H_\alpha, H_\alpha) > 0$.

Thus assuming the hermiticity let us define the "length" of a root α as $|\alpha| := \sqrt{(\alpha, \alpha)} = \sqrt{K(H_\alpha, H_\alpha)}$. We can now divide by (α, α). □

Lemma 3.12. *Let $J_+^{(\alpha)}$ ...*

$$K(J_+^{(\alpha)}, J_-^{(\alpha)}) = \frac{2}{(\alpha, \alpha)}.$$

We needed the previous lemma to be able to divide by (α, α) and thus to define the normalized generators $J_3^{(\alpha)}$ associated to the root α:

$$J_3^{(\alpha)} := \frac{H_\alpha}{(\alpha, \alpha)}.$$

Then...

The claim is that we have found a subalgebra of $\mathfrak{g}$ isomorphic to $\mathfrak{sl}(2; \mathbb{C})_\alpha$ associated to each root α, which is generated by the three elements $J_+^{(\alpha)}$, $J_-^{(\alpha)}$ and $J_3^{(\alpha)}$. This is called the ***root subalgebra*** *associated to the root α.*

Proof. By direct computation

$$\dots$$

Evaluating α on its dual vector H_α we get $\alpha(H_\alpha) = K(H_\alpha, H_\alpha) = (\alpha, \alpha)$ is just the norm; the commutator among H_α and $J_\pm^{(\alpha)}$ is just the eigenvalue of H_α on the root space $\mathfrak{g}_{\pm\alpha}$, since the latter ar eigenvalues of ad_{H_α} by definition.

We have in the end found the commutation relations of $\mathfrak{sl}(2; \mathbb{C})$ for the generators $J_3^{(\alpha)}$, $J_\pm^{(\alpha)}$. □

Every root of a semisimple lie algebra defines a subalgebra isomorphic to $\mathfrak{sl}(2;\mathbb{C})$, which is generated by the three elements $J_3^{(\alpha)}$, $J_\pm^{(\alpha)}$. This is a very important result, since it allows us to describe the lie algebra in terms of the representations of this subalgebra; in particular, taking the adjoint representation of the lie algebra, we can restrict it to the root subalgebra (spanned by J_+^α, J_-^α and J_3^α), and thus we can decompose the algebra into irreducible representations of $\mathfrak{sl}(2;\mathbb{C})$, on the space of endomorphisms of the algebra itself:

$$\mathrm{ad}^{(\mathfrak{g})}\big|_{\mathfrak{sl}(2;\mathbb{C})_\alpha} : \mathfrak{sl}(2;\mathbb{C})_\alpha \to \mathrm{End}(\mathfrak{g}).$$

Proposition 3.13. *Let $\alpha \in \mathcal{R}$ be a root of a complex semi-simple Lie algebra $\mathfrak{g}$. Then let the root subalgebra $\mathfrak{sl}(2;\mathbb{C})_\alpha = \mathrm{span}_\mathbb{C}\{J_3^{(\alpha)}, J_+^{(\alpha)}, J_-^{(\alpha)}\}$ associated to α. Then the subspace*

$$V \coloneqq \dots$$

this is a vector space which defines an irreducible representation of $\mathfrak{sl}(2;\mathbb{C})_\alpha$ under the adjoint action (induced by the restriction on the adjoint representation on the subspace): it is an $\mathfrak{sl}(2;\mathbb{C})$ irreducible representation under

$$\left\{ \mathrm{ad}^{(\mathfrak{g})}_{J_\pm^{(\alpha)}}\Big|_V, \mathrm{ad}^{(\mathfrak{g})}_{J_3^{(\alpha)}}\Big|_V \right\}.$$

Proof. First recall... spin... It is carachterized by

$$d(J_3)\left|j,m\right\rangle = \dots$$

Now, since $\mathfrak{g}_\alpha$ are orthogonal to each other (except for $\pm\alpha$), there can only be a finite set of $m \in \mathbb{Z}\backslash\{0\}$ for which $\mathfrak{g}_{m\alpha}$ is non-trivial, since $V \subset \mathfrak{g}$ must be finite dimensional.

We want to show that V is invariant under $\left\{ \mathrm{ad}^{(\mathfrak{g})}_{J_\pm^{(\alpha)}}\Big|_V, \mathrm{ad}^{(\mathfrak{g})}_{J_3^{(\alpha)}}\Big|_V \right\}$, in order to prove that it is a representation space of $\mathfrak{sl}(2;\mathbb{C})_\alpha$.

- ...
- ...the relevant cases are with $m = \pm 1$ since in the definition of V we are excluding 0 and if we act with $J_\pm^{(\alpha)}$ we get ± 1 in the eigenvalue, so we need to check the cases for $m = \pm 1$ to be sure of the result...

This shows that this operators restricted on V are endomorphisms of V, thus V is a representation space of $\mathfrak{sl}(2;\mathbb{C})_\alpha$.

From the first bullet point we know that the basis vectors of V are eigenvectors of $\mathrm{ad}_{J_3^{(\alpha)}}$ with eigenvalues $m \in \mathbb{Z}\backslash\{0\}$. Furthermore we know from this first point that the $J_3^{(\alpha)}$-eigenbasis in V has integer eigenvalues. If the subspace V splits in a direct sum of subspaces

$$\dots$$

i.e. it is reducible, then the irreducible components must be integer spin representations of $\mathfrak{sl}(2;\mathbb{C})$, which have a $J_3^{(\alpha)}$-eigenvalue 0 subspace, which is one dimensional, so that the $J_3^{(\alpha)}$-eigenvalue 0 subspace of V will have dimension N. However from the definition of V we have that this eigenspace is simply

$$\left\{ \mu J_3^{(\alpha)} \middle| \mu \in \mathbb{C} \right\},$$

which is one dimensional, so $N = 1$ and thus V is irreducible. $\square$

This leads to a set of practically useful facts about the roots of semi-simple Lie algebras:

Theorem 3.14. *Let $\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha\in\mathcal{R}} \mathfrak{g}_\alpha$ be the root decomposition of a complex semi-simple Lie algebra $\mathfrak{g}$ with respect to a Cartan subalgebra $\mathfrak{h}$. Then the following properties hold:*

(i) *a*

(ii) *Every root subspace is one dimensional, i.e.,* $\dim(\mathfrak{g}_\alpha) = 1$ *for all* $\alpha \in \mathcal{R}$.

(iii) *This particular pairing between vectors we are claiming that they are in* $\mathbb{Z}$ *i.e. they are integers...*

(iv) *On the dual space we can define a linear map called reflection which takes any vector... if we tae anothr root, its reflection is also a root... if we reflects two roots with respect to each other, we get a rotation in the plane spanned by these two roots, and this rotation maps roots to roots...*

(v) *For any* $\alpha \in \mathcal{R}$, *the only multiples of* α *which are roots are* $\pm\alpha$.

(vi) *For roots* α, $\beta \neq \alpha$, *then any*

$$W = \bigoplus_{k\in\mathbb{Z}} \mathfrak{g}_{\beta+k\alpha},$$

is an irreducible representation of the root subalgebra $\mathfrak{sl}(2;\mathbb{C})_\alpha$ *associated to* α.

Proof. We can prove these properties one by one.

(i) If $\mathcal{R}$ does not span $\mathfrak{h}^\vee$ then there exists a non-zero element $H \in \mathfrak{h}$ such that $\alpha(H) = 0$ for all $\alpha \in \mathcal{R}$. This means that H commutes with all the generators of the algebra, thus it is in the center of the algebra, which contradicts the assumption of semi-simplicity...

(ii) This follows from Prop. 3.4.8: since $\{\mu J_3^{(\alpha)} | \mu \in \mathbb{C}\} \bigoplus$ ...

(iii) $J_3^{(\alpha)}$ every eigenvalue has to be integer or half integer, thus this fraction must be an integer... the pairing is not symmetric and it is only linear in the second argument.

(iv) For any pair of roots their angular bracket is integer and from the previous proof we saw that $E_\beta \in \mathfrak{g}_\beta$ has $J_3^{(\alpha)}$ eigenvalue is $\frac{n}{2}$, then there must be.... we can think of it as ....

$$0 \neq \left|j, -\tfrac{n}{2}\right\rangle = \begin{cases} [\ldots] & n > 0, \\ [\ldots] & n < 0. \end{cases}$$

Thus in the end we get

$$\implies \ldots$$

(v) Assume that α and $\beta = c\alpha$, $c \in \mathbb{C}$, are both roots. By (iii), $\langle\alpha,\beta\rangle = 2c$ and$\langle\beta,\alpha\rangle = 2/c$ must be integer, so we have the possibilities $c = \pm1$, ±2, $\pm\frac{1}{2}$...

By part (ii) of the theorem, we know...

So 2α is not a root. Analogously, we can argue for $\mathfrak{g}_{-\alpha} = \{\mu J_-^{(\alpha)} | \mu \in \mathbb{C}\}$ to be the lowest weight subspace, and so $\mathfrak{g}_{m\alpha} = \{0\}$ for $m < 1$, i.e., -2α is not a root.

(vi) ...

$\square$

The next corollary will directly tell us how to construct the entire algebra from the root system, and thus it is the key to the classification of complex semi-simple Lie algebras.

Corollary 3.15. *We only need to know the finite set of root vectors and the pairings among them to reconstruct the entire algebra.*

As a vectors space, the algebra is a direct sum of one dimensional root spaces, each associated to a root, plus the Cartan subalgebra. The Lie bracket is determined by the root structure and the pairing among the root vectors.

- *For each root we find basis elements and generators for the cartan subspace which carries some symetric inner product, which along with the pairing directly defines the killing form. The norm is already defined*
- *For every* $X = J_3^{(\alpha)}$, E_α *we can define the adjoint action of every generator of the algebra on it (define the crresponding endomorphism* $\mathrm{ad}_X \in \mathrm{End}(\mathfrak{g})$*), which is determined by the root structure and the pairing, thus we can find the commutation relations among all the generators of the algebra:*
  - ...
  - ...
  - ...

the last is the only non trivial, but after showing that we have a irreducible representation of $\mathfrak{sl}(2;\mathbb{C})$ we can use the known representation theory of $\mathfrak{sl}(2;\mathbb{C})$ so that we can use E_α as a raising operator to find the commutation relations with all the other generators of the algebra, which are determined by the root structure and the pairing...

Proof. Almost everything follows straightforwardly from Theorem 3.4.9 and its proof. The only non-trivial points are the numerical factors in the last equation. To see these... ☐

Summary

Let us summarize what we have seen so far.

- By viewing a complex semi-simple $\mathfrak{g}$ as the representation space of the adjoint representation $(\mathrm{ad}, \mathfrak{g})$, we are naturally led to consider a vector space basis for $\mathfrak{g}$ consisting of simultaneous eigenvectors of the maximally commuting subalgebra (the Cartan subalgebra) $\mathfrak{h}$ of $\mathfrak{g}$.
- The standard diagonalization theorem directly implies the root decomposition Theorem 3.4.3, where the eigenspaces $\mathfrak{g}_\alpha$ are labelled by the eigenvalues $\alpha \in \mathcal{R} \subset \mathfrak{h}^\vee$ called roots.
- Every root $\alpha \in \mathcal{R}$ is associated with a subalgebra $\mathfrak{sl}(2;\mathbb{C})_\alpha \subset \mathfrak{g}$; through the adjoint action, $\mathfrak{g}$ becomes a (generally reducible) representation of $\mathfrak{sl}(2;\mathbb{C})_\alpha$. Using the known representation theory of $\mathfrak{sl}(2;\mathbb{C})$, we could deduce a series of properties of $\mathfrak{g}$ and the roots, which culminated in Theorem 3.4.9.
- These results allow us to essentially completely reconstruct the algebra $\mathfrak{g}$ from its roots. In turn, the stringent consistency conditions of roots make them much easier to quantify and categorize. This will become particularly clear in the following section.

3.5 | The Geometry of Roots

So far we have treated roots as elements of the dual vector space $\mathfrak{h}^\vee$ of the Cartan subalgebra $\mathfrak{h}$, which is a complex vector space. In this section we will see how they have a geometric meaning in real Euclidean space, which makes them (almost) visualizable.

Theorem 3.16. *This theorem gives two statements which let us embed the roots in a real Euclidean space.*

1. *Let $\mathfrak{h}_\mathbb{R} \subset \mathfrak{h}$ be the real vector space spanned by $H_\alpha \in \mathcal{R}$ for some $\alpha \in \mathcal{R}$. Then $\mathfrak{h} = \mathfrak{h}_\mathbb{R} \oplus i\mathfrak{h}_\mathbb{R} \cong (\mathfrak{h}_\mathbb{R})_\mathbb{C}$. The restriction of the Killing form to $\mathfrak{h}_\mathbb{R}$ is $\mathbb{R}$-valued and positive definite.*
2. *Let $\mathfrak{h}_\mathbb{R}^\vee \subset \mathfrak{h}^\vee$ be the real vector space spanned by $\alpha \in \mathcal{R} \subset \mathfrak{h}^\vee$. Then $\mathfrak{h}^\vee = (\mathfrak{h}_\mathbb{R}^\vee)_\mathbb{C}$. Also, $\mathfrak{h}_\mathbb{R}^\vee = \{\lambda \in \mathfrak{h}^\vee | \lambda(H) \in \mathbb{R} \forall H \in \mathfrak{h}_\mathbb{R}\} = (\mathfrak{h}_\mathbb{R})^\vee$.*

[...]

[...]

Example ($\mathfrak{g} = \mathfrak{sl}(2, \mathbb{C})$). [...]

[...]

Example ($\mathfrak{g} = \mathfrak{sl}(3, \mathbb{C}) = \mathfrak{su}(3)_\mathbb{C}$).

[...]

For an alternative approach, we quote a fact without proof, which is that for a simple

[...]

Thus we have the following Euclidean representation of the roots (up to an overall rescaling of their lengths):

[*drawing*]

The $\mathfrak{sl}(3, \mathbb{C})$ example shows that roots can be written as integer linear combinations of a subset of them. To make this precise, we have the following definitions.

Definition 3.6. Let $0 \neq t \in \mathfrak{h}_\mathbb{R}^\vee \subset \mathfrak{h}^\vee$ be a non-zero element of the real dual space of $\mathfrak{h}_\mathbb{R}$. ... Then we can split the roots in a set of positive and negative roots, classification justified by projection on...

In the previous example we are implicitly using the element t to split the roots into positive and negative ones, which is why we have the particular choice of positive roots. We can also choose a different t to get a different set of positive roots, but they will be related by a Weyl group transformation, so the choice of t does not affect the structure of the root system.

Positive/negative roots are up to a choice of t; for $\mathfrak{sl}(3, \mathbb{C})$, we have shown the standard choice in the figure above. The "symmetries" of roots ensure that two different choices are related again by a "change of basis". The same applies to the following

Definition 3.7. Fix a choice of positive/negative roots. A positive root $\alpha \in \mathcal{R}^+$ is called *simple* if it cannot be written as the sum of two positive roots. We denote the set of simple roots by Δ.

In the previous example, we have $\Delta = \{\alpha_1, \alpha_2\}$: we cannot write α_1 or α_2 as the sum of two positive roots, but we can write α_3 as a sum of positive roots. The simple roots are the "building blocks" of the root system, in the sense that every root can be written as an integer linear combination of simple roots with coefficients all positive or all negative.

Theorem 3.17. *Every $\alpha \in \mathcal{R}$ can be uniquely written as an integer linear combination of simple roots $\Delta = \{\alpha(1), \alpha(2), \ldots, \alpha(r)\}$:*

$$\alpha = \sum_{i=1}^{r} n_i \alpha^{(i)}, \quad n_i \in \mathbb{Z}, r = \mathrm{rank}(\mathfrak{g}) = \dim(\mathfrak{h}).$$

If $\alpha \in \mathcal{R}^+$, then all $n_i \geq 0$; if $\alpha \in \mathcal{R}^-$, then all $n_i \leq 0$.

Proof. We will just sketch the proof.

The first step is to prove that every positive root, i.e., every $\alpha \in \mathcal{R}$ such that $(\alpha, t) > 0$, is a sum of simple roots. So let $\alpha \in \mathcal{R}^+$, $\alpha = \alpha' + \alpha''$ be non-simple, and α', α'' be positive $\implies$ $(t, \alpha) > (t, \alpha')$ and $(t, \alpha) > (t, \alpha'')$. If one of these two are not simple, say α', we can then continue splitting it into other positive roots $\alpha' = \alpha'' + \alpha'''$ with $(t, \alpha''('')) < (t, \alpha') < (t, \alpha)$. Since there are only finitely many roots, (t, α) takes finitely many values, and so this splitting has to eventually terminate at some simple roots. But then we have effectively expressed α as a sum of simple roots.

The same argument can be given for negative roots, choosing $\alpha \in \mathcal{R}^-$ and splitting it into negative roots, which are just the negatives of positive roots (for the reflection), and then applying the same argument as above.

The second step is then to ... (just sketch)

□

Definition 3.8 (Root Lattice)**.** The $\mathbb{Z}$-span of $\mathcal{R}$ inside the Euclidean space $\mathfrak{h}_{\mathbb{R}}^{\vee}$ is called the root lattice $\Lambda_{\mathcal{R}}$. It is a finitely generated free Abelian group (under vector addition) with generators $\{\alpha^{(1)}, \alpha^{(2)}, \ldots, \alpha^{(r)}\}$, where $\alpha^{(i)}$ are the simple roots.

3.6 | Weights and Representations

We only treated the adjoint representation of $\mathfrak{g}$ so far, but we can also consider other representations. In this section we will see how the roots of $\mathfrak{g}$ are related to the weights of its representations, and how the geometry of the roots can be used to understand the structure of the representations.

We will study the eigenvalues of the Cartan subalgebra $\mathfrak{h}$ operator in other representations.

[...]

Thus we are basically going to repeat the same analysis we did for the adjoint representation with the roots, but now for a generic representation (d, V) of $\mathfrak{g}$, getting to the weights. We will see that the structure of the weights of (d, V) is closely related to the structure of the roots of $\mathfrak{g}$, and that we can use the geometry of the roots to understand the structure of the weights.

Definition 3.9. Given a complex representation (d, V) of $\mathfrak{g}$ with a fixed Cartan subalgebra $\mathfrak{h}$, an element $\omega \in \mathfrak{h}^\vee$ of the vector space dual to the Cartan subalgebra is called a weight of (d, V), if

$$\dots$$

Conversely...

$$\dots$$

[...]

$$\Omega \dots$$

This is the analogous construction to the roots, but now for a generic representation instead of the adjoint representation, defining the **weight system**.

By standard diagonalization theorems, this implies a **weight decomposition** of

$$V = \bigoplus_{\omega \in \Omega_d} V_\omega,$$

of the representation space, similarly to the root ecomposition of $\mathfrak{g}$. For a finite dimensional representation, it also means that the set Ω_d of weights is finite.

> **Example ($\mathfrak{g} = \mathfrak{sl}(2, \mathbb{C})$).** For $\mathfrak{g} = \mathfrak{sl}(2, \mathbb{C})$, the irreps of spin j have the obvious decomposition [...]
>
> $$d(J_3) |j, m\rangle = m |j, m\rangle$$
>
> ...

By diagonalizing the cartan subalgebra, we immediatly know how the ... act on the weight spaces, and we can use this to understand the structure of the representation.

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus \mathfrak{g}_\alpha,$$

where... By taking $E_\alpha \in \mathfrak{g}_\alpha$ and $y \in V_\omega$ a weight vector, we have by direct computation that

$$\dots$$

So the Lie algebra elements in $\mathfrak{g}_\alpha$ raise / lower weights by the positive / negative root α. Again, this is familiar from the $\mathfrak{sl}(2, \mathbb{C})$ example. Starting from a weight ω, we can raise or decrease the cartan eigenvalue by adding or subtracting roots, and we can use this to understand the structure of the representation.

Proposition 3.18. *If (d, V) is an irrep of $\mathfrak{g}$, then the difference of any two weights is in the root lattice:*

$$\dots$$

such that it is an integer linear combination of roots.

... invariant since cartan dont change anything and the other operators from the weight spaces just add or subtract weights, so we can only get weights that differ by integer linear combinations of weights. □

we can use the $\mathfrak{sl}(2, \mathbb{C})$ ...

$$\dots$$

where now we think of $\omega \in \mathfrak{h}_{\mathbb{R}}$. ...

Theorem 3.19. *The weights ω of any representation of $\mathfrak{g}$ satisfy*

$$\dots$$

the integrality condition.

[Definition and Proposition 3.6.4.]

Example ($\mathfrak{g} = \mathfrak{su}(2)_{\mathbb{C}} = \mathfrak{sl}(2, \mathbb{C})$). ... The foundamentl weight is the one sending J_3 to $\frac{1}{2}$. It is the highest weight of the spin $\frac{1}{2}$ representation, and it is the only weight of that representation. The spin j representation has highest weight j and weights $j, j-1, \dots, -j$.

Since the weight lattice the $\mathbb{Z}$-span of the fundamental weights, it means that any weight ω is an integer multiple of w

...

Example ($\mathfrak{g} = \mathfrak{su}(3)_{\mathbb{C}} = \mathfrak{sl}(3, \mathbb{C})$). We have already seen an explicit parametrization of the lie algebra $\mathfrak{sl}(3, \mathbb{C})$ in terms of the Gell-Mann matrices, and we can use that to find the roots and weights of the adjoint representation. We have already found the two roots, after computing the commutators with the Cartan subalgebra. ...

$$\Delta = \dots$$

...

$$K(H_i, H_j) = \mathrm{Tr}_{\mathfrak{g}}(\mathrm{ad}_{H_i} \mathrm{ad}_{H_j}) = \kappa_{ij}$$

and

$$\mathrm{Tr}_V(d(H_i) d(H_j)) = \lambda \kappa_{ij} = C_{ij}$$

where (d, V) is the defining representation of $\mathfrak{sl}(3, \mathbb{C})$ on $\mathbb{C}^3$, and λ is a constant that depends on the normalization of the trace. But when computing the weights, we are only interested in the ratio of these two traces, so the constant λ will not affect the result. ... Since we want to find the foundamental weight, we need to find the ...

We want to find the parametrization $\omega = a_1 \alpha^{(1)} + a_2 \alpha^{(2)}$ such that the following is a fundamental weight:

$$\left\langle \alpha^{(k)}, \omega \right\rangle = \dots$$

... think about raw vector space, and then we can use the inner product to find the coordinates of the fundamental weights in the root basis. Just algebra and basis change, we can find the coordinates of the fundamental weights in the root basis, and then we can use that to find the coordinates of the roots in the fundamental weight basis, which is what we want to plot them in. ...

[*drawing: plusses are previous hexagon roots, fundamental weights are w1 and w2 in blue and orange, then applying linear combinations of the roots to the fundamental weights we get the other weights of the defining representation, which are the vertices of the triangle (one blue and one orange)*]

What is the relation among this geometry and the structure of the representation?

$$w^{(1)} = (1, 0) \implies W^{(i)} \in \mathfrak{h}^\vee \text{ such that } \dots$$

def rep and action of H_1 and H_2 on canonical basis column vectors, looking at the eigenvalues we find the weights of the defining representation, then he rewrites as the sum of the fundamental weights and the roots.

[...]

The three blue dots then represented the weights of the defining representation, which are the vertices of the triangle, and the six plusses are the roots, which are the vertices of the hexagon. Then the orange ones related to the complex conjugate of the defining representation, which is the anti-fundamental representation, and its weights are the vertices of the triangle with orange vertices. They are indeed made by subtracting the roots from the fundamental weights, ... The complex conjugate and the foundamental representations are not the same since they have different weights. [...]

Appendices

A | Notation and Conventions

B | Computations and further developments